



Automatic Darkness Activated Alarm and Load Control System Using LDR, BC547 Transistor, Relay, Lamp, and Buzzer in Proteus

Proteus Based Simulation and Analysis of an LDR Controlled Alarm and Load Circuit

Aysel Ismayilova

School of Science and Engineering
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Aysel Ismayilova

Student No.

Supervisor: Telman Askeraliyev
Supervisor

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Abstract

This report presents the design, calculation, simulation, and troubleshooting analysis of an automatic darkness activated alarm and load control system using an LDR, a BC547 transistor, a relay, a lamp, and a buzzer in Proteus. The project addresses the need for automatic response to reduced light intensity without manual switching. The LDR senses the surrounding light condition, the resistor network converts the LDR resistance variation into a sensing voltage, the transistor operates as a switching stage, and the relay transfers the control action to the output load circuit. When sufficient light falls on the LDR, the sensing voltage remains low, the transistor remains off, and the lamp and buzzer stay inactive. When darkness occurs, the LDR resistance increases, the sensing voltage rises, the transistor turns on, the relay energizes, and the lamp and buzzer become active. The report also includes theoretical calculations for the voltage divider, base current, relay coil current, transistor saturation, LED current, flyback diode protection, and power dissipation. Simulation and troubleshooting sections verify the intended operation and identify practical hardware limits such as relay current, base drive margin, component rating, and load side safety. The project demonstrates a complete sensor to actuator pathway using basic electronic components.

Keywords: LDR, BC547 transistor, relay control, Proteus simulation, darkness activated alarm, lamp and buzzer.

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1

Description

1.1 Project Identity

1.1.1 Project Title

The title of this project is Automatic Darkness Activated Alarm and Load Control System Using LDR, BC547 Transistor, Relay, Lamp, and Buzzer in Proteus.

This title accurately represents the complete identity of the project. The word automatic indicates that the circuit performs its operation without manual switching after the required light condition occurs. The phrase darkness activated states that the circuit responds specifically to a reduction in light intensity. The term alarm and load control system shows that the circuit is not limited to sensing only, because it also activates output devices. The listed components identify the main electrical elements used in the circuit, namely the light dependent resistor, the BC547 transistor, the relay, the lamp, and the buzzer. The reference to Proteus identifies the design and simulation environment in which the project is implemented and tested.

Therefore, the project title is technically complete because it defines the input condition, the control method, the switching mechanism, the output devices, and the simulation platform.

1.1.2 Project Category

This project belongs to the category of electronic sensing and automatic switching systems.

It is an electronic sensing system because it uses an LDR to detect a physical environmental condition, namely light intensity. It is also an automatic switching system because the change in light intensity is converted into an electrical control signal that determines whether the output load remains inactive or becomes active. The circuit therefore connects a physical stimulus to an electrical response.

The project may also be classified under basic analog electronics, sensor based automation, relay control circuits, and introductory electronic security systems. Its structure is based on discrete electronic components rather than a microcontroller or programmable digital controller.

1.1.3 Project Type

The project is a Proteus based simulation project with possible real hardware implementation.

In its present form, the circuit is designed and tested in Proteus. Proteus allows the circuit to be assembled virtually, simulated under different light conditions, and observed before physical construction. This makes the project suitable for academic demonstration, circuit analysis, and functional verification.

The same circuit can be implemented in hardware, but physical construction requires additional verification. The relay coil current, transistor saturation condition, flyback diode rating, LED current, power supply capacity, and load side safety must be checked before real hardware use. Therefore, the project is primarily a simulation based academic electronics project, but it has a direct path toward hardware implementation after proper electrical validation.

1.1.4 Core Project Function

The core function of the project is to detect darkness and automatically activate a lamp and buzzer through a relay.

The circuit receives light intensity as its input condition. When the LDR is exposed to sufficient light, its resistance decreases and the transistor remains switched off. In this state, the relay is not energized, so the lamp and buzzer remain inactive. When the LDR is placed in darkness, its resistance increases. This raises the sensing voltage at the transistor base through the voltage divider and base resistor. When the transistor receives sufficient base drive, it switches on and energizes the relay coil. The relay contact then closes the load circuit and activates the lamp and buzzer.

The core function can be expressed as the following logical relation:

$$\begin{aligned} \text{Darkness detected} &\Rightarrow \text{relay energized} \\ &\Rightarrow \text{lamp and buzzer activated} \end{aligned}$$

This means the project converts darkness into a controlled electrical output.

1.2 Project Overview

1.2.1 General Overview

This project is an automatic light sensitive circuit that senses the surrounding light level using an LDR and activates output devices when the light level decreases. The LDR acts as the primary sensing element. Its resistance changes according to the amount of light falling on it. When the light level is high, the LDR resistance is low. When the light level is low, the LDR resistance becomes high.

The circuit uses this resistance variation to create a changing voltage signal. This voltage signal is applied to the base of a BC547 transistor through a current limiting resistor. The transistor then works as an electronic switch. When the transistor turns on, it allows current to flow through the relay coil. The energized relay changes the state of its contact and connects the lamp and buzzer to the output power source.

Thus, the project demonstrates the complete process of sensing, signal conversion, switching, isolation, and actuation.

1.2.2 System Level Explanation

At system level, the circuit operates through a clear sequence of functional stages:

Light or darkness → LDR → voltage divider
 → BC547 transistor → relay
 → lamp and buzzer

The first stage is the environmental input, which is the amount of light falling on the LDR. The second stage is the LDR itself, which converts light intensity into resistance variation. The third stage is the voltage divider formed by R1 and the LDR. This divider converts resistance variation into voltage variation. The fourth stage is the BC547 transistor, which receives the sensing voltage and acts as a switch. The fifth stage is the relay, which allows the low power control section to operate a separate load circuit. The final stage consists of the lamp and buzzer, which produce the visual and audible response.

The mathematical basis of the sensing node is the voltage divider relation:

$$V_{sense} = V_{CC} \frac{R_{LDR}}{R_1 + R_{LDR}}$$

In this project arrangement, when R_{LDR} increases in darkness, V_{sense} also increases. When V_{sense} becomes sufficient to forward bias the base emitter junction of the transistor, the transistor conducts and the relay is activated.

1.2.3 Control Logic

The control logic of the project is darkness activated. This means that the output is turned on when the LDR is in a dark condition.

The logic depends on the position of the LDR in the voltage divider. In the shown circuit, the LDR is connected toward the ground side of the divider. Under bright light, the LDR resistance decreases and pulls the sensing node voltage downward. This keeps the transistor below its switching condition. Under darkness, the LDR resistance increases and the sensing node voltage rises. This drives the transistor base and causes the relay to energize.

The control logic can be stated as follows:

High light intensity $\Rightarrow R_{LDR}$ low
 $\Rightarrow V_{sense}$ low $\Rightarrow Q_1$ off
 Low light intensity $\Rightarrow R_{LDR}$ high
 $\Rightarrow V_{sense}$ high $\Rightarrow Q_1$ on

Therefore, the circuit is not designed as a light activated output system. It is designed as a darkness activated output system.

1.2.4 Output Response

The output response of the system is both visual and audible. The lamp provides a visible indication when darkness is detected. The buzzer provides an audible alarm

signal. The use of both output devices improves the practical visibility and noticeability of the circuit response.

When the relay is not energized, the relay contact remains in its normal state and the output circuit remains inactive. When the relay energizes, the contact changes position and completes the electrical path for the lamp and buzzer. The lamp then turns on and the buzzer produces sound.

This dual output response makes the circuit suitable for alarm indication, automatic lighting demonstration, darkness warning, and simple security applications.

1.3 Background of the Project

1.3.1 Need for Automatic Light Based Control

Automatic light based control is useful because many systems must respond to environmental light conditions without human intervention. Manual switching requires a person to observe the environment and operate a switch. This is inefficient in situations where continuous monitoring is needed.

A darkness activated system solves this problem by allowing the circuit itself to detect the light condition and respond immediately. Such a system can turn on a lamp when the environment becomes dark or activate an alarm when light is blocked. This principle is widely useful in simple automation systems because the sensor replaces manual observation and the relay replaces manual switching.

The project therefore addresses the need for automatic response to darkness using a simple and understandable electronic design.

1.3.2 Role of Light Sensors in Electronics

Light sensors are used in electronics to detect changes in illumination and convert those changes into electrical signals. The LDR is one of the simplest light sensors. It is a passive component whose resistance varies with incident light.

In bright conditions, the LDR resistance decreases. In dark conditions, the LDR resistance increases. This property allows the LDR to be used in voltage divider circuits, automatic lamps, light interruption alarms, street lighting systems, object detection arrangements, and basic security circuits.

In this project, the LDR does not directly power the lamp or buzzer. Instead, it provides the sensing condition that controls the transistor and relay. This is an important design principle because a sensor signal is usually weak and should be used to control a driver stage rather than directly operate a load.

1.3.3 Role of Relays in Control Systems

A relay is an electromechanical switching device that allows a low power control signal to switch a separate load circuit. In this project, the relay coil is controlled by the transistor. The relay contact is used to control the lamp and buzzer.

The relay is important because it separates the control circuit from the load circuit. The LDR and transistor operate in the low power control section, while the lamp and buzzer are operated through the relay contact. This separation makes the circuit more practical than directly connecting the load to the sensing divider.

The relay also allows the output load to use a different source from the control circuit, provided that the relay contact rating is suitable. This makes relays useful in automation circuits where a small electrical signal must control a larger or separate electrical load.

1.3.4 Role of Proteus in Circuit Design

Proteus is used as the design and simulation environment for this project. It allows the circuit to be built virtually, tested under controlled conditions, and observed before hardware construction. In Proteus, the LDR response can be simulated by changing the light condition, and the resulting behavior of the transistor, relay, lamp, buzzer, and LED can be examined.

The use of Proteus is academically valuable because it allows the project to be tested without immediately requiring physical components. It also makes it possible to identify wiring mistakes, incorrect component values, and functional problems before practical construction.

However, Proteus simulation does not remove the need for real hardware verification. Real components have tolerances, current limits, voltage drops, and physical safety requirements. Therefore, Proteus verifies the functional concept, while hardware construction requires additional electrical and safety checks.

1.4 Problem Statement

1.4.1 Main Problem

The main problem addressed by this project is that manual activation of alarms or lights is inefficient when an automatic response to darkness is required. In many situations, it is useful for a system to respond immediately when light becomes absent or insufficient. Relying on a human operator to detect darkness and switch on an alarm or lamp is slower, less consistent, and less reliable.

This project solves the problem by using an LDR based circuit that automatically detects a dark condition and activates the required output devices.

1.4.2 Technical Problem

The technical problem is that a low power light sensor cannot directly drive a lamp or buzzer reliably. The LDR changes resistance according to light, but it is not designed to operate a relay coil, lamp, or buzzer directly. Its output must first be converted into a usable electrical signal and then passed into a driver circuit.

For this reason, the circuit uses the LDR in a voltage divider and applies the resulting voltage to the base of a BC547 transistor. The transistor increases the switching capability of the system by allowing the weak sensor signal to control the relay coil current.

1.4.3 Design Problem

The design problem is to convert a weak light dependent signal into a reliable switching action. This requires a correct sequence of stages. The LDR must sense the light condition. The resistor network must produce an appropriate voltage. The transistor

must switch properly. The relay must energize at the correct time. The relay contact must complete the output circuit. The lamp and buzzer must activate only under the desired condition.

If any stage is incorrectly designed, the project may fail. For example, if the base current is too weak, the transistor may not saturate. If the relay coil current is too high, the transistor may overheat. If the diode is missing or reversed, the transistor may be damaged by relay coil kickback. Therefore, the design problem is not only to create a circuit that appears connected, but to create a complete and logically consistent sensing to output system.

1.4.4 Safety and Reliability Problem

The safety and reliability problem has two main parts. First, the switching transistor must be protected from the inductive voltage spike produced by the relay coil when the coil is switched off. This is why D1 is placed across the relay coil as a flyback diode. Without this diode, the transistor may be exposed to a high reverse voltage that can damage it.

Second, the load side must be properly isolated and safely controlled. The relay contact separates the control section from the output section, but the load voltage, load current, insulation, and relay contact rating must still be verified. If the output side uses a high voltage source, proper safety measures are required before hardware implementation.

Therefore, the circuit must be evaluated not only for function, but also for protection, reliability, and safe operation.

1.5 Aim of the Project

1.5.1 Main Aim

The main aim of this project is to design and simulate an automatic darkness activated alarm and load control circuit in Proteus.

The circuit is intended to detect a decrease in light intensity and respond by activating a lamp and buzzer through a relay. The aim is achieved when the output devices remain inactive in bright light and become active in darkness.

1.5.2 Technical Aim

The technical aim is to use an LDR, voltage divider, BC547 transistor, relay, flyback diode, lamp, and buzzer in a complete control circuit.

Each component has a specific role. The LDR senses light. The voltage divider converts resistance variation into voltage variation. The base resistor limits transistor base current. The BC547 transistor drives the relay coil. The flyback diode protects the transistor. The relay switches the output side. The lamp and buzzer provide the final output response.

Thus, the technical aim is not only to connect components, but to organize them into a correct functional chain.

1.5.3 Educational Aim

The educational aim is to demonstrate the principles of sensor based switching, transistor relay driving, load isolation, output activation, and circuit simulation.

Through this project, the student can understand how a physical input condition is converted into an electrical signal, how a transistor can be used as a switch, why a relay is needed for load control, why a diode is required across an inductive coil, and how Proteus can be used to verify circuit behavior before hardware construction.

This makes the project suitable for learning fundamental electronics in a practical and structured way.

1.6 Objectives of the Project

1.6.1 Sensor Objective

The first objective is to detect light intensity using an LDR. The LDR is selected because its resistance changes according to the amount of light falling on it. This allows the circuit to sense whether the environment is bright or dark.

The successful achievement of this objective requires that the LDR be correctly placed in the voltage divider and that its resistance variation affects the sensing node voltage.

1.6.2 Signal Objective

The second objective is to convert LDR resistance variation into a voltage signal. A transistor cannot be controlled directly by resistance change alone. The resistance variation must be converted into a voltage that can influence the base emitter junction of the transistor.

This is achieved using R1 and the LDR as a voltage divider. The sensing node voltage changes as the LDR resistance changes. This voltage becomes the control signal for the transistor stage.

1.6.3 Switching Objective

The third objective is to use a BC547 transistor as an electronic switch. The BC547 is used to control current through the relay coil. When the transistor is off, the relay coil does not receive current. When the transistor is on, current flows through the relay coil and energizes it.

The transistor therefore acts as the bridge between the low power sensing signal and the relay coil.

1.6.4 Relay Objective

The fourth objective is to energize a 5 V relay using the transistor stage. The relay is necessary because the LDR divider cannot directly operate the output load. The relay allows the control circuit to switch a separate load circuit.

For this objective to be achieved correctly, the relay coil voltage must match the control supply, and the transistor must be able to provide enough collector current for the relay coil.

1.6.5 Protection Objective

The fifth objective is to protect the transistor using a flyback diode. Since the relay coil is an inductive component, it stores magnetic energy when energized. When the transistor turns off, the coil can produce a voltage spike. The flyback diode provides a safe path for this stored energy and protects the transistor.

The protection objective is achieved only when the diode is connected with correct polarity across the relay coil.

1.6.6 Output Objective

The sixth objective is to activate a lamp and buzzer when darkness is detected. The lamp provides visual indication, while the buzzer provides audible indication. These two outputs make the circuit response clear and observable.

The output objective is achieved when both the lamp and buzzer activate only after the relay is energized by the dark condition.

1.6.7 Simulation Objective

The seventh objective is to verify the full operation of the circuit in Proteus. The simulation must demonstrate two main states. In bright light, the output should remain off. In darkness, the output should turn on.

The simulation objective is achieved when the behavior of the LDR, transistor, relay, lamp, buzzer, and indicator agrees with the expected circuit operation.

1.7 Scope of the Project

1.7.1 Included in the Project

This project includes LDR sensing, voltage divider operation, transistor switching, relay activation, lamp and buzzer output, LED indication, and Proteus simulation.

The included scope covers the complete functional path from environmental light detection to output activation. It includes the core components required to demonstrate automatic darkness detection and load switching.

The project also includes a basic protection method through the flyback diode and a visual status indication through the LED and current limiting resistor.

1.7.2 Not Fully Included in the Basic Design

The basic design does not fully include precision lux measurement, microcontroller based control, digital calibration, hysteresis circuitry, printed circuit board fabrication, or certified mains safety implementation.

The circuit detects darkness qualitatively rather than measuring exact light intensity in lux. It does not include a microcontroller, so it does not perform programmed decision making or digital adjustment. It does not include hysteresis, so the relay may chatter near the switching threshold. It is not presented as a certified high voltage product. If the circuit is converted into hardware, additional safety design is necessary.

These exclusions do not invalidate the project. They define the academic and technical boundary of the present design.

1.7.3 Simulation Scope

The simulation scope is to verify the logical and electrical behavior of the circuit in Proteus. The simulation can confirm that the LDR state affects the transistor, that the transistor controls the relay, and that the relay activates the lamp and buzzer.

The simulation can also support measurement of sensing voltage, transistor base emitter voltage, relay coil voltage, and output load voltage.

However, simulation results should be interpreted as functional evidence rather than complete proof of hardware reliability. Real hardware requires physical component testing and safety verification.

1.7.4 Hardware Scope

The project can be built physically only after important electrical conditions are verified. The relay current must be known. The BC547 must be able to handle the relay coil current. The base current must be sufficient to saturate the transistor. The flyback diode must be rated for the relay coil. The LED resistor must limit current safely. The load side must be compatible with the relay contact rating.

If the output side uses a high voltage source, the hardware version must include safe insulation, correct wiring, suitable enclosure, and protective elements such as a fuse. Therefore, hardware construction is possible, but it requires engineering validation beyond the basic Proteus schematic.

1.8 Components Used

1.8.1 Input Component

The input component is LDR1. The LDR detects light intensity and changes its resistance according to the illumination level. It is the primary sensing element of the project.

In bright light, the resistance of the LDR decreases. In darkness, the resistance of the LDR increases. This resistance variation is the starting point of the entire control process.

1.8.2 Resistors

The circuit uses three resistors. R1 has a value of 10 k Ω . R2 has a value of 6.8 k Ω . R3 has a value of 330 Ω .

R1 forms the voltage divider with the LDR. R2 limits the base current entering the BC547 transistor. R3 limits the current through the green LED. Each resistor has a specific function, and none of them is arbitrary.

The resistor network is essential because it controls sensing voltage, transistor base current, and LED current.

1.8.3 Switching Component

The switching component is Q1, which is a BC547 NPN transistor. In this project, the transistor is used as a low side switch for the relay coil.

The base receives the control signal through R2. The collector is connected to the relay coil side. The emitter is connected to ground. When sufficient base current is present, the transistor conducts and allows relay coil current to flow.

1.8.4 Relay Component

The relay component is RL1, a 5 V relay. The relay has two important parts: the coil and the contact. The coil is controlled by the transistor. The contact is used to switch the output load.

The relay allows the low power control circuit to operate the lamp and buzzer circuit. It also provides separation between the sensing stage and the output stage.

1.8.5 Protection Component

The protection component is D1, a 1N4148 diode. It is used as a flyback diode across the relay coil.

Its purpose is to protect the transistor from the high voltage spike that appears when the relay coil is switched off. The diode must be connected with correct polarity so that it does not conduct during normal relay operation but conducts when the coil releases its stored energy.

1.8.6 Indicator Component

The indicator component is D2, a green LED. The LED provides visual indication of circuit state, depending on its exact connection in the schematic.

The LED must be used with a current limiting resistor. In this circuit, R3 performs that role. Without a series resistor, the LED may draw excessive current and become damaged.

1.8.7 Output Components

The output components are L1 and BUZ1. L1 is the lamp, and BUZ1 is the buzzer.

The lamp provides a visual output when the relay activates. The buzzer provides an audible output. Together, these components make the circuit response clear and practical. Their operation depends on the relay contact completing the output circuit.

1.8.8 Power and Source Components

The circuit uses a control supply for the sensing and relay driver section and V1 as the sine source for the output load side.

The control supply powers the LDR divider, transistor switching stage, relay coil, and LED indicator. The output side source supplies the lamp and buzzer through the relay contact. The separation of these two source functions is important because the relay allows one circuit section to control another section.

1.9 Functional Block Description

1.9.1 Sensor Block

The sensor block consists of the LDR. Its function is to detect the presence or absence of light. The LDR provides the first electrical change in the system by changing its resistance in response to illumination.

This block is the input layer of the project. Without the sensor block, the circuit would not have an environmental condition to respond to.

1.9.2 Voltage Divider Block

The voltage divider block consists of R1 and the LDR. Its function is to convert the LDR resistance change into a voltage change.

The sensing voltage is determined by the supply voltage, R1, and the LDR resistance:

$$V_{sense} = V_{CC} \frac{R_{LDR}}{R_1 + R_{LDR}}$$

This voltage is the decision signal that determines whether the transistor remains off or turns on.

1.9.3 Base Drive Block

The base drive block consists mainly of R2. R2 connects the sensing node to the base of Q1 and limits the base current.

This block is necessary because the base emitter junction of a transistor behaves like a diode when forward biased. Without a current limiting resistor, excessive base current could flow. R2 therefore protects the transistor base and controls the input drive.

1.9.4 Transistor Switching Block

The transistor switching block consists of the BC547 transistor. Its function is to switch the relay coil current.

When the base voltage and base current are insufficient, Q1 remains off and no relay coil current flows. When the base voltage and current become sufficient, Q1 conducts and provides a current path from the relay coil to ground. This energizes the relay.

The transistor therefore converts a small control signal into a switching action capable of operating the relay coil.

1.9.5 Relay Isolation Block

The relay isolation block consists of the relay coil and relay contact. The relay separates the low power control side from the load side.

This is important because the sensing and transistor circuit should not directly carry the output load current. The relay contact performs the load switching while the control side only energizes the coil. This arrangement improves functional separation and makes the project more suitable for practical load control.

1.9.6 Protection Block

The protection block consists of D1, the flyback diode. Its function is to protect Q1 from inductive voltage spikes produced by the relay coil.

When the relay coil is energized, it stores magnetic energy. When Q1 turns off, the coil attempts to keep current flowing. The flyback diode provides a safe current path and prevents a damaging voltage spike from appearing across the transistor.

1.9.7 Output Block

The output block consists of the lamp and buzzer. These devices activate when the relay contact closes the load circuit.

The lamp provides the visible part of the response, and the buzzer provides the audible part. This makes the output block easy to observe during simulation and useful for alarm applications.

1.9.8 Indicator Block

The indicator block consists of D2 and R3. D2 provides visual indication, while R3 limits LED current.

The indicator block helps the observer understand the circuit state during operation. Its exact meaning should be stated clearly in the final report. Depending on connection, the LED may indicate relay activation, control power presence, or another circuit state.

1.10 Working Principle

1.10.1 Bright Light Condition

In bright light, the LDR receives sufficient illumination. Its resistance decreases. Since the LDR is connected toward the ground side of the voltage divider, the sensing node voltage becomes low.

Because the sensing voltage is low, the base emitter junction of Q1 is not sufficiently forward biased. The transistor remains off. Since Q1 is off, no effective current path exists through the relay coil to ground. The relay remains de energized. As a result, the relay contact does not close the output circuit. The lamp and buzzer remain off.

The bright condition can be represented as follows:

$$\begin{aligned} \text{Bright light} &\Rightarrow R_{LDR} \text{ low} \Rightarrow V_{sense} \text{ low} \\ &\Rightarrow Q_1 \text{ off} \Rightarrow \text{relay off} \Rightarrow \text{output off} \end{aligned}$$

1.10.2 Dark Condition

In darkness, the LDR receives little or no illumination. Its resistance increases. Since the LDR resistance is now high, the sensing node voltage rises.

When the sensing voltage becomes high enough to forward bias the base emitter junction of Q1, base current flows through R2. The transistor turns on and allows current to flow through the relay coil. The relay energizes and changes the state of its contact. The output circuit becomes complete, causing the lamp and buzzer to activate.

The dark condition can be represented as follows:

$$\begin{aligned}\text{Darkness} &\Rightarrow R_{LDR} \text{ high} \Rightarrow V_{sense} \text{ high} \\ &\Rightarrow Q_1 \text{ on} \Rightarrow \text{relay on} \Rightarrow \text{output on}\end{aligned}$$

1.10.3 Switching Chain

The switching chain is the central operating structure of the project. Darkness does not directly power the lamp and buzzer. Instead, darkness changes the LDR resistance. The LDR resistance changes the divider voltage. The divider voltage controls the transistor. The transistor energizes the relay. The relay activates the output.

This layered structure is important because each stage performs a separate function. The LDR senses. The divider forms a control voltage. The transistor switches. The relay isolates and transfers control. The output devices respond.

The switching chain is therefore a complete sensor to actuator pathway.

1.10.4 Protection Action

When the relay is switched off, the current through the relay coil cannot stop instantly because the coil is inductive. The collapsing magnetic field can generate a high voltage spike. This voltage spike can damage the transistor if it is not controlled.

D1 suppresses this effect by providing a path for the relay coil current when Q1 turns off. During normal relay operation, D1 is reverse biased and does not conduct. During switch off, the coil voltage reverses and D1 becomes forward biased, allowing the stored energy to dissipate safely.

This protection action is essential for reliable transistor operation.

1.11 Applications

1.11.1 Automatic Night Alarm

The circuit can be used as an automatic night alarm. When darkness occurs, the relay activates and turns on the buzzer. The lamp can also provide a visual indication. This application is useful where a simple automatic warning is required when light is absent.

1.11.2 Light Failure Alarm

The circuit can be used as a light failure alarm. If a light source that normally illuminates the LDR fails, the LDR enters a dark condition. The circuit then activates the output devices. This makes the system useful for detecting loss of illumination in a monitored area.

1.11.3 Security Alarm

The circuit can be used in a simple security alarm arrangement. If the light falling on the LDR is blocked by an object or person, the resistance of the LDR increases and the circuit may activate. This principle can be used to demonstrate basic light interruption detection.

The circuit is not a complete professional security system, but it demonstrates the basic sensing and switching principle used in simple alarm designs.

1.11.4 Automatic Lamp Control

The circuit can be used for automatic lamp control. In darkness, the relay activates and the lamp turns on. This is similar to the principle used in automatic night lamps and basic street lighting demonstrations.

The project therefore shows how a light sensitive circuit can control illumination without manual switching.

1.11.5 Educational Electronics Demonstration

The project is highly suitable as an educational electronics demonstration. It combines several important concepts in one circuit: LDR behavior, voltage division, transistor switching, relay operation, flyback diode protection, LED current limiting, output control, and Proteus simulation.

Because of this, the project can be used to teach both theoretical and practical electronics principles.

1.12 Limitations of the Basic Design

1.12.1 No Adjustable Threshold

The basic design does not include an adjustable threshold. The trigger level depends mainly on the fixed value of R1 and the resistance behavior of the LDR.

This means the circuit may not switch at the exact darkness level desired in every environment. Real LDRs also have tolerance and variation, so two physical LDRs may not produce identical switching behavior. A potentiometer could be added to allow threshold adjustment.

1.12.2 No Hysteresis

The basic design does not include hysteresis. Hysteresis means having different turn on and turn off thresholds. Without hysteresis, the circuit may become unstable near the light and dark boundary.

If the light level is close to the switching point, small variations in illumination can cause the transistor and relay to switch repeatedly. This can produce relay chattering. A comparator with positive feedback or another hysteresis method would improve stability.

1.12.3 Possible Weak Base Drive

The BC547 may not saturate strongly if the relay coil current is high and the available base current is low. The base drive depends on the sensing voltage and the value of R2. If the base current is insufficient, the transistor may operate partially on rather than fully saturated.

This is important because a transistor that is not saturated may heat and may fail to energize the relay reliably. The condition for strong switching can be checked using:

$$I_B \geq \frac{I_C}{10}$$

In this circuit, I_C is approximately the relay coil current. Therefore, the relay current and transistor base current must be verified.

1.12.4 Diode Rating Must Be Verified

D1 is shown as a 1N4148 diode. This diode may be acceptable for small relay coils, but it is not automatically suitable for every relay. The relay coil current and stored inductive energy should be considered.

If the relay coil current is high, a stronger diode may be required. The diode must also be connected with correct polarity. If the diode is reversed, it may create an incorrect current path and prevent proper circuit operation.

1.12.5 Load Side Safety Must Be Verified

The load side safety must be verified, especially if the output side uses high voltage or alternating current. The relay contact must be rated for the load voltage and current. The wiring must be insulated. The circuit should include proper protection if implemented physically.

Proteus simulation can show functional switching, but it does not prove physical safety. Real hardware must follow safe electrical construction practices.

1.13 Chapter Summary

1.13.1 Summary of Project Identity

This project is a darkness activated automatic relay control system. It uses an LDR to detect light conditions and a BC547 transistor to drive a relay. The relay then controls a lamp and buzzer.

The project identity is clear because the input condition, sensing element, switching device, relay mechanism, output devices, and simulation platform are all explicitly defined.

1.13.2 Summary of Main Function

The main function of the project is to detect darkness and activate a lamp and buzzer automatically. In bright light, the output remains off. In darkness, the LDR resistance increases, the sensing voltage rises, the transistor turns on, the relay energizes, and the output devices activate.

The complete functional statement is:

Darkness $\Rightarrow$ sensing voltage rises
 $\Rightarrow$ transistor switches $\Rightarrow$ relay energizes
 $\Rightarrow$ lamp and buzzer activate

1.13.3 Summary of Engineering Meaning

The engineering meaning of this project is that it demonstrates sensor to actuator conversion using discrete electronic components. It shows how a physical environmental condition can be detected, converted into a voltage signal, amplified through a transistor switching stage, transferred through a relay, protected by a flyback diode, and finally expressed through visible and audible output devices.

The chapter establishes the project as a complete automatic control system rather than an isolated component demonstration. It also identifies the main limitations that must be considered before physical construction, including threshold adjustability, hysteresis, transistor drive margin, diode rating, and load side safety.

2

Calculation

2.1 Given Data

This section presents the numerical and component data required for the theoretical analysis of the automatic darkness activated alarm and load control circuit. The purpose of listing the given data before calculation is to establish a clear electrical basis for every later formula, substitution, interpretation, and design judgement. The circuit contains resistive, semiconductor, switching, protection, indication, and load components. Each of these components contributes to the final operation of the system, therefore their values and functions must be stated before the mathematical analysis is performed.

2.1.1 Resistor Values

The circuit uses three resistors with different functions. The first resistor is R_1 , with a value of $10\text{ k}\Omega$. This resistor forms the upper part of the light dependent voltage divider together with the LDR. Its value strongly affects the sensitivity of the circuit and helps determine the light level at which the transistor begins to conduct.

The second resistor is R_2 , with a value of $6.8\text{ k}\Omega$. This resistor is connected between the sensing node and the base of the BC547 transistor. Its function is to limit the current entering the base terminal of the transistor. Without this resistor, the base current could become uncontrolled under certain conditions. Therefore, R_2 acts as a current limiting and protection element for the transistor input.

The third resistor is R_3 , with a value of $330\text{ }\Omega$. This resistor is connected in series with the green LED. Its purpose is to limit the LED current to a safe operating value. Since an LED is a current sensitive device, it must not be connected directly to the supply without a current limiting resistor.

The resistor values used in the circuit are therefore:

$$R_1 = 10\text{ k}\Omega$$

$$R_2 = 6.8\text{ k}\Omega$$

$$R_3 = 330 \Omega$$

These values form the foundation for the voltage divider calculation, transistor base current calculation, and LED current calculation.

2.1.2 Semiconductor Components

The main semiconductor switching component in the circuit is Q_1 , which is a BC547 NPN transistor. The BC547 is used as an electronic switch. Its role is to control the relay coil current according to the voltage produced by the LDR voltage divider. When the base of the transistor receives sufficient voltage and current, the transistor conducts from collector to emitter and energizes the relay coil.

The protection diode is D_1 , which is a 1N4148 diode. This diode is connected across the relay coil. Its function is to suppress the reverse voltage spike generated by the relay coil when the transistor switches OFF. Since the relay coil is an inductive component, it stores magnetic energy during operation. When the transistor interrupts the coil current, this stored energy can produce a high voltage spike. The diode provides a safe discharge path for that stored energy.

The indicator diode is D_2 , which is a green LED. It provides visual indication of circuit state. The LED must always be used with the series resistor R_3 , because its current must remain within a safe operating range.

The semiconductor components used in the circuit are:

$$Q_1 = BC547$$

$$D_1 = 1N4148$$

$$D_2 = \text{Green LED}$$

2.1.3 Relay Value

The relay used in the circuit is RL_1 , rated at 5 V. The relay is an electromechanical switching device that allows the low power transistor control circuit to switch a separate load circuit. The relay coil belongs to the control side of the circuit, while the relay contact belongs to the load switching side.

The relay value is:

$$RL_1 = 5 V \text{ relay}$$

This means that the relay coil is intended to operate when approximately 5 V is applied across it. If the coil voltage is significantly lower than its rated value, the relay may fail to energize. If the applied voltage is too high, the relay coil may overheat or become damaged. Therefore, the relay voltage rating must be matched with the circuit supply.

2.1.4 Load Components

The output load consists of a lamp and a buzzer. The lamp is represented as L_1 , and the buzzer is represented as BUZ_1 . These two components provide the final response of the system after darkness is detected.

The lamp gives a visual output. It can represent illumination, warning, or state indication. The buzzer gives an audible output. It can represent an alarm, warning signal, or attention signal. Together, the lamp and buzzer create a combined visual and audible response.

The load components are:

$$L_1 = \text{Lamp}$$

$$BUZ_1 = \text{Buzzer}$$

The load components are not driven directly by the LDR. They are controlled through the relay contact. This is important because the LDR and transistor stage are part of the low power control circuit, while the lamp and buzzer may require a different current or voltage level.

2.1.5 Assumed Supply Value

For the theoretical calculation, the control supply voltage is assumed to be:

$$V_{CC} = 5\text{ V}$$

This assumption is used because the relay shown in the project is a 5 V relay. Therefore, a 5 V control supply is a reasonable theoretical reference for calculation. However, if the actual Proteus circuit uses a different supply voltage, every calculation that depends on V_{CC} must be recalculated using the actual value.

This condition is important because the voltage divider output, transistor base current, LED current, relay coil voltage, and power dissipation all depend on the supply voltage. A calculation based on 5 V is valid only when the control circuit supply is actually 5 V.

2.2 LDR Voltage Divider Calculation

The first major calculation in the circuit is the LDR voltage divider calculation. The LDR does not directly produce voltage by itself. Instead, it changes resistance according to the light intensity falling on it. This resistance variation is converted into a voltage by placing the LDR in a voltage divider network with R_1 .

In this circuit, R_1 is connected toward the positive supply, and the LDR is connected toward ground. The output of the divider is taken from the junction between R_1 and the LDR. This output voltage is called the sensing voltage, represented as V_{sense} . The sensing voltage is then transferred through R_2 to the base of the transistor.

2.2.1 Voltage Divider Formula

For the given arrangement, the sensing voltage is calculated using the standard voltage divider relation:

$$V_{sense} = V_{CC} \frac{R_{LDR}}{R_1 + R_{LDR}}$$

In this expression, V_{sense} is the voltage at the sensing node, V_{CC} is the supply voltage, R_1 is the fixed resistor, and R_{LDR} is the resistance of the LDR at a particular light level.

This formula shows that the sensing voltage depends directly on the LDR resistance. Since the LDR is the lower resistor in the divider, an increase in R_{LDR} increases V_{sense} , while a decrease in R_{LDR} decreases V_{sense} .

2.2.2 Meaning of (V_{sense})

The value V_{sense} represents the electrical decision voltage of the sensing stage. It is the voltage that determines whether the transistor will remain OFF or begin conducting.

If V_{sense} is below the base emitter turn ON requirement of the transistor, the transistor remains OFF. In that condition, no sufficient collector current flows through the relay coil, and the relay remains de energized.

If V_{sense} rises high enough to drive the base emitter junction of the transistor, the transistor begins to conduct. If the base current is strong enough, the transistor enters saturation and energizes the relay coil.

Therefore, V_{sense} is the central control voltage of the circuit. It converts the physical condition of light intensity into an electrical switching condition.

2.2.3 Bright Condition Effect

In bright light, the resistance of the LDR decreases. This behavior is expressed as:

$$R_{LDR} \downarrow$$

Since the LDR is connected to ground, a lower LDR resistance pulls the sensing node closer to ground potential. Therefore, the sensing voltage also decreases:

$$V_{sense} \downarrow$$

This means that under bright light, the base of the transistor receives little or insufficient voltage. As a result, the transistor remains OFF, the relay coil remains de energized, and the lamp and buzzer remain inactive.

The bright condition therefore produces the inactive state of the system.

2.2.4 Dark Condition Effect

In darkness, the resistance of the LDR increases. This behavior is expressed as:

$$R_{LDR} \uparrow$$

Since the LDR is connected toward ground, a higher LDR resistance weakens the pull down effect at the sensing node. Therefore, the sensing voltage rises:

$$V_{sense} \uparrow$$

When V_{sense} rises sufficiently, it can drive the base of the BC547 transistor through R_2 . The transistor then conducts, the relay coil is energized, and the lamp and buzzer are activated.

The dark condition therefore produces the active state of the system. This proves that the circuit is darkness activated.

2.3 Bright State Calculation

The bright state calculation explains what happens when the LDR receives light. This condition is important because it represents the normal non alarm condition of the system. In this state, the output devices should remain inactive.

2.3.1 Bright State LDR Resistance

When light falls on the LDR, its resistance becomes low. The exact resistance depends on the LDR model and the intensity of the light. In a theoretical explanation, this condition is represented by a decrease in R_{LDR} .

The electrical consequence is that the lower branch of the voltage divider becomes more conductive. Since the LDR is connected to ground, the sensing node is pulled downward. This prevents the transistor from receiving sufficient base voltage.

2.3.2 Bright State Sense Voltage

Using the voltage divider formula:

$$V_{sense} = V_{CC} \frac{R_{LDR}}{R_1 + R_{LDR}}$$

a low value of R_{LDR} produces a low value of V_{sense} . For example, if the LDR resistance becomes much smaller than R_1 , the fraction

$$\frac{R_{LDR}}{R_1 + R_{LDR}}$$

becomes small. Therefore, V_{sense} becomes small.

This low sensing voltage is not enough to forward bias the base emitter junction of the transistor. Therefore, the transistor does not conduct significantly.

2.3.3 Bright State Transistor Result

In the bright state, V_{sense} is lower than the required base emitter turn ON voltage. The transistor base therefore does not receive sufficient voltage and current. As a result, Q_1 remains OFF.

When Q_1 is OFF, the current path through the relay coil is interrupted. The relay coil cannot energize because the transistor does not provide a path from the coil to ground.

Thus, in bright light:

$$Q_1 = OFF$$

2.3.4 Bright State Output Result

Since the relay coil is not energized, the relay contact remains in its normal state. If the lamp and buzzer are connected through the normally open contact, the output circuit remains open. Therefore, the lamp and buzzer do not receive operating power.

The bright state output result is:

$$Relay = OFF$$

$$Lamp = OFF$$

$$Buzzer = OFF$$

This result is correct for a darkness activated alarm circuit, because the alarm should not activate while sufficient light is present.

2.4 Dark State Calculation

The dark state calculation explains what happens when the LDR receives little or no light. This is the main activation condition of the circuit. In this state, the output devices should become active.

2.4.1 Dark State LDR Resistance

In darkness, the resistance of the LDR increases. The exact value depends on the LDR type and the amount of darkness, but the general behavior is that R_{LDR} becomes much larger than it is in bright light.

This increase in resistance changes the voltage divider condition. Since the LDR is connected toward ground, a higher LDR resistance reduces the ability of the LDR branch to pull the sensing node downward.

2.4.2 Dark State Sense Voltage

Using the same voltage divider formula:

$$V_{sense} = V_{CC} \frac{R_{LDR}}{R_1 + R_{LDR}}$$

a high value of R_{LDR} produces a higher value of V_{sense} . If R_{LDR} becomes much larger than R_1 , the fraction

$$\frac{R_{LDR}}{R_1 + R_{LDR}}$$

approaches (1). Therefore, V_{sense} approaches V_{CC} .

This means that in darkness, the transistor base receives a higher voltage through R_2 . If this voltage is high enough to forward bias the base emitter junction, the transistor begins to conduct.

2.4.3 Dark State Transistor Result

When the base emitter junction receives sufficient forward voltage, Q_1 turns ON. In a switching circuit, the ideal goal is not only to make the transistor begin conducting, but to drive it strongly into saturation. Saturation means the transistor behaves like a closed switch with a low collector emitter voltage.

In the dark condition:

$$Q_1 = ON$$

If the base current is sufficient, the relay coil current flows through the collector emitter path of Q_1 . The relay coil is then energized.

2.4.4 Dark State Output Result

When the relay coil is energized, the relay contact changes state. If the lamp and buzzer are connected through the normally open contact, the contact closes. This completes the load circuit and allows current to flow through the lamp and buzzer.

The dark state output result is:

$$Relay = ON$$

$$Lamp = ON$$

$$Buzzer = ON$$

This confirms the intended darkness activated operation of the project.

2.5 Transistor Turn On Calculation

The transistor turn ON calculation determines the minimum electrical condition required for the BC547 transistor to begin conducting. Since the transistor is used as a switch, the base emitter junction must be forward biased before current can flow from collector to emitter.

2.5.1 Base Emitter Threshold

For a silicon NPN transistor such as the BC547, the base emitter junction typically begins conducting when the base emitter voltage reaches approximately 0.6 V to 0.7 V. This condition is expressed as:

$$V_{BE} \approx 0.6 V \text{ to } 0.7 V$$

For calculation purposes, the commonly used approximation is:

$$V_{BE} \approx 0.7 V$$

This approximation is suitable for basic circuit analysis. However, the exact value can vary depending on current, temperature, and the specific transistor.

2.5.2 Minimum Sense Voltage Condition

Since the base of the transistor is driven from the sensing node through R_2 , the sensing voltage must be high enough to forward bias the base emitter junction. Therefore, the minimum approximate condition for transistor conduction is:

$$V_{sense} \geq 0.7 V$$

This condition represents the beginning of transistor conduction. It does not automatically prove that the transistor is fully saturated. Full relay driving requires enough base current, not only enough base voltage.

2.5.3 Interpretation

The transistor begins to conduct when the LDR voltage divider produces sufficient voltage at the base. In this circuit, that happens when darkness increases the LDR resistance and raises the sensing node voltage.

Therefore, the transistor turn ON condition links the sensor stage to the switching stage. The LDR does not directly energize the relay. Instead, the LDR changes V_{sense} , and V_{sense} controls whether the transistor can conduct.

This establishes the mathematical connection between light intensity and relay activation.

2.6 Approximate LDR Trigger Resistance

The approximate LDR trigger resistance calculation estimates the LDR resistance at which the transistor begins to turn ON. This is a simplified calculation because it assumes that the base current loading effect is small. The result is useful for understanding the approximate switching threshold of the circuit.

2.6.1 Starting Equation

The transistor begins to conduct when:

$$V_{sense} \approx 0.7 V$$

Using the voltage divider equation and assuming:

$$V_{CC} = 5 V$$

$$R_1 = 10000 \Omega$$

the equation becomes:

$$0.7 = 5 \frac{R_{LDR}}{10000 + R_{LDR}}$$

This equation states that the LDR resistance must rise enough to produce approximately 0.7 V at the sensing node.

2.6.2 Expansion

The next step is to remove the denominator by multiplying both sides by:

$$10000 + R_{LDR}$$

This gives:

$$0.7(10000 + R_{LDR}) = 5R_{LDR}$$

This expanded equation prepares the expression for algebraic solving.

2.6.3 Solving

Expanding the left side gives:

$$7000 + 0.7R_{LDR} = 5R_{LDR}$$

Move the $0.7R_{LDR}$ term to the right side:

$$7000 = 5R_{LDR} - 0.7R_{LDR}$$

$$7000 = 4.3R_{LDR}$$

Divide both sides by (4.3):

$$R_{LDR} = \frac{7000}{4.3}$$

$$R_{LDR} \approx 1628 \Omega$$

Therefore, the approximate LDR resistance required to begin transistor conduction is:

$$R_{LDR} \approx 1.628 k\Omega$$

2.6.4 Meaning

This result means that under the simplified no load approximation, the transistor begins to turn ON when the LDR resistance rises above approximately 1.6 kΩ. However, this value represents only the beginning of conduction. It does not prove that the transistor is saturated, and it does not prove that the relay will operate reliably in physical hardware.

The calculation is useful because it shows the direction and approximate threshold of the circuit. As darkness increases, R_{LDR} increases. When R_{LDR} becomes large

enough, V_{sense} reaches the transistor base emitter threshold and the transistor begins to conduct.

A complete hardware design must also include base current, relay coil current, and saturation margin.

2.7 Base Current Calculation

The base current calculation determines how much current is available to drive the base of the BC547 transistor. This is one of the most important calculations in the circuit because a relay driver transistor must receive enough base current to switch the relay coil reliably.

2.7.1 Approximate Base Current Formula

In deep darkness, the LDR resistance becomes high. Under this condition, the sensing node approaches the supply voltage. The base current is limited mainly by R_1 , R_2 , and the base emitter junction voltage.

The approximate base current is calculated as:

$$I_B \approx \frac{V_{CC} - V_{BE}}{R_1 + R_2}$$

In this formula, I_B is the base current, V_{CC} is the supply voltage, V_{BE} is the base emitter voltage of the transistor, R_1 is the upper divider resistor, and R_2 is the base current limiting resistor.

This approximation gives a conservative understanding of the available base drive.

2.7.2 Substitution

Using:

$$V_{CC} = 5\text{ V}$$

$$V_{BE} = 0.7\text{ V}$$

$$R_1 = 10000\ \Omega$$

$$R_2 = 6800\ \Omega$$

the base current is:

$$I_B \approx \frac{5 - 0.7}{10000 + 6800}$$

$$I_B \approx \frac{4.3}{16800}$$

2.7.3 Result

Dividing (4.3) by (16800) gives:

$$I_B \approx 0.000256 \text{ A}$$

Converting amperes to milliamperes gives:

$$I_B \approx 0.256 \text{ mA}$$

Therefore, the approximate available base current is:

$$I_B \approx 0.256 \text{ mA}$$

2.7.4 Interpretation

The calculated base current is small. A small base current may be enough to make the transistor begin conducting, but it may not be enough to force the transistor into saturation if the relay coil requires a relatively high current.

This distinction is essential. A transistor that merely conducts is not necessarily a good relay switch. For relay driving, the transistor should normally operate in saturation. If it does not saturate, it may drop more voltage, dissipate more heat, and fail to provide the full relay coil current.

Therefore, the base current must be checked against the relay coil current requirement.

2.8 Relay Coil Current Calculation

The relay coil current calculation determines whether the BC547 transistor can safely drive the relay coil. Since the relay coil current becomes the collector current of the transistor, this calculation is necessary for reliability.

2.8.1 Relay Coil Resistance Requirement

To calculate the relay coil current, the coil resistance must be known. This resistance may be obtained from the relay datasheet, measured directly with a multimeter, or observed from the relay model used in Proteus.

Without the relay coil resistance, the relay current cannot be determined precisely. Therefore, the relay coil resistance is a required value for complete hardware verification.

2.8.2 Relay Coil Current Formula

The relay coil current is calculated using Ohm's law:

$$I_{relay} = \frac{V_{relay}}{R_{coil}}$$

In this formula, I_{relay} is the relay coil current, V_{relay} is the voltage across the relay coil, and R_{coil} is the resistance of the relay coil.

Since the relay is rated at 5 V, the ideal relay coil voltage is:

$$V_{relay} = 5 \text{ V}$$

Therefore:

$$I_{relay} = \frac{5}{R_{coil}}$$

2.8.3 Current Rating Check

The relay coil current must be within the safe collector current capability of the BC547 transistor. In this circuit, the relay coil current flows through the collector emitter path of Q_1 . Therefore:

$$I_C \approx I_{relay}$$

If I_{relay} is too high, the transistor may overheat or fail. If I_{relay} is within the transistor current rating but the base current is too low, the transistor may not saturate properly. Therefore, both current rating and saturation condition must be checked.

2.8.4 Relay Voltage Check

The relay coil voltage must match the rated relay voltage. Since the relay is rated at 5 V, the voltage across the relay coil should be close to 5 V during the ON state.

If the coil voltage is too low, the relay may not click or may chatter. If the coil voltage is too high, the coil may overheat. Therefore, the relay voltage check is a necessary part of the calculation chapter.

The correct relay condition is:

$$V_{coil} \approx 5 V$$

2.9 Transistor Saturation Check

The transistor saturation check determines whether the BC547 operates as a reliable switch rather than as a weak amplifier. This is one of the most important design checks in the entire project.

2.9.1 Conservative Saturation Rule

For a bipolar junction transistor used as a switch, a common conservative design rule is:

$$I_B \geq \frac{I_C}{10}$$

This rule means that the base current should be at least one tenth of the collector current. This is known as forced beta design. It is used to make sure the transistor enters saturation even when transistor gain varies from one device to another.

2.9.2 Collector Current Meaning

In this circuit, the collector current of Q_1 is approximately equal to the relay coil current, because the relay coil is connected in series with the collector emitter path of the transistor.

Therefore:

$$I_C \approx I_{relay}$$

This means that the required base current depends directly on the relay coil current. A relay with a larger coil current requires a larger base current for reliable saturation.

2.9.3 Saturation Verification

To verify saturation, the calculated base current must be compared with the required base current. The required base current is:

$$I_{B(required)} = \frac{I_C}{10}$$

Since:

$$I_C \approx I_{relay}$$

the expression becomes:

$$I_{B(required)} = \frac{I_{relay}}{10}$$

The available base current from section 2.7 is:

$$I_B \approx 0.256 \text{ mA}$$

Therefore, the relay can be considered strongly driven only if:

$$0.256 \text{ mA} \geq \frac{I_{relay}}{10}$$

Multiplying both sides by (10):

$$I_{relay} \leq 2.56 \text{ mA}$$

This means that under the conservative saturation rule, the calculated base current can strongly saturate the transistor only for a very small relay coil current of approximately 2.56 mA or less.

2.9.4 Engineering Judgment

The calculation shows an important design limitation. The circuit logic is correct, but the base drive may be weak for many real 5 V relays. Many physical relays require more than 2.56 mA of coil current. If the relay current is higher, the transistor may still turn ON, but it may not enter strong saturation.

This means that the circuit may work in Proteus while still being marginal in real hardware. The correct engineering judgement is that the design should be verified by measuring or obtaining the actual relay coil current. If the required current is higher than the saturation condition allows, the driver stage should be improved.

Possible improvements include reducing the base resistance, using a transistor with stronger drive arrangement, using a logic level MOSFET, or using a dedicated relay

driver such as a transistor array. The calculation therefore does not reject the project. It identifies the exact condition required for reliable physical implementation.

2.10 LED Current Calculation

The LED current calculation determines whether the indicator LED operates safely. The LED is connected in series with resistor R_3 , and the resistor limits the current flowing through the LED.

2.10.1 LED Resistor Formula

The LED current is calculated by subtracting the LED forward voltage from the supply voltage and dividing by the series resistance:

$$I_{LED} = \frac{V_{CC} - V_F}{R_3}$$

In this equation, I_{LED} is the LED current, V_{CC} is the supply voltage, V_F is the forward voltage of the LED, and R_3 is the LED current limiting resistor.

For a green LED, a typical forward voltage approximation is:

$$V_F \approx 2\text{ V}$$

The exact value may vary depending on the LED type and operating current.

2.10.2 Substitution for 5 V Supply

Using:

$$V_{CC} = 5\text{ V}$$

$$V_F = 2\text{ V}$$

$$R_3 = 330\ \Omega$$

the LED current is:

$$I_{LED} = \frac{5 - 2}{330}$$

$$I_{LED} = \frac{3}{330}$$

2.10.3 Result

Solving the expression gives:

$$I_{LED} = 0.00909\text{ A}$$

Converting amperes to milliamperes gives:

$$I_{LED} \approx 9.09 \text{ mA}$$

Therefore, the LED current is approximately:

$$I_{LED} \approx 9.09 \text{ mA}$$

2.10.4 Interpretation

An LED current of approximately 9.09 mA is generally suitable for a normal indicator LED. It is high enough to make the LED visible and low enough to remain within a typical safe operating range.

However, this conclusion is valid only if the LED is actually powered from a 5 V supply. If the supply voltage is different, the LED current must be recalculated. For example, a higher supply voltage would increase the LED current unless R_3 is also increased.

Thus, $R_3 = 330 \Omega$ is reasonable for a 5 V indicator path, but the value must always be checked against the actual circuit supply.

2.11 Flyback Diode Calculation

The flyback diode calculation explains why a diode is required across the relay coil. Although the diode is not part of the normal sensing calculation, it is essential for transistor protection and circuit reliability.

2.11.1 Relay Coil Energy Formula

A relay coil is an inductive component. When current flows through it, magnetic energy is stored in the coil. The stored energy is expressed as:

$$E = \frac{1}{2}LI^2$$

In this formula, E is the energy stored in the relay coil, L is the inductance of the coil, and I is the current flowing through the coil.

This formula shows that the stored energy increases with inductance and with the square of the current.

2.11.2 Meaning of Coil Energy

When the relay is energized, current flows through the coil and creates a magnetic field. This magnetic field causes the relay contact to change position. While the relay remains energized, the coil continues to store magnetic energy.

The important point is that this energy cannot disappear instantly when the transistor turns OFF. The coil attempts to maintain current flow even after the switching path is interrupted.

2.11.3 Turn Off Problem

When Q_1 turns OFF, the current path through the relay coil is suddenly interrupted. Because the coil resists sudden current change, it generates a voltage spike in an at-

tempt to keep current flowing. This voltage spike can become much higher than the supply voltage.

If no protection component is present, this spike may appear across the collector emitter terminals of the transistor. The result can be transistor stress, breakdown, overheating, unreliable operation, or permanent damage.

Therefore, the relay coil requires a safe discharge path during transistor turn OFF.

2.11.4 Diode Function

The flyback diode D_1 provides the required safe current path. During normal relay operation, the diode is reverse biased and does not conduct. When the transistor turns OFF and the relay coil voltage reverses polarity, the diode becomes forward biased and allows the coil current to circulate safely until the stored energy is dissipated.

The diode therefore protects Q_1 from the relay coil voltage spike. Its function is not optional. It is a necessary protection element in an inductive relay driver circuit.

The correct flyback diode connection is that the diode cathode must be connected to the positive relay coil side, and the diode anode must be connected to the transistor collector side. With this polarity, the diode remains inactive during normal operation and becomes active only during coil turn OFF.

2.12 Power Dissipation Check

The power dissipation check determines whether the circuit components operate within safe thermal limits. Even if a circuit functions logically, it can still be unsafe if any component dissipates more power than it can tolerate.

2.12.1 Transistor Power

When the transistor is saturated, the transistor power dissipation is approximately:

$$P_Q = V_{CE(sat)} I_C$$

In this expression, P_Q is the transistor power dissipation, ($V_{CE(sat)}$) is the collector emitter saturation voltage, and I_C is the collector current.

Since the collector current is approximately the relay coil current:

$$I_C \approx I_{relay}$$

the transistor power depends directly on the relay coil current. If the relay current is high, the transistor power increases. If the transistor is not saturated, V_{CE} becomes larger than ($V_{CE(sat)}$), which increases power dissipation and heating.

Therefore, transistor saturation is important not only for relay operation but also for transistor thermal safety.

2.12.2 Resistor Power

The power dissipated in a resistor is calculated by:

$$P_R = I^2 R$$

In this expression, P_R is the resistor power, I is the current through the resistor, and R is the resistance value.

This formula should be applied to R_1 , R_2 , and R_3 if the actual current values are known. The calculated power should be lower than the resistor power rating.

For many small signal circuits, resistors rated at 0.25 W are commonly sufficient, but this must not be assumed without checking the actual current and voltage conditions.

2.12.3 LED Resistor Power

The power dissipated by the LED resistor R_3 can be calculated using the LED current:

$$P_{R3} = I_{LED}^2 R_3$$

Using the LED current calculated earlier:

$$I_{LED} = 0.00909 \text{ A}$$

and:

$$R_3 = 330 \Omega$$

the power is:

$$P_{R3} = (0.00909)^2 (330)$$

$$P_{R3} \approx 0.0273 \text{ W}$$

Therefore, the power dissipated in R_3 is approximately:

$$P_{R3} \approx 27.3 \text{ mW}$$

This is below 0.25 W, so R_3 is safe under the assumed 5 V condition.

2.12.4 Safety Interpretation

All component power values must remain below their rated limits. The transistor must safely handle the relay coil current. The resistors must not exceed their power ratings. The LED current must remain within safe limits. The relay coil must receive its rated voltage without overheating. The diode must withstand the relay coil current during flyback operation.

The calculation chapter therefore proves not only whether the circuit can operate, but whether it can operate safely and reliably.

2.13 Calculation Result Table

This section should collect the important calculation results in formal tables. The tables make the calculation chapter easier to verify and allow the reader to compare theoretical values, simulation values, and safety decisions.

2.13.1 Voltage Result Table

The voltage result table should include the main voltage quantities of the circuit. It should include the sensing voltage, transistor base emitter voltage, relay coil voltage, lamp voltage, and buzzer voltage.

A suitable voltage result table is shown below.

Table 2.1: *Voltage result table*

Quantity	Symbol	Expected condition	Required interpretation
Sensing voltage in bright light	V_{sense}	Low	Q1 should remain OFF
Sensing voltage in darkness	V_{sense}	High	Q1 should turn ON
Base emitter voltage	V_{BE}	About 0.7 V when active	Q1 begins conducting
Relay coil voltage	V_{coil}	About 5 V when active	Relay should energize
Lamp voltage	V_L	Present when relay contact closes	Lamp should operate
Buzzer voltage	V_{BUZ}	Present when relay contact closes	Buzzer should operate

This table should be completed using actual Proteus measurements after simulation.

2.13.2 Current Result Table

The current result table should include the current values that determine whether the circuit is safe and reliable.

Table 2.2: *Current result table*

Quantity	Symbol	Calculation method	Required interpretation
Base current	I_B	$\frac{V_{CC}-V_{BE}}{R_1+R_2}$	Must be sufficient for saturation
Relay coil current	I_{relay}	$\frac{V_{relay}}{R_{coil}}$	Must be within transistor capability
Collector current	I_C	Approximately I_{relay}	Must be safe for Q1
LED current	I_{LED}	$\frac{V_{CC}-V_F}{R_3}$	Must be safe for D2
Load current	I_{load}	Determined by lamp and buzzer rating	Must be within relay contact rating

The most important comparison in this table is between I_B and the required base current for saturation.

2.13.3 Safety Result Table

The safety result table should classify each major component as safe, marginal, or requiring verification.

Table 2.3: *Safety result table*

Component	Required check	Expected judgement
R_1	Divider current and power	Usually safe under 5 V
R_2	Base current limitation	Electrically safe but may limit drive too much
R_3	LED current and power	Safe under 5 V assumption
Q_1	Collector current and saturation	Must be verified using relay current
RL_1	Coil voltage and coil current	Must match 5 V supply
D_1	Flyback current capability	Must match relay coil current
D_2	LED current	Safe if current is near 9.09 mA
L_1	Load voltage and current	Must match relay contact rating
BUZ_1	Load voltage and current	Must match relay contact rating

This table makes the engineering conclusion precise. The circuit principle is correct, but some hardware quantities must be verified before physical construction.

2.14 Chapter Summary

The calculation chapter provides the mathematical proof of the circuit operation. It shows how the LDR resistance is converted into a sensing voltage, how the sensing voltage controls the transistor base, how the transistor energizes the relay, how the relay switches the lamp and buzzer, and how the LED and diode support indication and protection.

2.14.1 Main Calculation Result

The main calculation result is that the circuit principle is correct. The voltage divider is arranged so that the sensing voltage decreases in bright light and increases in darkness. This means the circuit is darkness activated.

In bright light, the LDR resistance is low, the sensing voltage is low, the transistor remains OFF, the relay remains de energized, and the lamp and buzzer remain OFF.

In darkness, the LDR resistance is high, the sensing voltage rises, the transistor turns ON, the relay energizes, and the lamp and buzzer activate.

This confirms the intended functional logic of the project.

2.14.2 Main Weakness Found

The main weakness found through calculation is the limited base current. The approximate base current is:

$$I_B \approx 0.256 \text{ mA}$$

Using the conservative saturation rule:

$$I_B \geq \frac{I_C}{10}$$

this base current can strongly saturate the transistor only if the relay coil current is approximately 2.56 mA or less. Many real relays require more current than this. Therefore, the relay may work in simulation but may be unreliable in physical hardware unless the relay coil current is low enough or the transistor driver stage is improved.

This is not a failure of the project concept. It is a hardware margin issue that must be verified and corrected if necessary.

2.14.3 Main Design Requirement

The main design requirement is that every component must match the actual supply and load conditions. The relay coil voltage must match the control supply. The transistor must safely handle the relay coil current. The base current must be sufficient for saturation. The flyback diode must be correctly connected and properly rated. The LED resistor must limit the LED current safely. The relay contacts must handle the lamp and buzzer load current.

Therefore, the final engineering conclusion of Chapter 2 is that the circuit has a correct theoretical operating principle, but reliable hardware implementation requires verification of relay current, transistor saturation, diode rating, LED current, resistor power, and load safety.

3

Scheme

3.1 Complete Circuit Diagram

3.1.1 Proteus Schematic Placement

The complete circuit diagram of the automatic darkness activated alarm and load control system is designed and arranged in the Proteus simulation environment. The schematic represents the full electrical structure of the project, including the light sensing section, voltage divider section, transistor switching section, relay driving section, protection section, indicator section, and output load section.

The Proteus schematic must be inserted at the beginning of this chapter because it is the primary visual evidence of the circuit design. It allows the reader to understand how every component is connected and how the signal moves from the sensing stage to the output stage. The schematic must be clear, centered, and large enough to show all component names, values, connection points, relay terminals, supply nodes, and ground references.

The circuit diagram must show that the project is based on an LDR controlled switching mechanism. The LDR detects the surrounding light condition. The voltage divider converts the LDR resistance variation into a voltage change. The BC547 transistor receives this voltage through the base resistor and controls the relay coil. The relay contact then switches the lamp and buzzer load. Therefore, the schematic is not only a drawing of components. It is the complete structural representation of the project logic.

For academic correctness, the schematic must be placed after a short explanatory sentence such as the following:

The complete Proteus schematic of the automatic darkness activated alarm and load control circuit is shown in Figure 3.1. This diagram presents the full connection of the LDR sensor, resistor network, BC547 transistor, relay, flyback diode, LED indicator, lamp, buzzer, and supply sources.

3.1.2 Component Labels

All components in the schematic must be clearly labeled so that the reader can identify each part without confusion. The resistor connected with the LDR must be labeled as R1. The base resistor must be labeled as R2. The LED current limiting resistor must be

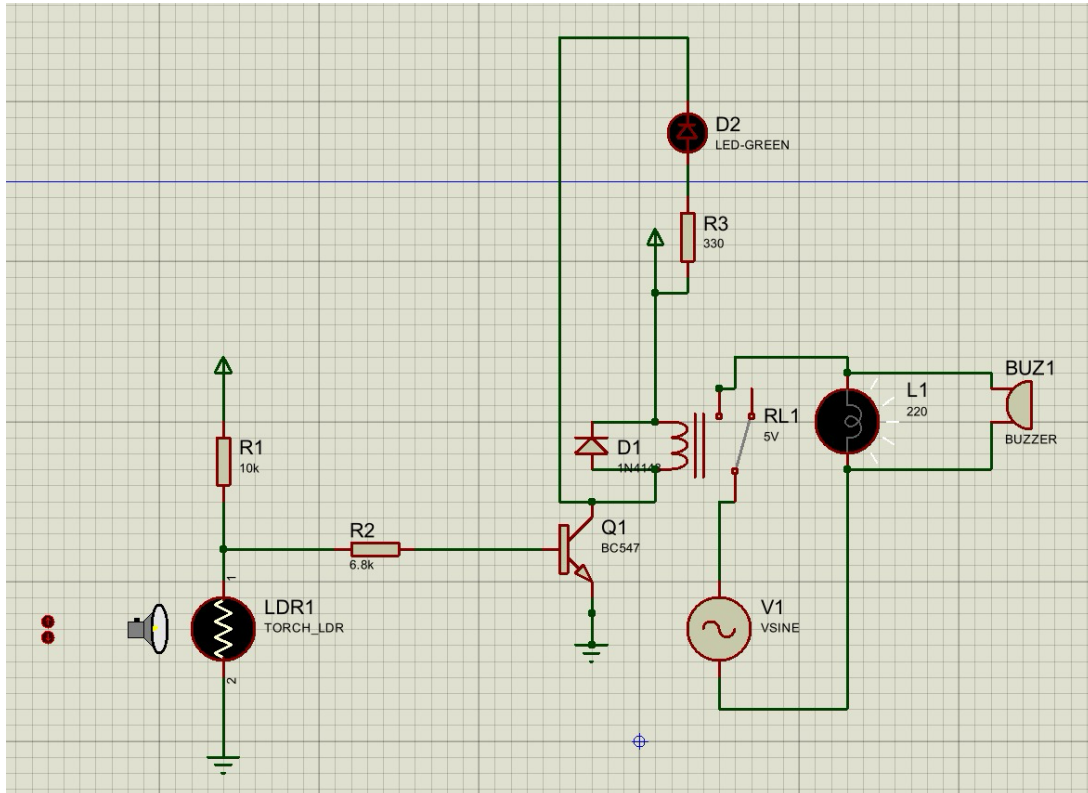


Figure 3.1: Complete Proteus schematic of the automatic darkness activated alarm and load control circuit

labeled as R3. The light dependent resistor must be labeled as LDR1. The transistor must be labeled as Q1 and identified as a BC547 NPN transistor. The flyback diode must be labeled as D1. The indicator LED must be labeled as D2. The relay must be labeled as RL1. The lamp must be labeled as L1. The buzzer must be labeled as BUZ1. The load side source must be labeled as V1.

Clear labeling is necessary because the circuit contains several different functional regions. Without labels, the reader may not be able to distinguish between the sensor side, the driver side, the relay side, and the output side. In an academic report, each component label must match the explanation in the written text. If the text refers to R2 as the base resistor, then the schematic must also show R2 in the base path of Q1. If the text refers to D1 as the flyback diode, then the schematic must show D1 directly across the relay coil. If the text refers to D2 as the indicator LED, then the schematic must show D2 with its current limiting resistor R3.

The labels also help the calculation chapter and troubleshooting chapter. For example, when the report later discusses the base current, the reader can immediately locate R2 and Q1. When the report discusses the LED current, the reader can locate R3 and D2. When the report discusses relay protection, the reader can locate D1 and RL1. Therefore, component labeling is not decorative. It is a necessary part of technical traceability.

3.1.3 Supply Labels

The schematic must clearly identify the control supply and the load side source. The control supply powers the sensing and switching side of the circuit. This includes the LDR voltage divider, the transistor base circuit, the relay coil, the flyback diode path, and the indicator LED path. The load side source powers the lamp and buzzer through the relay contact.

The separation between the control supply and the load supply is important because the circuit uses a relay. A relay allows a low power control circuit to switch a separate load circuit. In this project, the LDR and transistor do not directly power the lamp and buzzer. Instead, they energize the relay coil, and the relay contact switches the load side. This is a correct engineering arrangement because it separates the sensing circuit from the output load.

The supply labels must make it clear which voltage belongs to the relay coil and which source belongs to the lamp and buzzer. If the relay is rated at 5 V, the relay coil supply must be compatible with 5 V operation. If V1 is used as the source for the lamp and buzzer, the load voltage and current must be compatible with the relay contact rating and the load component ratings.

In the final academic diagram, the supply labels should be written in a way that prevents ambiguity. The control side supply may be labeled as control voltage. The load side source may be labeled as load source. The relay coil supply and the load side source must not be confused because they perform different functions in the system.

3.1.4 Ground Reference

The ground reference must be shown clearly in the schematic. The ground point is essential for the correct operation of the transistor switching stage. Since Q1 is used as a low side NPN switch, its emitter is connected to ground. The base voltage of Q1 is meaningful only when measured with respect to this ground reference. The relay coil current also completes its path through Q1 to ground when the transistor is turned on.

A missing or unclear ground reference can cause simulation failure and hardware failure. If the emitter of Q1 is not properly connected to ground, the transistor cannot operate as a defined low side switch. If the LDR voltage divider does not share the correct reference with the transistor base and emitter, the base emitter voltage cannot be correctly established. Therefore, the ground symbol must be visible and correctly connected to the control side of the circuit.

The ground reference also supports measurement. In simulation, the sensing voltage, base voltage, emitter voltage, relay coil voltage, and transistor collector voltage must all be interpreted relative to the circuit reference. A clearly shown ground makes these voltage measurements meaningful.

3.2 Block Diagram

3.2.1 Input Block

The input block of the system is the LDR sensing block. The input physical quantity is light intensity. The LDR responds to this physical input by changing its electrical resistance. When the surrounding illumination is high, the resistance of the LDR decreases.

When the surrounding illumination is low, the resistance of the LDR increases.

The input block is the first stage of the project because it converts an environmental condition into an electrical property. The circuit does not begin with a digital signal or a manual switch. It begins with a physical light condition. The LDR is therefore the sensory interface between the environment and the electrical control system.

In this project, the input block determines whether the circuit remains inactive or activates the relay. The LDR itself does not decide the final output alone. Instead, it provides resistance variation that becomes meaningful through the voltage divider. Therefore, the input block must be understood as the physical sensing foundation of the entire system.

3.2.2 Signal Conditioning Block

The signal conditioning block is formed by R1 and LDR1. These two components form a voltage divider. The purpose of the voltage divider is to convert the resistance change of the LDR into a voltage change at the sensing node.

The sensing node voltage may be represented as:

$$V_{\text{sense}} = V_{\text{CC}} \times R_{\text{LDR}} \div (R_1 + R_{\text{LDR}})$$

This expression applies when R1 is connected to the positive control supply and the LDR is connected toward ground. Under this arrangement, a low LDR resistance produces a low sensing voltage, while a high LDR resistance produces a higher sensing voltage. Therefore, darkness raises the sensing voltage.

The signal conditioning block is necessary because a transistor base cannot use resistance alone as a direct control signal. The transistor responds to base emitter voltage and base current. The voltage divider creates the voltage condition that can be applied to the transistor base through R2.

3.2.3 Switching Block

The switching block is formed by R2 and Q1. R2 transfers the sensing voltage to the base of Q1 while limiting base current. Q1 is the BC547 NPN transistor that controls the relay coil.

The transistor acts as an electronic switch. When the base emitter voltage is below the required turn on level, the transistor remains off. When the base emitter voltage rises sufficiently, the transistor conducts and allows current to pass through the relay coil. This current energizes the relay.

The switching block is structurally necessary because the LDR voltage divider cannot directly drive the relay coil in a reliable and safe way. The LDR and resistor network produce a weak control signal. The relay coil requires a stronger current path. Q1 provides the transition from low power sensing to higher current relay activation.

3.2.4 Relay Block

The relay block includes the relay coil and the relay contacts. The relay coil belongs to the control side. The relay contacts belong to the load switching side. When the relay coil is energized by the transistor, the internal magnetic mechanism changes the

contact state. This contact movement allows the lamp and buzzer circuit to be switched on.

The relay block is the isolation and transfer stage of the system. It transfers the control decision from the low power electronic circuit to the load circuit. This is an important structural advantage because it allows the sensor circuit to control a load without directly carrying the load current.

The relay has two different meanings in the circuit. Electrically, the coil is an inductive load driven by Q1. Mechanically, the contact is a switch that controls the lamp and buzzer. These two meanings must not be confused. The coil causes switching. The contact performs switching.

3.2.5 Output Block

The output block consists of the lamp and buzzer. The lamp provides visual output. The buzzer provides audible output. When the relay contact closes, the load side source supplies power to these output components.

The output block is the final response stage of the project. Its purpose is to make the darkness detection visible and audible. The project therefore does not stop at detection. It completes the full chain from sensing to action.

The output block also proves the success of the circuit operation during simulation. If the lamp and buzzer turn on when the LDR is dark and remain off when the LDR is illuminated, the basic functional aim of the circuit is achieved.

3.2.6 Protection and Indicator Block

The protection and indicator block consists of the flyback diode D1 and the LED indicator D2 with resistor R3. D1 protects Q1 from the high voltage transient produced by the relay coil when the transistor turns off. Since the relay coil is inductive, it stores energy while current flows through it. When the current is interrupted, the coil attempts to maintain current flow and can produce a damaging voltage spike. D1 provides a safe path for this stored energy.

D2 and R3 provide visual indication. R3 limits the current through D2 so that the LED operates safely. The exact meaning of the LED indication must be defined in the final report. It may indicate relay activation, control side activity, or another defined state, depending on the exact node connection in the schematic.

This block strengthens the project because it adds protection and observability. Protection improves reliability. Indication improves interpretability.

3.3 Sensor Section

3.3.1 LDR Function

The LDR is the primary sensing component of the circuit. Its function is to detect the intensity of light falling on its surface and convert that light condition into a change in electrical resistance. The term LDR means light dependent resistor. Its resistance is not fixed like an ordinary resistor. Instead, its resistance changes according to illumination.

In this project, the LDR is used as a darkness sensor. The circuit is arranged so that the output becomes active when the LDR is in a dark condition. The LDR therefore

acts as the environmental input element of the system.

The LDR does not produce energy by itself. It does not directly power the relay, lamp, or buzzer. It modifies the voltage condition of the circuit through its resistance. This resistance change becomes useful only because the LDR is combined with R1 in a voltage divider.

3.3.2 LDR in Bright Light

When bright light falls on the LDR, its resistance decreases. Since the LDR is placed toward ground in the voltage divider, this lower resistance pulls the sensing node closer to ground potential. As a result, the voltage at the sensing node becomes low.

A low sensing voltage means that the base of Q1 does not receive sufficient forward bias. The base emitter junction of the transistor remains below the level required for strong conduction. Therefore, Q1 remains off.

When Q1 remains off, no complete relay coil current path exists through the transistor to ground. The relay coil remains de energised. Since the relay contact does not switch into the active output state, the lamp and buzzer remain off.

Thus, the bright light condition corresponds to the inactive output condition.

3.3.3 LDR in Darkness

When the light falling on the LDR decreases, the resistance of the LDR increases. Since the LDR is connected toward ground, a higher LDR resistance reduces the pull toward ground. The sensing node voltage therefore rises toward the control supply voltage.

As the sensing voltage rises, current can pass through R2 into the base of Q1. If the base emitter voltage reaches the required level, Q1 begins to conduct. When Q1 conducts sufficiently, it provides a current path for the relay coil.

The relay coil then energizes, and the relay contact changes state. This contact change connects the load side source to the lamp and buzzer. Therefore, darkness causes the lamp and buzzer to turn on.

Thus, the dark condition corresponds to the active output condition.

3.3.4 Sensor Output Meaning

The output of the sensor section is not the final lamp or buzzer output. The sensor output is the electrical voltage produced at the divider node. This voltage is called the sensing voltage. It represents the light condition detected by the LDR.

The sensing voltage is an intermediate control signal. It must pass through R2 to the transistor base. The transistor then converts this weak signal into a stronger switching action. Therefore, the LDR section should be understood as an information generating section, not as a power driving section.

This distinction is important in academic explanation. The LDR senses. The divider converts. The transistor switches. The relay transfers. The lamp and buzzer respond.

3.4 Voltage Divider Section

3.4.1 R1 Function

R1 forms the upper part of the voltage divider. It is connected between the positive control supply and the sensing node. Its purpose is to work with the LDR to produce a variable voltage at the sensing node.

R1 gives the circuit a fixed reference resistance. Without R1, the LDR alone would not create a defined divider voltage. The voltage at the transistor base would not be properly controlled. Therefore, R1 is necessary for converting the LDR resistance variation into a usable voltage signal.

The value of R1 influences the switching threshold. If R1 is changed, the light level required to turn on Q1 will also change. Therefore, R1 is not only a passive resistor. It is part of the decision level of the circuit.

3.4.2 LDR Function in Divider

In the voltage divider, the LDR forms the lower part of the divider. It is connected between the sensing node and ground. Because its resistance changes with light intensity, the sensing node voltage also changes.

The divider relation is:

$$V_{\text{sense}} = V_{\text{CC}} \times R_{\text{LDR}} \div (R_1 + R_{\text{LDR}})$$

This relation shows that the sensing voltage depends directly on the LDR resistance when the LDR is the lower resistor. If R_{LDR} increases, V_{sense} increases. If R_{LDR} decreases, V_{sense} decreases.

This mathematical relation proves that the circuit is darkness activated under the shown arrangement.

3.4.3 Divider Output Node

The middle node between R1 and the LDR is the divider output node. This node provides V_{sense} . The sensing voltage is then applied to the base of Q1 through R2.

The divider output node is one of the most important measurement points in the circuit. It determines whether Q1 will remain off or turn on. During simulation, this node should be measured under both bright and dark conditions.

In bright light, the expected value of V_{sense} is low. In darkness, the expected value of V_{sense} is higher. If the simulation does not show this behavior, then the LDR placement, R1 connection, or ground reference may be incorrect.

3.4.4 Dark Activated Logic

The circuit is dark activated because the LDR is connected toward ground. In bright light, the LDR resistance becomes low and pulls the sensing node down. In darkness, the LDR resistance becomes high and allows the sensing node voltage to rise.

This logic may be written as:

$$\begin{aligned} \text{Light} \uparrow &\Rightarrow R_{\text{LDR}} \downarrow \\ &\Rightarrow V_{\text{sense}} \downarrow \Rightarrow Q_1 \text{ off} \end{aligned}$$

$$\begin{aligned} \text{Light} \downarrow &\Rightarrow R_{\text{LDR}} \uparrow \\ &\Rightarrow V_{\text{sense}} \uparrow \Rightarrow Q_1 \text{ on} \end{aligned}$$

Therefore, the position of the LDR in the divider determines the activation behavior. If R1 and the LDR were exchanged, the circuit behavior would become opposite. That alternate arrangement would make the transistor more likely to turn on under bright light instead of darkness.

3.5 Base Control Section

3.5.1 R2 Function

R2 is connected between the sensing node and the base of Q1. Its primary function is to limit the current entering the base of the transistor. A transistor base emitter junction behaves similarly to a forward biased diode when the transistor turns on. Without a base resistor, excessive current could flow into the base, which may damage the transistor or disturb the divider operation.

R2 therefore acts as a controlled boundary between the voltage divider and the transistor base. It allows the sensing voltage to influence Q1 while preventing uncontrolled base current.

The value of R2 affects the strength of transistor switching. If R2 is too small, the base current may become unnecessarily high. If R2 is too large, the base current may be too weak to saturate Q1. Therefore, R2 must be selected with both protection and switching reliability in mind.

3.5.2 Base Protection Role

The base protection role of R2 is essential because the base emitter junction of Q1 must not be connected directly to a low impedance voltage source without current limiting. The resistor protects the transistor by limiting current to a safe value.

The approximate base current can be estimated using:

$$I_B \approx (V_{\text{sense}} - V_{\text{BE}}) \div R_2$$

This expression applies when V_{sense} is greater than V_{BE} . The base emitter voltage V_{BE} is usually about (0.6V) to (0.7V) for a silicon transistor in conduction.

If V_{sense} is lower than the base emitter threshold, the base current is very small and Q1 remains off. If V_{sense} is sufficiently higher than the threshold, base current flows and Q1 turns on.

3.5.3 Signal Transfer Role

R2 also has a signal transfer role. It transfers the voltage condition created by the LDR divider into the base of Q1. The transistor base does not directly sense light. It senses the electrical effect of light through the voltage divider and R2.

This means that the LDR signal is not lost. It is translated into base drive. The voltage at the sensing node becomes a control input for the transistor. R2 permits this control action while limiting current.

The signal chain at this point is:

$$\begin{aligned} R_{\text{LDR}} \text{ changes} &\Rightarrow V_{\text{sense}} \text{ changes} \\ &\Rightarrow I_B \text{ changes} \Rightarrow Q_1 \text{ changes state} \end{aligned}$$

This chain explains how the sensor section influences the switching section.

3.5.4 Design Weakness

The possible weakness of this section is that R2 may limit the base current too much. If the base current is not sufficient, Q1 may not enter deep saturation. In that condition, the transistor may conduct partially, the relay may receive insufficient current, and the relay may chatter or fail to energize.

For reliable relay driving, the base current must be compared with the relay coil current. A conservative transistor switching design often checks the condition:

$$I_B \geq I_C \div 10$$

In this circuit, the collector current I_C is approximately the relay coil current when the relay is energized. If the relay coil requires a large current and the base current is small, the BC547 may not switch the relay reliably in real hardware.

Therefore, this design is structurally correct, but its real hardware reliability depends on base drive verification.

3.6 Transistor Switching Section

3.6.1 Q1 Component

Q1 is a BC547 NPN bipolar junction transistor. It is used as a switching device rather than as a linear amplifier. In this circuit, its role is to control the relay coil current according to the voltage produced by the LDR divider.

The transistor is necessary because the LDR voltage divider cannot directly supply the relay coil current. The transistor provides a controlled current path for the relay coil. When Q1 is off, the relay coil current is interrupted. When Q1 is on, the relay coil current flows.

Thus, Q1 is the electrical decision actuator between the sensing section and the relay section.

3.6.2 Q1 Connection

Q1 is connected as a low side switch. The emitter is connected to ground. The collector is connected to one side of the relay coil. The other side of the relay coil is connected to the positive control supply. The base is connected to the sensing node through R2.

This arrangement means that Q1 controls the connection between the relay coil and ground. When the transistor is off, the relay coil does not have a complete path to ground. When the transistor is on, the relay coil current flows from the positive supply through the coil, through the collector emitter path of Q1, and then to ground.

The low side switching arrangement is suitable for an NPN transistor because the emitter is referenced to ground and the base can be driven positive relative to the emitter.

3.6.3 OFF State

In the off state, the base emitter voltage is not high enough to forward bias the base emitter junction. The base current is insufficient or nearly zero. Because of this, the collector emitter path remains non conducting.

When the collector emitter path is not conducting, the relay coil current cannot flow. The relay remains de energised. The relay contact stays in its resting position. If the load is connected through the normally open contact, the lamp and buzzer remain off.

The off state occurs when the LDR is illuminated and its resistance is low. This low resistance pulls the sensing node voltage down, preventing Q1 from turning on.

3.6.4 ON State

In the on state, the sensing voltage rises enough to drive base current through R2. The base emitter junction becomes forward biased. Q1 then conducts between collector and emitter.

When Q1 conducts strongly enough, the relay coil receives current. The magnetic field inside the relay coil builds up and causes the relay contact to change state. If the lamp and buzzer are connected through the normally open contact, they turn on when the contact closes.

For reliable operation, Q1 should operate in saturation when driving the relay coil. In saturation, the collector emitter voltage is low, and most of the supply voltage appears across the relay coil. This allows the relay to receive the voltage and current required for proper activation.

3.7 Relay Coil Section

3.7.1 Relay Coil Function

The relay coil is the electromagnetic control part of the relay. When current flows through the coil, a magnetic field is produced. This magnetic field moves the internal contact mechanism of the relay. As a result, the relay contact changes from its resting state to its energized state.

In this project, the relay coil allows a low power transistor circuit to control a separate output circuit. The coil belongs to the control side, while the relay contact belongs to the load side.

The relay coil is therefore the conversion point between electronic switching and mechanical contact movement.

3.7.2 Relay Coil Current Path

When Q1 turns on, the relay coil current path is completed. The current path is:

$$\begin{aligned} V_{CC} &\Rightarrow \text{relay coil} \Rightarrow \text{collector of } Q_1 \\ &\Rightarrow \text{emitter of } Q_1 \Rightarrow \text{ground} \end{aligned}$$

This path exists only when Q1 conducts. If Q1 is off, the path is open and the relay coil cannot energize.

The relay coil current may be calculated if the coil resistance is known:

$$I_{\text{relay}} = V_{\text{relay}} \div R_{\text{coil}}$$

This current must be within the safe collector current capability of the BC547. It must also be low enough for the control supply to provide without voltage collapse.

3.7.3 Relay Activation

When the relay coil receives sufficient current, the relay activates. The internal magnetic field pulls the contact from its normal position to its energized position. This causes the normally open contact to close and the normally closed contact to open.

In this project, relay activation is the direct result of darkness detection. The LDR detects darkness, the divider raises the sensing voltage, Q1 turns on, and the relay coil energizes. The contact movement then allows the output load to receive power.

Relay activation is therefore the central mechanical event that connects the control decision to the load response.

3.7.4 Relay Rating Requirement

The relay coil voltage must match the control supply. Since the relay shown in the project is a 5 V relay, the coil should receive approximately 5 V during activation. If the coil receives significantly less voltage, the relay may not energize. If the coil receives too much voltage, the relay may overheat or fail.

The relay coil current must also be checked. If the coil current is too high for Q1, the transistor may overheat or fail. If the supply cannot provide the required relay current, the circuit may behave unpredictably.

Therefore, the relay is not selected only by voltage. Its coil resistance, coil current, contact rating, and switching requirement must all be compatible with the circuit.

3.8 Flyback Diode Section

3.8.1 D1 Function

D1 is the flyback diode connected across the relay coil. Its function is to protect Q1 from the high voltage spike generated when the relay coil is switched off.

The relay coil is an inductive component. An inductor resists sudden changes in current. When Q1 turns off, the current through the relay coil cannot instantly become zero. The coil therefore produces a voltage that attempts to keep current flowing. This voltage can become large enough to damage the transistor.

D1 provides a safe path for the coil current during switch off. It clamps the voltage across the coil and protects Q1 from destructive reverse voltage stress.

3.8.2 Correct Diode Orientation

The flyback diode must be connected with correct polarity. The cathode of D1 must be connected to the positive side of the relay coil. The anode of D1 must be connected to the transistor collector side of the relay coil.

This orientation keeps the diode reverse biased during normal relay operation. While the relay is on, current flows through the coil and Q1, but D1 does not conduct. When Q1 turns off, the coil voltage reverses in order to keep current flowing. D1 then becomes forward biased and conducts the stored coil energy.

If D1 is reversed, it may create a direct short path across the supply during relay activation. This can damage the diode, overload the supply, or prevent the circuit from operating correctly.

3.8.3 Normal Operation

During normal relay on operation, D1 is reverse biased. This means that the diode does not conduct while the relay coil is being energized by the supply through Q1.

The normal relay current path is through the relay coil and transistor, not through the diode. D1 remains inactive during this stage, but it is ready to conduct when the transistor switches off.

This inactive normal state is important because a flyback diode must not interfere with relay energization. It must protect the circuit only during the turn off transient.

3.8.4 Switch Off Operation

When Q1 turns off, the relay coil current path through the transistor is interrupted. The magnetic field in the coil collapses. The collapsing field produces a voltage that attempts to continue current flow.

At this moment, D1 becomes forward biased and allows current to circulate through the coil and diode. The stored magnetic energy is then dissipated safely as heat in the coil resistance and diode path.

The energy stored in the coil may be represented as:

$$E = 0.5 \times L \times I^2$$

In this expression, L is the inductance of the relay coil and I is the coil current before switch off. This equation shows that higher coil current stores more energy. Therefore, the diode must be rated appropriately for the relay coil.

3.9 Relay Contact Section

3.9.1 Contact Function

The relay contact is the load switching part of the relay. It is physically moved by the magnetic action of the relay coil. Its function is to open or close the load circuit.

In this project, the contact controls the lamp and buzzer. The LDR and transistor do not directly power the output devices. They only energize the relay coil. The relay contact then completes or interrupts the output load circuit.

This makes the relay contact the final switching boundary between the control decision and the output action.

3.9.2 Common Terminal

The common terminal is the moving contact terminal of the relay. It is the terminal that changes connection depending on whether the relay is energized or de energized.

In the load circuit, the common terminal should be connected in such a way that it can transfer the load side source to the output components when the relay activates. The common terminal must be correctly identified in the Proteus relay symbol because incorrect contact wiring can cause the relay coil to work while the lamp and buzzer remain off.

The common terminal is therefore a critical connection point for the output side.

3.9.3 Normally Open Terminal

The normally open terminal is open when the relay coil is not energized. It closes when the relay coil is energized. For a darkness activated alarm, the normally open terminal is the correct choice if the lamp and buzzer must turn on only in darkness.

When the LDR is illuminated, Q1 remains off and the relay coil remains de energised. The normally open contact remains open, so the lamp and buzzer remain off. When darkness occurs, Q1 turns on and energizes the relay. The normally open contact closes, completing the load circuit and activating the lamp and buzzer.

This contact behavior matches the desired project function.

3.9.4 Normally Closed Terminal

The normally closed terminal is closed when the relay coil is not energized. It opens when the relay coil is energized. If the lamp and buzzer were connected through the normally closed terminal, the behavior would be opposite. The output would be active in the relay resting state and inactive when the relay energizes.

For this project, the normally closed terminal is not the preferred output terminal if the goal is to activate the lamp and buzzer in darkness. It is useful only if the project were redesigned to create an opposite switching behavior.

The report should identify the normally closed terminal so that the examiner understands why the normally open terminal was selected.

3.9.5 Correct Contact Selection

For the required darkness activated behavior, the output load should be connected through the normally open contact. This ensures that the lamp and buzzer remain off until the relay coil energizes.

The correct sequence is:

$$\begin{aligned} \text{darkness} &\Rightarrow Q_1 \text{ on} \Rightarrow \text{relay coil energized} \\ &\Rightarrow \text{normally open contact closed} \Rightarrow \text{lamp and buzzer on} \end{aligned}$$

This contact selection makes the relay behavior consistent with the project aim. If the contact selection is wrong, the output behavior may become inverted or may not operate at all.

3.10 Output Load Section

3.10.1 Lamp Function

The lamp is the visual output component of the project. Its function is to provide visible indication or illumination when the circuit detects darkness. When the relay contact closes, the lamp receives power from the load side source and turns on.

The lamp makes the output state observable. In a darkness activated system, this is especially meaningful because the lamp can represent automatic lighting, warning indication, or visual alarm response.

The lamp must be selected according to the voltage and current available from the load side source. It must also be compatible with the relay contact rating.

3.10.2 Buzzer Function

The buzzer is the audible output component of the project. Its function is to produce sound when the circuit detects darkness and the relay contact activates the output path.

The buzzer gives the project an alarm function. While the lamp provides visual output, the buzzer provides sound output. Together, they create a stronger response than either output alone.

The buzzer must be rated for the load side voltage. If the buzzer is polarity sensitive, its positive and negative terminals must be connected correctly. In Proteus, the buzzer model must also be compatible with the selected source type.

3.10.3 Load Activation

The lamp and buzzer activate when the relay contact closes. The relay contact completes the path from the load side source to the output components. This means the output activation does not occur directly from the LDR or transistor. It occurs through the relay contact.

The load activation sequence is:

relay coil energized $\Rightarrow$ contact closes
 $\Rightarrow$ load circuit completes $\Rightarrow$ lamp and buzzer activate

This sequence confirms that the relay is the switching interface between the control circuit and the output circuit.

3.10.4 Load Rating Requirement

The lamp and buzzer must match the load side source. Their voltage ratings must be compatible with the source voltage. Their current requirements must be within the relay contact rating. If the load current exceeds the relay contact rating, the relay contacts may overheat, weld, or fail.

The relay contact rating must be checked for both voltage and current. If the load is alternating current, the contact rating must support that type of load. If the load has inrush current, such as some lamps, the relay must tolerate the starting current as well as the steady operating current.

Therefore, the output section must be designed not only for functional operation but also for electrical safety.

3.11 LED Indicator Section

3.11.1 D2 Function

D2 is the green LED used as a visual indicator in the circuit. Its purpose is to provide a visible signal related to the operating state of the control circuit. Depending on its exact connection, it may indicate relay activation, control side power, or another defined circuit condition.

An indicator LED improves the readability of the circuit during simulation. It allows the user to observe a state change without relying only on the relay movement, lamp, or buzzer. In an academic project, the indicator also helps explain the internal condition of the circuit.

However, the meaning of D2 must be stated precisely. A report should not simply say that the LED is present. It must explain what the LED indicates.

3.11.2 R3 Function

R3 is the current limiting resistor for D2. LEDs must not be connected directly across a voltage source without current limiting. Without R3, excessive current may flow through D2 and damage it.

The LED current may be estimated by:

$$I_{LED} = (V_{CC} - V_F) \div R_3$$

In this expression, V_F is the forward voltage of the LED. If the control supply is 5 V and the green LED forward voltage is approximately 2 V, then for $R_3 = 330\Omega$:

$$I_{LED} = (5 - 2) \div 330$$

$$I_{LED} \approx 0.00909A$$

$$I_{LED} \approx 9.09mA$$

This current is generally reasonable for an indicator LED under a 5 V supply. If the supply voltage is higher, R3 must be recalculated.

3.11.3 Indicator Current Path

The indicator current path exists when the LED branch is forward biased and connected to an active voltage difference. Current flows through R3 and D2 in series. R3 limits the magnitude of that current.

The exact path depends on the schematic connection. If D2 and R3 are connected across the relay coil supply path, the LED may indicate relay coil activation. If they are connected directly across the control supply, the LED may indicate supply presence. If they are connected to the output side, the LED may indicate output activation.

Therefore, the report must describe the current path according to the actual Proteus net connections. The written explanation must match the circuit exactly.

3.11.4 Indicator Meaning

The meaning of D2 must be explicitly defined in the report. The most academically precise statement is that D2 provides visual status indication for the node to which it is connected. If it is wired to the relay activation node, it indicates relay activation. If it is wired to the control supply, it indicates that the control side is powered. If it is wired to the output path, it indicates output activation.

For the strongest report, the indicator should be described after verifying its connection in Proteus. The report should avoid vague language. It should state:

D2 is used as the visual indicator for the selected circuit state. R3 limits the LED current and protects D2 from excessive current.

This prevents ambiguity and makes the indicator explanation academically defensible.

3.12 Supply Section

3.12.1 Control Supply

The control supply powers the sensing, switching, protection, and relay coil sections of the circuit. It provides voltage to R1, the LDR divider, the relay coil, and the indicator branch. It also provides the electrical potential needed to forward bias the base emitter junction of Q1 through R2.

The control supply must be compatible with the relay coil rating. Since the relay is identified as a 5 V relay, the control supply should provide the correct voltage for that coil. If the control voltage is too low, the relay may not energize. If it is too high, the relay coil may overheat.

The control supply must also be stable. When the relay coil turns on, it draws current. If the supply voltage drops significantly during this event, the transistor and relay may behave unstably.

3.12.2 Load Supply

The load supply provides power to the lamp and buzzer. It is switched by the relay contact. This means the load supply is not controlled directly by the LDR or transistor. Instead, the relay contact determines whether the load circuit is open or closed.

The load supply may be different from the control supply, provided that the relay contact is rated for the load voltage and current. This is one of the advantages of using a relay. The control side can operate at a low voltage, while the load side can use a separate source suitable for the lamp and buzzer.

The load supply must be clearly labeled in the schematic and described in the report.

3.12.3 Grounding Requirement

The transistor control side requires a correct ground reference. The emitter of Q1 must be connected to ground. The LDR divider must also be referenced correctly so that the base voltage is meaningful relative to the emitter.

The base emitter voltage is:

$$V_{BE} = V_B - V_E$$

Since the emitter is connected to ground, V_E is approximately zero in normal operation. Therefore, the base voltage directly determines whether the transistor turns on.

If the ground reference is missing or incorrect, the base emitter voltage cannot be established properly. The transistor may remain off, behave unpredictably, or fail to drive the relay.

3.12.4 Isolation Requirement

The relay provides electrical isolation between the control side and the load side. The control side includes the LDR, resistors, transistor, relay coil, diode, and indicator. The load side includes the relay contact, lamp, buzzer, and load source.

This isolation is important because the low power sensing circuit does not need to carry the load current. The relay contact carries the load current, while the transistor carries only the relay coil current.

The isolation also improves safety when the load side uses a different voltage source. However, relay isolation does not automatically make the circuit physically safe. If the load side uses high voltage, proper insulation, spacing, enclosure, and protection are still required.

3.13 Bright Condition Current Path

3.13.1 Sensor State

In the bright condition, light falls on the LDR. The resistance of the LDR becomes low. Since the LDR is connected toward ground, this low resistance pulls the divider output node toward ground.

The sensing voltage is therefore low:

$$R_{LDR} \downarrow \Rightarrow V_{sense} \downarrow$$

This is the first electrical effect of the bright condition.

3.13.2 Base State

Because V_{sense} is low, the base of Q1 does not receive enough voltage through R2. The base emitter voltage remains below the turn on threshold.

The transistor condition may be written as:

$$V_{BE} < 0.6V \text{ to } 0.7V$$

Therefore, the base current is insufficient for transistor conduction.

3.13.3 Transistor State

Since the base current is insufficient, Q1 remains off. The collector emitter path is open or nearly non conducting. The relay coil cannot complete its path to ground through

Q1.

This means that the transistor blocks relay coil current in the bright condition.

3.13.4 Relay State

Because Q1 is off, the relay coil is de energised. No sufficient magnetic field is produced inside the relay coil. The relay contact remains in its resting state.

If the lamp and buzzer are connected through the normally open contact, the output circuit remains open.

3.13.5 Output State

Since the relay contact remains open in the bright condition, the lamp and buzzer do not receive load power. The lamp remains off, and the buzzer remains silent.

The complete bright condition chain is:

$$\begin{aligned}\text{bright light} &\Rightarrow R_{\text{LDR}} \downarrow \\ &\Rightarrow V_{\text{sense}} \downarrow \Rightarrow Q_1 \text{ off} \\ &\Rightarrow \text{relay off} \Rightarrow \text{lamp and buzzer off}\end{aligned}$$

This confirms that the circuit remains inactive when the LDR is illuminated.

3.14 Dark Condition Current Path

3.14.1 Sensor State

In the dark condition, the light falling on the LDR decreases. The resistance of the LDR increases. Since the LDR is connected toward ground, this higher resistance weakens the pull toward ground.

The sensing voltage increases:

$$R_{\text{LDR}} \uparrow \Rightarrow V_{\text{sense}} \uparrow$$

This is the first electrical effect of darkness.

3.14.2 Base State

As V_{sense} rises, current flows through R2 into the base of Q1. When the base emitter voltage reaches the required conduction level, the transistor begins to turn on.

The approximate turn on condition is:

$$V_{\text{BE}} \approx 0.6V \text{ to } 0.7V$$

The base current is controlled by R2. This base current allows Q1 to conduct between collector and emitter.

3.14.3 Transistor State

When Q1 receives sufficient base current, it turns on. Ideally, it enters saturation so that it can drive the relay coil efficiently. In the on state, the collector emitter path becomes conductive.

The relay coil current path becomes complete:

$$V_{CC} \Rightarrow \text{relay coil} \Rightarrow Q_1 \Rightarrow \text{ground}$$

This current energizes the relay coil.

3.14.4 Relay State

When the relay coil energizes, the relay contact changes state. The normally open contact closes, and the load circuit becomes complete.

The relay therefore converts the transistor switching action into load switching action. It is the mechanical transfer stage between the control side and the output side.

3.14.5 Output State

When the relay contact closes, the lamp and buzzer receive power from the load side source. The lamp turns on and provides visual indication. The buzzer sounds and provides audible indication.

The complete dark condition chain is:

$$\begin{aligned} \text{darkness} &\Rightarrow R_{LDR} \uparrow \\ &\Rightarrow V_{\text{sense}} \uparrow \Rightarrow Q_1 \text{ on} \\ &\Rightarrow \text{relay on} \Rightarrow \text{lamp and buzzer on} \end{aligned}$$

This confirms that the circuit performs the intended darkness activated function.

3.15 Scheme Safety Notes

3.15.1 Relay Contact Rating

The relay contact must be rated for the voltage and current of the lamp and buzzer load. The contact rating must not be exceeded. If the load current is higher than the relay contact rating, the contacts may heat, arc, weld, or fail.

The contact rating must be checked separately from the relay coil rating. A relay may have a 5 V coil but still have contact ratings for different load voltages and currents. The coil rating tells how to activate the relay. The contact rating tells what load the relay can safely switch.

Therefore, the report must distinguish between relay coil rating and relay contact rating.

3.15.2 Transistor Current Rating

Q1 must be able to handle the relay coil current. Since Q1 is in series with the relay coil during activation, its collector current is approximately the relay coil current.

The condition is:

$$I_C \approx I_{\text{relay}}$$

The relay coil current must be lower than the safe collector current rating of the transistor. In addition, Q1 must receive enough base current to operate in saturation. If Q1 is not saturated, it may dissipate more power and heat up.

The transistor power during saturation may be estimated as:

$$P_Q = V_{CE\text{ sat}} \times I_C$$

This value must remain within the safe power dissipation of the transistor.

3.15.3 Diode Polarity

D1 must not be reversed. The cathode must be connected to the positive side of the relay coil, and the anode must be connected to the transistor collector side.

Correct polarity allows D1 to remain reverse biased during normal operation and conduct only during the relay turn off transient. Reversed polarity may cause a short circuit across the relay supply when Q1 turns on. This can damage the diode, the supply, or the transistor.

Therefore, diode orientation is a critical safety and reliability requirement.

3.15.4 LED Resistor Safety

R3 must be selected according to the control supply voltage and the forward voltage of the LED. The LED current must remain within a safe operating range.

The LED current is:

$$I_{LED} = (V_{CC} - V_F) \div R_3$$

If V_{CC} increases, the same value of R3 will produce higher LED current. Therefore, R3 must be recalculated if the supply voltage changes.

This safety note is necessary because the LED may appear simple, but incorrect current limiting can damage it.

3.15.5 AC Load Safety

If V1 represents an alternating current load source or a high voltage source, additional safety measures are required before physical construction. The relay contact must be rated for the load voltage and current. The wiring must be insulated. The circuit must be enclosed. A fuse should be used in the load path. Proper spacing must be maintained between the control side and the high voltage load side.

Proteus simulation can show logical operation, but it does not prove physical safety. Real hardware involving high voltage requires careful design, safe construction practice, and proper protective components.

Therefore, the scheme is acceptable as a simulation model, but a real high voltage implementation requires a stricter safety design.

3.16 Chapter Summary

3.16.1 Scheme Function Summary

The schematic implements the complete function of the automatic darkness activated alarm and load control system. The LDR senses the light condition. R1 and the LDR form the voltage divider. R2 transfers and limits the base signal. Q1 switches the relay coil. The relay contact switches the lamp and buzzer. D1 protects the transistor from relay coil kickback. D2 and R3 provide visual indication.

The scheme therefore contains all essential stages of the project:

sensing $\Rightarrow$ signal conditioning $\Rightarrow$ switching
 $\Rightarrow$ relay isolation $\Rightarrow$ output actuation
 $\Rightarrow$ protection and indication

This confirms that the circuit is a complete sensor controlled switching system.

3.16.2 Scheme Strength Summary

The main strength of the scheme is its correct separation of functional layers. The LDR does not directly drive the lamp or buzzer. The weak sensor signal is first converted into a voltage, then applied to a transistor, then used to energize a relay, and finally used to switch the output load.

This layered structure is electrically sound because each component performs a defined role. The LDR senses. The divider generates a control voltage. R2 limits base current. Q1 switches the relay coil. D1 protects Q1. The relay isolates and switches. The lamp and buzzer provide output response.

The scheme is also strong because it includes both visual and audible output. It is therefore suitable as an educational model for automatic alarm and light control.

3.16.3 Scheme Weakness Summary

The main weakness of the scheme is that real hardware margins must be verified before physical construction. The base current through R2 may be too small for strong relay driving if the relay coil current is high. The relay contact rating must be checked against the lamp and buzzer load. The flyback diode must be correctly oriented and properly rated. The LED resistor must match the supply voltage. If the load side source is high voltage, additional safety protection is required.

The scheme is therefore structurally correct and academically valid, but its physical implementation requires verification of current, voltage, component ratings, saturation margin, and load safety.

A stronger version of the scheme would include an adjustable threshold, a comparator, hysteresis, and a stronger relay driver. However, the existing scheme already demonstrates the essential operating principle of a darkness activated relay controlled alarm system.

4

Simulation

The simulation video is available at <https://youtu.be/4wlEyLrvDhk>.

4.1 Proteus Simulation Environment

The simulation stage represents the experimental verification part of the project. After the circuit has been described, calculated, and arranged schematically, it is necessary to test whether the theoretical operation is reproduced under controlled simulation conditions. In this project, the simulation is used to verify that the LDR-based sensing stage, transistor switching stage, relay driver stage, and lamp and buzzer output stage operate as one complete automatic darkness-activated system.

The simulation chapter is not only a visual demonstration of the circuit. It is the formal verification layer of the report. It proves whether the designed circuit performs the intended operation under two opposite environmental conditions: bright light and darkness. Therefore, this chapter must show the simulation environment, setup procedure, expected behavior, observed behavior, measurement points, result comparison, evidence screenshots, acceptance criteria, and limitations of the simulation.

4.1.1 Software Used

The circuit was simulated using Proteus Design Suite. Proteus was selected because it allows electronic circuits to be designed, connected, powered, simulated, and tested in a virtual environment before physical implementation. The software provides a schematic workspace where components such as resistors, LDRs, transistors, relays, diodes, LEDs, lamps, buzzers, voltage sources, and measurement instruments can be placed and interconnected.

In this project, Proteus provides a suitable platform for verifying the operation of the darkness-activated alarm circuit. The complete circuit can be assembled in the schematic editor, and its behavior can be observed while the light condition applied to the LDR is changed. This makes it possible to confirm whether the circuit responds correctly to bright and dark conditions.

4.1.2 Simulation Type

The simulation type used in this project is electronic circuit simulation. The circuit is simulated as an analog and electromechanical system because it includes resistive sensing, transistor switching, relay coil activation, relay contact movement, lamp operation, buzzer operation, LED indication, and diode protection.

The simulation contains both low-power control behavior and switched-load behavior. The low-power section includes the LDR voltage divider, base-current path, transistor, relay coil, indicator LED, and flyback diode. The output section includes the relay contact, lamp, buzzer, and load-side source. Because the relay creates a controlled connection between the sensing circuit and the output circuit, the simulation must confirm not only voltage changes but also relay switching action.

4.1.3 Simulation Purpose

The main purpose of the simulation is to verify the automatic switching behavior of the circuit under bright and dark conditions. The circuit is expected to remain inactive when light falls on the LDR and to become active when the LDR is placed in darkness. This behavior must be proven through observed circuit states and electrical measurements.

The simulation specifically verifies the following chain of operation:

$$\begin{aligned} & \text{Light condition} \rightarrow \text{LDR resistance} \rightarrow V_{\text{sense}} \rightarrow V_{BE} \\ & \rightarrow Q_1 \text{ state} \rightarrow \text{Relay state} \rightarrow \text{Output state} \end{aligned}$$

This chain is the complete functional path of the project. If any stage fails, the final output will not correctly represent the light condition. Therefore, the simulation must confirm each stage, not only the final lamp and buzzer behavior.

4.1.4 Simulation Advantage

The main advantage of simulation is that the circuit can be tested before hardware construction. This reduces the risk of wiring errors, incorrect component selection, unsafe load connection, and component damage. Simulation also allows the designer to observe voltages, currents, and switching states without physically measuring the circuit with external instruments.

In this project, simulation is especially useful because the circuit contains a relay and an LDR. The LDR response can be changed virtually using the light source control in Proteus, while the relay behavior can be observed as the transistor turns on and off. This allows the designer to verify the intended darkness-activated response before building the circuit on a breadboard or printed circuit board.

4.2 Simulation Setup

The simulation setup defines how the circuit is arranged in Proteus before testing begins. A correct setup is necessary because a simulation result is only valid when the component values, component positions, supply connections, relay contact wiring, and ground references are correctly configured.

The setup stage therefore establishes the exact virtual conditions under which the circuit is tested. It ensures that the sensor stage, switching stage, protection stage, relay stage, indicator stage, and output stage are all connected according to the circuit design.

4.2.1 Component Placement

The required components were placed on the Proteus schematic workspace. The sensing section includes the LDR and the resistor R1. The switching section includes resistor R2 and transistor Q1. The relay-driving section includes the relay coil and the transistor collector path. The protection section includes diode D1 connected across the relay coil. The indicator section includes LED D2 and resistor R3. The output section includes the lamp L1, buzzer BUZ1, relay contact, and output-side source.

Each component must be placed so that its function is clear in the schematic. The LDR must be placed in the sensing section. The transistor must be placed between the relay coil and ground as a low-side switch. The flyback diode must be placed directly across the relay coil. The lamp and buzzer must be connected through the relay contact so that they operate only when the relay switches to the active state.

4.2.2 Component Values

The component values used in the simulation are taken from the designed circuit. The main resistor values are:

$$R_1 = 10k\Omega$$

$$R_2 = 6.8k\Omega$$

$$R_3 = 330\Omega$$

The relay used in the simulation is a 5 V relay. The transistor used is a BC547 NPN transistor. The flyback diode is a 1N4148 diode. The indicator is a green LED. The output devices are a lamp and a buzzer.

These values determine the behavior of the circuit. R1 and the LDR determine the sensing voltage. R2 determines the base-current limitation. R3 determines the LED current. The relay voltage rating determines whether the relay coil can energize correctly. Therefore, the simulation must use these values consistently so that the result corresponds to the actual circuit design.

4.2.3 LDR Light Source

The LDR is controlled using the Proteus light source or torch control. This control allows the light intensity applied to the LDR to be changed during simulation. When the torch is applied to the LDR, the LDR represents a bright-light condition. When the torch is removed or reduced, the LDR represents a dark condition.

The purpose of this control is to simulate the environmental input of the project. Since the project is based on darkness detection, the LDR is the primary input com-

ponent. The simulation is meaningful only if the LDR condition is changed and the response of the remaining circuit is observed.

4.2.4 Relay Contact Wiring

The lamp and buzzer are connected through the relay contact. The relay coil belongs to the control side of the circuit, while the relay contact belongs to the output side. This separation allows the low-power transistor circuit to control the external output load.

For darkness activation, the output devices should be connected through the normally open contact of the relay. In the inactive state, the contact remains open and the lamp and buzzer remain off. When darkness causes the transistor to energize the relay coil, the relay contact closes and supplies power to the lamp and buzzer.

The relay contact wiring must be checked carefully because incorrect use of the normally open or normally closed terminal can reverse the intended behavior of the circuit. If the load is connected to the wrong contact, the lamp and buzzer may activate in bright light instead of darkness.

4.3 Simulation Aim

The simulation aim defines what the simulation must prove. The purpose is not only to show that the lamp and buzzer can turn on. The purpose is to prove that the entire sensing and switching chain behaves correctly according to the project design.

The simulation must prove that the circuit is dark-activated. It must also prove that the LDR affects the transistor base voltage, that the transistor can energize the relay, and that the relay contact can supply the lamp and buzzer.

4.3.1 Main Aim

The main aim of the simulation is to prove that darkness activates the output. When the LDR is exposed to bright light, the lamp and buzzer should remain off. When the LDR is placed in darkness, the lamp and buzzer should turn on.

The main simulation claim can be written as:

$$\text{Darkness} \Rightarrow \text{Output ON}$$

$$\text{Bright light} \Rightarrow \text{Output OFF}$$

This is the primary functional requirement of the project. If this requirement is not satisfied, the project does not achieve its main purpose.

4.3.2 Sensor Aim

The sensor aim is to prove that changes in LDR resistance affect the transistor base voltage. The LDR and R1 form a voltage divider. The output of this voltage divider is the sensing voltage, represented as:

$$V_{sense} = V_{CC} \frac{R_{LDR}}{R_1 + R_{LDR}}$$

In bright light, the LDR resistance decreases. Since the LDR is connected toward ground, the sensing voltage decreases. In darkness, the LDR resistance increases, so the sensing voltage rises. The simulation must confirm that this voltage change occurs when the light condition is changed.

4.3.3 Relay Aim

The relay aim is to prove that Q1 can energize the relay. The relay coil is controlled by the transistor. When the base-emitter voltage of Q1 becomes high enough, the transistor conducts and allows current to pass through the relay coil.

The switching condition can be described by the approximate base-emitter threshold:

$$V_{BE} \approx 0.6V \text{ to } 0.7V$$

When V_{BE} reaches this approximate value and sufficient base current is available, Q1 turns on. The relay coil should then energize. The simulation must verify that the relay changes state only when Q1 conducts.

4.3.4 Output Aim

The output aim is to prove that relay activation powers the lamp and buzzer. The output devices are not directly controlled by the LDR. They are controlled by the relay contact. Therefore, the simulation must verify that the output devices activate only when the relay contact closes.

The output relation can be expressed as:

$$\text{Relay coil energized} \Rightarrow \text{Relay contact closed} \Rightarrow \text{Lamp ON and Buzzer ON}$$

This proves that the circuit performs a complete sensor-to-actuator operation.

4.4 Bright-Light Test

The bright-light test verifies the inactive state of the circuit. This test is necessary because a darkness-activated system must remain off when light is present. The circuit cannot be considered correct if it activates during bright conditions.

In this test, light is applied to the LDR using the Proteus torch control. The expected result is that the LDR resistance decreases, the sensing voltage becomes low, Q1 remains off, the relay remains de-energized, and the lamp and buzzer remain off.

4.4.1 Test Condition

The test condition for this part of the simulation is bright light applied to the LDR. The Proteus torch or light source is directed toward the LDR so that the LDR behaves as a low-resistance component.

This represents a condition in which the environment is sufficiently illuminated. Under this condition, the circuit should not trigger the alarm output.

4.4.2 Expected LDR Behavior

When light is applied to the LDR, its resistance decreases. This is the physical basis of the sensing function. Since the LDR is connected in the lower part of the voltage divider, a lower LDR resistance pulls the sensing node closer to ground potential.

The expected relationship is:

$$\text{Light increases} \Rightarrow R_{LDR} \text{ decreases}$$

$$R_{LDR} \text{ decreases} \Rightarrow V_{sense} \text{ decreases}$$

Thus, under bright light, the sensing voltage should be too low to drive the transistor into conduction.

4.4.3 Expected Transistor Behavior

In the bright-light condition, Q1 is expected to remain off. The reason is that the base-emitter voltage should remain below the required turn-on level. If the transistor base does not receive enough voltage and current, the transistor cannot allow relay coil current to flow from collector to emitter.

The expected condition is:

$$V_{BE} < 0.6V \text{ to } 0.7V$$

Therefore:

$$Q_1 = OFF$$

4.4.4 Expected Relay Behavior

Since Q1 is expected to remain off, the relay coil should not receive sufficient current. As a result, the relay should remain de-energized. The relay contact should remain in its normal state.

For a normally open output connection, the open contact prevents current from reaching the lamp and buzzer. Therefore, the relay must remain inactive during the bright-light test.

4.4.5 Expected Output Behavior

The lamp and buzzer are expected to remain off during the bright-light condition. This is the correct output response because the circuit is designed to activate under darkness, not under illumination.

The expected output state is:

$$\text{Lamp} = OFF$$

$$\text{Buzzer} = OFF$$

If the lamp or buzzer turns on during bright light, the circuit behavior is incorrect or the relay contact wiring must be checked.

4.5 Bright-Light Simulation Result

The bright-light simulation result records what actually happens in Proteus when light is applied to the LDR. This section must not depend only on visual observation. It should include measured electrical quantities that prove the state of the circuit.

The most important values to record are the sensing voltage, base-emitter voltage, relay state, and output state. These values confirm whether the circuit remains inactive under bright conditions.

4.5.1 Observed (V_{sense})

The sensing voltage V_{sense} is measured at the junction between R1, the LDR, and the input path to R2. In the bright-light condition, this voltage should be low because the LDR resistance is low.

The observed value should be recorded as:

$$V_{sense\ bright} = \text{measured value}$$

For correct operation, the measured value should be low enough that it does not turn on Q1. The expected condition is:

$$V_{sense\ bright} < V_{BE\ on}$$

where $V_{BE\ on}$ is approximately between 0.6 V and 0.7 V for a silicon BJT.

4.5.2 Observed (V_{BE})

The base-emitter voltage of Q1 is measured between the base and emitter terminals of the BC547 transistor. In the bright-light condition, this voltage should remain below the turn-on threshold.

The observed value should be recorded as:

$$V_{BE\ bright} = \text{measured value}$$

The correct condition is:

$$V_{BE\ bright} < 0.6V\ to\ 0.7V$$

If this condition is satisfied, Q1 remains off and the relay remains de-energized.

4.5.3 Observed Relay State

The relay state is observed during the bright-light test. The relay should remain off because Q1 is not conducting enough current through the relay coil.

The observed relay state should be recorded as:

$$Relay_{bright} = OFF$$

A correct bright-light result means that the relay contact does not close the output path.

4.5.4 Observed Output State

The output state is observed by checking the lamp and buzzer. In the bright-light condition, the lamp should not glow and the buzzer should not sound.

The observed output state should be recorded as:

$$Lamp_{bright} = OFF$$

$$Buzzer_{bright} = OFF$$

If these results are obtained, the simulation confirms that the circuit correctly remains inactive when the LDR receives light.

4.6 Dark Condition Test

The dark condition test verifies the active state of the circuit. This is the central test of the project because the circuit is designed to activate when light is removed from the LDR.

In this test, the torch or light source is removed from the LDR. The expected result is that the LDR resistance increases, the sensing voltage rises, Q1 turns on, the relay energizes, and the lamp and buzzer activate.

4.6.1 Test Condition

The test condition for this part of the simulation is darkness applied to the LDR. This is done by removing or reducing the Proteus torch or light source. The LDR then behaves as a high-resistance component.

This condition represents the absence of sufficient environmental light. Since the project is a darkness-activated system, the output devices should activate under this condition.

4.6.2 Expected LDR Behavior

When light is removed from the LDR, its resistance increases. Because the LDR is connected in the lower part of the voltage divider, an increase in LDR resistance raises the sensing voltage at the divider output.

The expected relationship is:

$$Light\ decreases \Rightarrow R_{LDR}\ increases$$

$$R_{LDR}\ increases \Rightarrow V_{sense}\ increases$$

This rise in sensing voltage is the electrical trigger for the transistor stage.

4.6.3 Expected Transistor Behavior

In the dark condition, Q1 is expected to turn on. The sensing voltage should provide enough base-emitter voltage through R2. When the base-emitter voltage reaches the required threshold, base current flows and the transistor conducts.

The expected condition is:

$$V_{BE\,dark} \approx 0.6V \text{ to } 0.7V$$

The transistor state should then be:

$$Q_1 = ON$$

When Q1 turns on, it provides a current path for the relay coil.

4.6.4 Expected Relay Behavior

When Q1 turns on, current flows through the relay coil. The relay coil should energize and change the position of the relay contact. The normally open contact should close, allowing the output-side source to power the lamp and buzzer.

The expected relay state is:

$$Relay_{dark} = ON$$

This state confirms that the transistor stage successfully drives the relay.

4.6.5 Expected Output Behavior

When the relay contact closes, the lamp and buzzer should receive power from the output-side source. The lamp should glow and the buzzer should sound.

The expected output state is:

$$Lamp_{dark} = ON$$

$$Buzzer_{dark} = ON$$

This confirms that darkness has successfully produced the intended visual and audible response.

4.7 Dark Condition Simulation Result

The dark condition simulation result records the actual behavior of the circuit when the LDR is not illuminated. This section proves whether the circuit successfully activates under darkness.

The result must include the measured sensing voltage, measured base-emitter voltage, observed relay state, and observed output state.

4.7.1 Observed (V_{sense})

The sensing voltage is measured at the output node of the LDR voltage divider. In the dark condition, this voltage should rise because the LDR resistance is high.

The observed value should be recorded as:

$$V_{sense\,dark} = \text{measured value}$$

For correct operation, the sensing voltage must be high enough to drive Q1 through R2. The expected condition is:

$$V_{sense\ dark} > V_{sense\ bright}$$

This confirms that the LDR divider is responding correctly to the light change.

4.7.2 Observed (V_{BE})

The base-emitter voltage of Q1 is measured during the dark condition. This value should reach approximately the turn-on voltage of the transistor.

The observed value should be recorded as:

$$V_{BE\ dark} = \text{measured value}$$

The expected condition is:

$$V_{BE\ dark} \approx 0.6V \text{ to } 0.7V$$

If this condition is satisfied and relay current flows, the transistor is operating as the switching element of the circuit.

4.7.3 Observed Relay State

The relay state is observed during the dark condition. If the transistor conducts properly, the relay coil energizes and the contact changes state.

The observed relay state should be recorded as:

$$Relay_{dark} = ON$$

This confirms that the transistor stage is capable of driving the relay coil in the simulation.

4.7.4 Observed Output State

The lamp and buzzer are observed during the dark condition. When the relay contact closes, both output devices should activate.

The observed output state should be recorded as:

$$Lamp_{dark} = ON$$

$$Buzzer_{dark} = ON$$

This result confirms that the circuit successfully performs its intended function in the dark condition.

4.8 Voltage Measurement Points

Voltage measurements are necessary because they prove the internal operation of the circuit. Visual observation alone is not sufficient for a formal academic report. The voltages show whether each stage receives the correct electrical condition.

The most important voltages are the LDR divider output voltage, transistor base-emitter voltage, relay coil voltage, lamp voltage, and buzzer voltage.

4.8.1 LDR Divider Output Voltage

The LDR divider output voltage is measured at the junction of R1, LDR, and R2. This voltage is represented as:

$$V_{sense}$$

The mathematical expression for the ideal divider output is:

$$V_{sense} = V_{CC} \frac{R_{LDR}}{R_1 + R_{LDR}}$$

This voltage must be measured in both bright and dark conditions. A correct simulation should show:

$$V_{sense\ dark} > V_{sense\ bright}$$

This proves that the LDR stage correctly converts the light condition into a voltage change.

4.8.2 Transistor Base-Emitter Voltage

The base-emitter voltage is measured between the base and emitter of Q1. It is represented as:

$$V_{BE} = V_B - V_E$$

Since the emitter of Q1 is connected to ground, the base-emitter voltage is approximately equal to the base voltage with respect to ground:

$$V_{BE} \approx V_B$$

The transistor begins to turn on when:

$$V_{BE} \approx 0.6V \text{ to } 0.7V$$

Therefore, V_{BE} is the most direct measurement for confirming whether Q1 should be off or on.

4.8.3 Relay Coil Voltage

The relay coil voltage is measured across the relay coil terminals. It is represented as:

$$V_{relay}$$

In the bright-light condition, the relay coil voltage should be too low to energize the relay. In the dark condition, the relay coil voltage should approach the required operating voltage of the relay.

For a 5 V relay, the expected active condition is:

$$V_{relay\ dark} \approx 5V$$

This measurement proves whether the relay coil is actually receiving enough voltage.

4.8.4 Lamp Voltage

The lamp voltage is measured across the terminals of L1. This measurement shows whether the relay contact has supplied the lamp during darkness.

The lamp voltage should be approximately zero or inactive during bright light:

$$V_{L1\ bright} \approx 0V$$

During darkness, the lamp should receive its operating voltage:

$$V_{L1\ dark} = \text{load-side operating voltage}$$

This proves that the relay contact is correctly controlling the lamp path.

4.8.5 Buzzer Voltage

The buzzer voltage is measured across the terminals of BUZ1. This measurement confirms whether the buzzer receives power when the relay contact closes.

The expected condition is:

$$V_{BUZ1\ bright} \approx 0V$$

$$V_{BUZ1\ dark} = \text{load-side operating voltage}$$

This proves that the audible output is activated only during the intended dark condition.

4.9 Current Measurement Points

Current measurements are necessary to verify whether the components are operating within safe and functional limits. Voltage may show that a node is active, but current proves whether the component is actually conducting and whether the current path is complete.

The most important current measurements are base current, relay coil current, LED current, and load current.

4.9.1 Base Current

The base current is measured through R2 into the base of Q1. It is represented as:

$$I_B$$

A simplified estimate for the base current in the dark condition is:

$$I_B \approx \frac{V_{CC} - V_{BE}}{R_1 + R_2}$$

Using the assumed values:

$$I_B \approx \frac{5 - 0.7}{10000 + 6800}$$

$$I_B \approx 0.256mA$$

This current should be sufficient to switch Q1 in the Proteus simulation. However, for real hardware, it must also be compared with the required relay coil current to check whether Q1 is strongly saturated.

4.9.2 Relay Coil Current

The relay coil current is measured through the coil of RL1. It is represented as:

$$I_{relay}$$

If the relay coil resistance is known, it can be estimated by:

$$I_{relay} = \frac{V_{relay}}{R_{coil}}$$

The relay coil current must be high enough to energize the relay, but it must also remain within the safe collector-current capability of the BC547. The transistor collector current is approximately equal to the relay coil current:

$$I_C \approx I_{relay}$$

For reliable switching, the base current should be compared with the collector current using the saturation condition:

$$I_B \geq \frac{I_C}{10}$$

This check is important because a circuit may operate in simulation while still having weak hardware margin.

4.9.3 LED Current

The LED current is measured through D2 and R3. It is represented as:

$$I_{LED}$$

The LED current can be calculated approximately as:

$$I_{LED} = \frac{V_{CC} - V_F}{R_3}$$

For a 5 V supply and an approximate green LED forward voltage of 2 V:

$$I_{LED} = \frac{5 - 2}{330}$$

$$I_{LED} \approx 9.09mA$$

This is normally suitable for an indicator LED. If the supply voltage is different, the LED current must be recalculated.

4.9.4 Load Current

The load current is the current flowing through the lamp and buzzer when the relay contact closes. This current depends on the voltage source used on the output side and the electrical ratings of L1 and BUZ1.

The total output current can be represented as:

$$I_{load} = I_{lamp} + I_{buzzer}$$

The relay contact must be rated to handle this current. In simulation, the load current confirms whether the lamp and buzzer are actually receiving power. In physical hardware, the load current also determines relay contact safety and wire safety.

4.10 Simulation Observation Table

The simulation observation table organizes the results in a clear academic form. It allows the reader to compare the bright-light condition and the dark condition without confusion.

The table should include the LDR state, sensing voltage, transistor state, relay state, lamp state, and buzzer state. The values should be recorded from the Proteus simulation.

4.10.1 Bright-State Table

The bright-state table records the circuit behavior when light is applied to the LDR.

Recommended table structure:

Table 4.1: *Bright-state simulation observation table*

Parameter	Expected Bright-State Result	Observed Bright-State Result
LDR condition	Illuminated	To be recorded
LDR resistance	Low	To be recorded
V_{sense}	Low	To be recorded
V_{BE}	Below turn-on threshold	To be recorded
Q1 state	OFF	To be recorded
Relay state	OFF	To be recorded
Lamp state	OFF	To be recorded
Buzzer state	OFF	To be recorded

This table proves that the circuit remains inactive when light is present.

4.10.2 Dark-State Table

The dark-state table records the circuit behavior when light is removed from the LDR.

Recommended table structure:

Table 4.2: *Dark-state simulation observation table*

Parameter	Expected Dark-State Result	Observed Dark-State Result
LDR condition	Dark	To be recorded
LDR resistance	High	To be recorded
V_{sense}	High	To be recorded
V_{BE}	Approximately 0.6 V to 0.7 V	To be recorded
Q1 state	ON	To be recorded
Relay state	ON	To be recorded
Lamp state	ON	To be recorded
Buzzer state	ON	To be recorded

This table proves that the circuit becomes active when darkness is detected.

4.10.3 Comparison Table

The comparison table directly compares the expected results with the actual simulation results.

Recommended comparison structure:

Table 4.3: *Expected and actual simulation comparison table*

Test Condition	Expected Result	Actual Result	Status
Bright light on LDR	Lamp and buzzer OFF	To be recorded	Pass or fail
Darkness on LDR	Lamp and buzzer ON	To be recorded	Pass or fail
Q1 in bright light	OFF	To be recorded	Pass or fail
Q1 in darkness	ON	To be recorded	Pass or fail
Relay in bright light	De-energized	To be recorded	Pass or fail
Relay in darkness	Energized	To be recorded	Pass or fail

The comparison table is important because it converts observation into evaluation.

4.11 Expected vs Actual Result Analysis

This section interprets the simulation results. It is not enough to list observed states. The report must explain whether the observed behavior agrees with the expected theoretical behavior.

The analysis must cover the bright-light state, dark state, relay response, and output response.

4.11.1 Bright-State Analysis

In the bright-light condition, the LDR resistance is expected to decrease. As a result, the sensing voltage should remain low. If the measured V_{sense} is low and the measured V_{BE} is below the transistor turn-on threshold, Q1 remains off.

If Q1 remains off, the relay coil is not energized. The relay contact remains in its normal state, so the lamp and buzzer remain off. Therefore, a successful bright-state simulation proves that the circuit does not falsely activate when light is present.

The bright-state result is accepted if:

$$V_{sense\ bright} < V_{sense\ dark}$$

$$V_{BE\ bright} < 0.6V\ to\ 0.7V$$

$$Relay_{bright} = OFF$$

$$Lamp_{bright} = OFF$$

$$Buzzer_{bright} = OFF$$

4.11.2 Dark-State Analysis

In the dark condition, the LDR resistance is expected to increase. This should raise the sensing voltage. If the sensing voltage becomes high enough, base current flows through R2 into Q1.

When Q1 turns on, relay coil current flows. The relay energizes and closes the output contact. The lamp and buzzer then receive power from the output-side source. Therefore, a successful dark-state simulation proves that the circuit correctly responds to darkness.

The dark-state result is accepted if:

$$V_{sense\ dark} > V_{sense\ bright}$$

$$V_{BE\ dark} \approx 0.6V\ to\ 0.7V$$

$$Relay_{dark} = ON$$

$$Lamp_{dark} = ON$$

$$Buzzer_{dark} = ON$$

4.11.3 Relay Response Analysis

The relay response is the key transition between the low-power control side and the output side. The relay must remain de-energized in bright light and must energize in darkness.

A correct relay response proves that Q1 is functioning as a switch and that the relay coil receives current only when the LDR indicates darkness. If the relay does not switch, the possible causes include insufficient base current, wrong relay coil voltage, incorrect transistor connection, or incorrect supply connection.

The relay response is considered correct when:

$$Q_1 = OFF \Rightarrow Relay = OFF$$

$$Q_1 = ON \Rightarrow Relay = ON$$

4.11.4 Output Response Analysis

The output response is the final proof of the project function. The lamp and buzzer must remain off in bright light and turn on in darkness. This confirms that the relay contact is wired correctly and that the output-side circuit has a complete current path.

A correct output response proves that the project performs its intended real-world action: visual and audible activation when darkness is detected.

The output response is considered correct when:

$$Relay = OFF \Rightarrow Lamp = OFF \text{ and } Buzzer = OFF$$

$$Relay = ON \Rightarrow Lamp = ON \text{ and } Buzzer = ON$$

4.12 Simulation Screenshots

Screenshots provide visual evidence of the simulation results. They are necessary because they show the actual Proteus circuit and its operating states. The screenshots should be clear, labeled, and placed near the relevant explanation.

A complete simulation chapter should include a full circuit screenshot, a bright-state screenshot, a dark-state screenshot, and a measurement screenshot if probes or meters are used.

4.12.1 Full Circuit Screenshot

The full circuit screenshot should show the complete Proteus schematic. It must include the LDR, resistors, BC547 transistor, relay, flyback diode, LED, lamp, buzzer, sources, and ground connections.

This screenshot proves that the circuit was assembled as a complete system. The labels of the components should be visible so that the schematic can be compared with the project description and scheme chapter.

4.12.2 Bright-State Screenshot

The bright-state screenshot should show the circuit when light is applied to the LDR. In this screenshot, the lamp and buzzer should be off. The relay should appear inactive. If measurement probes are included, they should show a low sensing voltage and a low base-emitter voltage.

This screenshot visually supports the claim that the system does not activate during bright conditions.

4.12.3 Dark-State Screenshot

The dark-state screenshot should show the circuit when light is removed from the LDR. In this screenshot, the relay should be energized and the lamp and buzzer should be active. If measurement probes are included, they should show an increased sensing voltage and a base-emitter voltage near the transistor turn-on level.

This screenshot visually supports the claim that the system activates during darkness.

4.12.4 Measurement Screenshot

The measurement screenshot should show voltage probes, current probes, or virtual meters used in the simulation. The most useful measurements are V_{sense} , V_{BE} , relay coil voltage, relay coil current, LED current, lamp voltage, and buzzer voltage.

This screenshot strengthens the academic quality of the report because it provides numerical evidence instead of relying only on visible lamp and buzzer states.

4.13 Simulation Acceptance Criteria

The acceptance criteria define the conditions that must be satisfied for the simulation to be considered successful. Without acceptance criteria, the simulation remains descriptive but not formally verified.

The criteria must test each essential layer of the circuit: sensor response, transistor switching, relay operation, output activation, and protection connection.

4.13.1 Sensor Criterion

The LDR must change the circuit state according to light. The sensing voltage must be lower during bright light and higher during darkness.

The required condition is:

$$V_{sense\ dark} > V_{sense\ bright}$$

This criterion confirms that the sensor stage is functioning correctly.

4.13.2 Transistor Criterion

Q1 must remain off in the bright-light condition and must turn on in the dark condition. This criterion confirms that the LDR voltage divider successfully controls the transistor base.

The required conditions are:

$$V_{BE\ bright} < 0.6V\ to\ 0.7V$$

$$V_{BE\ dark} \approx 0.6V\ to\ 0.7V$$

The transistor criterion is satisfied only if the transistor state changes correctly when the LDR condition changes.

4.13.3 Relay Criterion

The relay must energize only when Q1 turns on. If the relay energizes in bright light, the circuit behavior is incorrect. If the relay does not energize in darkness, the circuit also fails its intended purpose.

The required conditions are:

$$Q_1 = OFF \Rightarrow Relay = OFF$$

$$Q_1 = ON \Rightarrow Relay = ON$$

This criterion confirms that the transistor stage successfully controls the relay coil.

4.13.4 Output Criterion

The lamp and buzzer must activate only when the relay contact closes. They must remain off during bright light and turn on during darkness.

The required conditions are:

$$Relay = OFF \Rightarrow Lamp = OFF \text{ and } Buzzer = OFF$$

$$Relay = ON \Rightarrow Lamp = ON \text{ and } Buzzer = ON$$

This criterion confirms that the circuit achieves the final project objective.

4.13.5 Protection Criterion

D1 must be correctly placed across the relay coil. The diode must be reverse-biased during normal relay operation and must conduct only when the relay coil is switched off and produces a reverse voltage spike.

The correct orientation is:

$$Cathode \text{ of } D1 \rightarrow \text{Positive side of relay coil}$$

$$Anode \text{ of } D1 \rightarrow \text{Transistor collector side of relay coil}$$

This criterion confirms that the transistor is protected from relay coil kickback during switching.

4.14 Simulation Limitations

Although simulation is useful, it does not fully replace real hardware testing. A simulation can confirm logical behavior and approximate electrical operation, but real components may behave differently because of tolerance, temperature, manufacturing variation, wiring resistance, contact resistance, supply instability, and load safety constraints.

Therefore, this section states the limits of the Proteus simulation and explains why physical verification is still required before real-world use.

4.14.1 Ideal Component Models

Proteus component models may behave more ideally than real components. A simulated transistor, relay, LDR, LED, lamp, or buzzer may not perfectly match the tolerance and non-linear behavior of real devices.

For this reason, successful simulation proves that the circuit principle is correct, but it does not automatically prove that every real component will operate with the same margin. Real hardware must still be tested with actual component values and measured currents.

4.14.2 Real LDR Variation

Actual LDR resistance can vary significantly between components. One LDR may have a different dark resistance and bright resistance compared with another LDR of the same general type. This means that the switching threshold in hardware may not exactly match the Proteus simulation.

The practical effect is that the circuit may turn on too early, too late, or inconsistently unless the threshold is adjusted. A potentiometer can improve the real design by allowing the trigger point to be calibrated.

4.14.3 Relay Current Difference

The relay coil current in real hardware may differ from the simulated relay model. The real relay coil resistance determines the actual current required for activation.

The relay current is calculated as:

$$I_{relay} = \frac{V_{relay}}{R_{coil}}$$

If the real relay requires more current than Q1 can safely provide, the transistor may heat, fail to saturate, or fail to energize the relay. Therefore, the relay coil current must be measured or obtained from the actual relay specification before hardware construction.

4.14.4 Hardware Safety Not Fully Proven

Simulation does not prove safe physical wiring for AC or high-voltage loads. If the output-side source represents a high-voltage supply, the physical circuit must include correct relay contact rating, insulation, spacing, enclosure, fuse protection, and safe wiring practice.

Proteus can show that the relay contact opens and closes, but it cannot prove that the physical construction is safe for human handling. Therefore, hardware implementation requires additional safety verification beyond simulation.

4.15 Chapter Summary

This chapter presented the simulation process for the automatic darkness-activated alarm and load-control circuit. The simulation was performed in Proteus to verify the behavior of the LDR sensing stage, BC547 switching stage, relay driving stage, protection diode stage, LED indicator stage, and lamp and buzzer output stage.

The chapter verified the circuit under two main conditions: bright light and darkness. In bright light, the LDR resistance is expected to decrease, causing a low sensing voltage, an off transistor, a de-energized relay, and inactive output devices. In darkness, the LDR resistance is expected to increase, causing a higher sensing voltage, an active transistor, an energized relay, and active lamp and buzzer output.

4.15.1 Simulation Result Summary

The simulation result is considered successful if the circuit remains off during bright light and turns on during darkness. The correct bright-state result is:

$$Lamp = OFF$$

$$Buzzer = OFF$$

The correct dark-state result is:

$$Lamp = ON$$

$$Buzzer = ON$$

This behavior confirms that the circuit performs as an automatic darkness-activated system.

4.15.2 Verification Summary

The successful operation of the circuit is verified by measuring and observing the following:

$$V_{sense\ bright} < V_{sense\ dark}$$

$$V_{BE\ bright} < 0.6V\ to\ 0.7V$$

$$V_{BE\ dark} \approx 0.6V\ to\ 0.7V$$

$$Relay_{bright} = OFF$$

$$Relay_{dark} = ON$$

$$Lamp_{bright} = OFF$$

$$Lamp_{dark} = ON$$

$$Buzzer_{bright} = OFF$$

$$Buzzer_{dark} = ON$$

These measurements and observations prove that the circuit changes state correctly when the light condition changes.

4.15.3 Limitation Summary

The Proteus simulation confirms the functional logic of the circuit, but it does not fully prove hardware reliability. Real hardware must still be checked for LDR resistance variation, relay coil current, transistor saturation, flyback diode rating, LED current, load current, relay contact rating, and electrical safety.

Therefore, the final conclusion of the simulation chapter is that the circuit is functionally correct in Proteus, while physical implementation requires additional component verification and safety validation.

5

Troubleshooting

The troubleshooting video is available at <https://youtu.be/4wLEyLrvDhk>.

5.1 Troubleshooting Overview

5.1.1 Purpose of Troubleshooting

Troubleshooting is an essential part of the project because a circuit is not considered complete merely because it has been drawn correctly in Proteus. A circuit must also be examined according to its real electrical behavior, its switching logic, its component ratings, its current paths, and its response under different operating conditions. The purpose of troubleshooting in this project is to identify, isolate, measure, and correct any fault that prevents the automatic darkness activated alarm and load control system from operating as intended.

In this project, troubleshooting focuses on the complete chain of operation. The LDR must correctly sense the light condition. The voltage divider must generate a suitable sensing voltage. The base resistor must transfer sufficient but safe current to the transistor base. The BC547 transistor must switch the relay coil reliably. The relay must change contact state when energized. The lamp and buzzer must receive power through the relay contact. The flyback diode must protect the transistor from inductive voltage stress. The LED indicator must operate with correct polarity and safe current. Therefore, troubleshooting is not limited to one component. It is a complete structural verification of the entire sensing, switching, protection, isolation, and output system.

The troubleshooting process also separates simulation success from engineering reliability. A circuit may appear to work in Proteus while still containing weak base drive, incorrect real component assumptions, insufficient relay current margin, reversed polarity, unstable threshold behavior, or unsafe load side design. For that reason, the troubleshooting chapter provides a systematic method for proving that every fault has a measurable cause and every correction has an electrical justification.

5.1.2 Troubleshooting Domains

The troubleshooting domains of this project are the sensor domain, transistor domain, relay domain, diode protection domain, LED indicator domain, output load domain, supply domain, and simulation setup domain.

The sensor domain includes the LDR and the resistor R1. This domain determines whether light and darkness are being converted into a useful voltage signal. If this domain fails, the transistor will receive an incorrect control signal.

The transistor domain includes R2 and Q1. This domain determines whether the weak sensing voltage is converted into a switching action. If this domain fails, the relay may not energize, or the transistor may heat because it is not operating in proper saturation.

The relay domain includes the relay coil and relay contacts. This domain determines whether the low voltage control side can switch the separate output load side. If this domain fails, the relay may not click, or the relay may click while the lamp and buzzer still remain inactive.

The diode protection domain includes D1. This domain determines whether the transistor is protected from the inductive voltage spike generated by the relay coil when the transistor turns off. If this domain fails, the circuit may work temporarily but may damage the transistor during repeated switching.

The LED indicator domain includes D2 and R3. This domain determines whether the visual indicator is correctly connected, correctly polarized, and safely current limited.

The output load domain includes the lamp and buzzer. This domain determines whether the relay contact is actually providing a complete power path for the final output devices.

The supply domain includes the control supply, the load side source, and the ground reference. If the supply domain is incorrect, every other part of the circuit may behave incorrectly.

The simulation setup domain includes Proteus component selection, component values, relay configuration, light source configuration for the LDR, and measurement placement. A wrong model or an incomplete virtual connection can create a fault that does not belong to the theoretical circuit itself.

5.1.3 Diagnostic Method

The diagnostic method must follow a strict order. The correct method is not to replace components randomly. The correct method is to measure the circuit from the input condition toward the final output response.

First, the light condition must be verified. The LDR must be tested in bright light and darkness. In bright light, the LDR resistance should decrease. In darkness, the LDR resistance should increase. This confirms that the sensing component is behaving correctly.

Second, the sensing voltage must be measured at the voltage divider output node. This voltage is represented as V_{sense} . For the given arrangement, where R1 is connected toward the positive supply and the LDR is connected toward ground, the sensing voltage is given by

$$V_{sense} = V_{CC} \frac{R_{LDR}}{R_1 + R_{LDR}}$$

This formula shows that V_{sense} decreases when R_{LDR} is low and increases when R_{LDR} is high. Therefore, the circuit should turn on in darkness, not in bright light.

Third, the base emitter voltage of Q1 must be measured. The transistor begins to conduct when

$$V_{BE} \approx 0.6V \text{ to } 0.7V$$

If V_{BE} remains below this range in darkness, the transistor will not switch correctly. If V_{BE} reaches this range but the relay does not energize, the fault is likely related to insufficient base current, transistor pinout, relay coil rating, or supply weakness.

Fourth, the base current must be checked. An approximate base current expression is

$$I_B \approx \frac{V_{CC} - V_{BE}}{R_1 + R_2}$$

Using $V_{CC} = 5V$, $V_{BE} = 0.7V$, $R_1 = 10k\Omega$, and $R_2 = 6.8k\Omega$, the approximate maximum base current is

$$I_B \approx \frac{5 - 0.7}{10000 + 6800}$$

$$I_B \approx 0.256mA$$

This value must be compared with the relay coil current requirement. A conservative switching design often uses

$$I_B \geq \frac{I_C}{10}$$

Since the collector current is approximately the relay coil current, the relay can be considered reliably driven only if the available base current is sufficient for transistor saturation.

Fifth, the relay coil voltage and current must be measured. The relay coil current is calculated as

$$I_{relay} = \frac{V_{relay}}{R_{coil}}$$

The relay must receive its rated voltage, and the transistor must be able to conduct the required coil current safely.

Sixth, the relay contact path must be checked. A relay coil can energize correctly while the lamp and buzzer remain off if the load is connected to the wrong contact terminal or if the load side power path is incomplete.

Seventh, the protection diode must be inspected. D1 must be placed directly across the relay coil. Its cathode must be connected to the positive relay coil side, and its anode must be connected to the transistor collector side. This orientation keeps the diode reverse biased during normal relay operation and allows it to conduct only when the coil releases stored magnetic energy.

Eighth, the output load must be verified. The voltage across the lamp and buzzer must be measured when the relay is energized. If the relay changes state but no voltage appears across the load, the problem is in the contact wiring or load side supply.

This diagnostic method proves each stage in order. It prevents confusion between sensor faults, switching faults, relay faults, protection faults, and output faults.

5.2 Relay Does Not Activate

5.2.1 Possible Cause. Low Base Voltage

The relay may fail to activate if the transistor base does not receive sufficient voltage from the LDR voltage divider. The relay coil depends on Q1, and Q1 depends on the base emitter junction reaching the required turn on voltage. If the voltage at the base of Q1 remains below approximately (0.6V) to (0.7V), the transistor will remain off or only weakly conduct. In that condition, the relay coil cannot receive enough current to energize.

The first measurement is V_{sense} , which is the voltage at the divider node between R1 and the LDR. In darkness, this voltage should rise because the LDR resistance increases. If V_{sense} does not rise in darkness, the problem may be an incorrect LDR model, wrong LDR connection, incorrect R1 value, missing supply, or faulty ground reference.

The second measurement is V_{BE} , which is the voltage between the base and emitter of Q1. Even if V_{sense} rises, the transistor will not switch if the base emitter voltage does not reach the required threshold. A low V_{BE} value means the transistor is not being driven correctly.

The expected relationship is

$$V_{sense} \uparrow \Rightarrow V_{BE} \uparrow \Rightarrow Q1 \text{ ON} \Rightarrow \text{Relay ON}$$

If this chain is broken at the first or second step, the relay will not activate.

5.2.2 Possible Cause. R2 Too Large

The resistor R2 limits the current entering the base of Q1. This is necessary because the transistor base must not be connected directly to the divider node without current limitation. However, if R2 is too large, the base current becomes too small. A small base current may allow the transistor to begin conducting, but it may not drive the transistor into strong saturation.

For reliable relay operation, Q1 must usually act as a saturated switch. In saturation, the collector emitter voltage is low, and the relay coil receives nearly the full supply voltage. If the transistor is not saturated, part of the supply voltage is lost across Q1, and the relay coil may receive insufficient voltage.

The approximate base current is

$$I_B \approx \frac{V_{CC} - V_{BE}}{R_1 + R_2}$$

With the given values, the approximate base current is

$$I_B \approx 0.256mA$$

If the relay coil current is high, this base current may be insufficient. The saturation condition should be checked using

$$I_B \geq \frac{I_C}{10}$$

If this condition is not satisfied, the relay may not activate reliably. The correction may require reducing R2, reducing the divider resistance values, using a Darlington

relay driver, using a transistor with higher gain and current capacity, using a logic level MOSFET, or using a ULN2003 driver.

5.2.3 Possible Cause. Wrong BC547 Pinout

The BC547 has three terminals: collector, base, and emitter. If these terminals are connected incorrectly, the transistor will not operate as intended. A wrong pin connection can prevent the relay from activating, cause abnormal current flow, or heat the transistor.

In this project, the correct structural connection is as follows. The emitter must be connected to ground. The collector must be connected to the lower side of the relay coil. The upper side of the relay coil must be connected to the positive control supply. The base must receive the sensing signal through R2.

If the emitter and collector are swapped, the transistor may still show weak behavior in simulation or in rare physical cases, but it will not provide reliable switching. The relay may fail to energize because the transistor cannot conduct the required coil current properly.

The diagnostic method is to inspect the Proteus symbol pin labels and compare them with the intended circuit function. If building the circuit physically, the datasheet pin order for the exact BC547 package must be checked. The physical pin order must never be assumed from another transistor type.

5.2.4 Possible Cause. Wrong Relay Coil Voltage

The relay may not activate if its coil voltage rating does not match the available control supply. A 5 V relay requires approximately 5 V across its coil to energize reliably. If the supply is lower than the relay rating, or if the transistor is dropping too much voltage because it is not saturated, the relay coil may not receive enough voltage.

The coil voltage must be measured directly across the relay coil when the transistor is supposed to be on. If the measured relay coil voltage is much lower than the rated value, the relay will not switch correctly. The cause may be weak base drive, supply voltage drop, incorrect relay model, incorrect wiring, or an overloaded control supply.

The coil current must also be considered. If the relay coil resistance is R_{coil} , then

$$I_{relay} = \frac{V_{relay}}{R_{coil}}$$

This current must remain within the safe collector current range of Q1. If the relay requires more current than the BC547 can safely provide, the correct solution is not to force the transistor harder without analysis. The correct solution is to choose a suitable driver stage or a relay with a compatible coil current.

5.2.5 Correction

The correction begins with measurement, not replacement. The first value to measure is V_{sense} in bright and dark conditions. The second value to measure is V_{BE} of Q1. The third value to measure is the voltage across the relay coil. The fourth value to verify is the transistor pinout. The fifth value to estimate or measure is relay coil current.

If V_{sense} does not change correctly, the LDR and R1 arrangement must be corrected. If V_{BE} is below the turn on threshold in darkness, the divider or R2 must be adjusted.

If V_{BE} is correct but the relay still does not activate, the relay coil voltage, transistor saturation, and pinout must be inspected. If the relay coil current is too high for the BC547, a stronger driver stage must be used.

The final corrected chain must be

$$\begin{aligned} \text{Darkness} &\Rightarrow V_{sense} \text{ rises} \Rightarrow V_{BE} \text{ reaches threshold} \\ &\Rightarrow I_B \text{ sufficient} \Rightarrow Q_1 \text{ saturates} \Rightarrow \text{Relay energizes} \end{aligned}$$

5.3 Lamp Does Not Turn On

5.3.1 Possible Cause. Wrong Relay Contact

The lamp may fail to turn on even when the relay coil activates if the lamp is connected to the wrong relay contact. A relay contains a coil side and a contact side. The coil side only creates magnetic motion. The contact side performs the actual switching of the external load. Therefore, relay activation does not automatically mean that the lamp has a complete current path.

The contact terminals must be identified correctly. The common terminal is the moving contact. The normally open terminal is open when the relay is not energized and closes when the relay is energized. The normally closed terminal is closed when the relay is not energized and opens when the relay is energized.

For this project, if the goal is to turn the lamp on only in darkness, the lamp should be connected through the normally open contact. When the relay energizes in darkness, the normally open contact closes and allows current to flow through the lamp.

If the lamp is accidentally connected to the normally closed contact, the lamp may operate in the opposite condition. If the lamp is connected to a terminal that does not form a complete path with the supply, it will not turn on at all.

5.3.2 Possible Cause. Missing Load Supply

The lamp requires a complete load side circuit. The relay contact does not generate power by itself. It only opens or closes an existing power path. If the output side source is missing, incorrectly connected, or not referenced correctly for the selected load model, the lamp will remain off even if the relay works perfectly.

The load side circuit must contain a source, the relay contact, the lamp, and a return path. The basic load current path should be traceable from one side of the load source, through the relay contact, through the lamp, and back to the other side of the source. If any part of this path is open, the lamp will not glow.

The diagnostic test is to measure voltage across the lamp when the relay is energized. If the relay is on but the voltage across the lamp is zero, the fault is in the relay contact path or the load supply. If voltage is present but the lamp does not glow, the lamp model, voltage rating, or connection may be incorrect.

5.3.3 Possible Cause. Lamp Model Issue

In Proteus, some lamp models require specific voltage and current conditions. If the lamp model is not compatible with the selected source, it may not produce visible light

even though the circuit is electrically connected. A lamp selected for one voltage level may not behave correctly when connected to another voltage level.

The lamp voltage rating must therefore match the source used on the output side. If the source is a sine source, the voltage amplitude and frequency must be appropriate for the lamp model. If the lamp expects direct current but receives an unsuitable alternating signal, or if the voltage is too low, the simulation result may not show the expected output.

The lamp should be tested independently if necessary. Connecting the lamp directly to its intended source in a separate simple test circuit can confirm whether the lamp model works before reconnecting it to the relay contact path.

5.3.4 Correction

The correction process for a lamp that does not turn on begins by proving whether the relay contact is actually closing. The relay common terminal and normally open terminal must be checked. The lamp must be connected in the load path that becomes complete only when the relay energizes.

The load source must then be verified. The voltage across the lamp must be measured when the relay is activated. If the voltage is absent, the relay contact wiring or source connection must be corrected. If voltage is present but the lamp remains off, the lamp model and rating must be checked.

The final correct operation is

$$\begin{aligned} \text{Relay energized} &\Rightarrow \text{Contact closed} \Rightarrow \text{Lamp voltage present} \\ &\Rightarrow \text{Lamp ON} \end{aligned}$$

5.4 Buzzer Does Not Sound

5.4.1 Possible Cause. Wrong Buzzer Polarity

Some buzzers are polarity sensitive, especially active direct current buzzers. If the positive and negative terminals are reversed, the buzzer may not sound. In simulation, polarity sensitivity depends on the selected buzzer model, but in physical hardware it is a common cause of failure.

The buzzer terminals must be checked carefully. The positive terminal should be connected to the positive side of the load path, and the negative terminal should return to the negative side or ground of the load source, depending on the circuit arrangement. If the buzzer is placed on an alternating current load side, the chosen model must support that type of supply.

A polarity fault is diagnosed by measuring the voltage across BUZ1 and checking its sign and magnitude. If a voltage is present but the buzzer does not sound, polarity or model compatibility should be suspected.

5.4.2 Possible Cause. Wrong Buzzer Voltage

A buzzer operates only within its specified voltage range. If the applied voltage is too low, the buzzer may remain silent. If the applied voltage is too high, the buzzer may be damaged in real hardware or behave incorrectly in simulation.

The voltage across BUZ1 must be measured when the relay is energized. This voltage must match the required buzzer operating voltage. If the source voltage is not suitable, the source must be changed or a buzzer with a matching voltage rating must be selected.

The report should clearly state the buzzer voltage rating. A missing buzzer rating weakens the engineering proof because the output function cannot be fully verified without confirming that the load is compatible with the source.

5.4.3 Possible Cause. Open Load Path

The buzzer may not sound because the relay contact does not close the circuit path that includes the buzzer. This can occur even if the relay coil is energized and even if the lamp works, depending on how the buzzer is connected.

The load path must be traced from the source, through the relay contact, through the buzzer, and back to the source. If the buzzer is connected in parallel with the lamp, it must have the same switched supply available across its terminals. If it is connected in series with another load, the total voltage distribution and current path must be checked.

An open load path is diagnosed by measuring voltage directly across BUZ1. If there is no voltage when the relay is energized, the buzzer is not actually connected into a complete switched circuit.

5.4.4 Correction

The correction begins by measuring the voltage across BUZ1 while the relay is energized. If the voltage is missing, the relay contact path and wiring must be corrected. If voltage is present but the buzzer does not sound, the buzzer polarity and voltage rating must be checked.

The proper result should be

$$\text{Relay contact closed} \Rightarrow \text{Voltage across BUZ1} \Rightarrow \text{Buzzer ON}$$

If the lamp turns on but the buzzer does not, the fault is local to the buzzer branch. If both lamp and buzzer fail, the fault is probably in the relay contact path or the load supply.

5.5 LED Does Not Glow

5.5.1 Possible Cause. LED Reversed

The LED D2 may fail to glow if it is connected in reverse polarity. An LED conducts current primarily in the forward direction. If the anode and cathode are reversed, the LED blocks current and remains off.

The correct polarity depends on how the indicator path is intended to operate. If D2 is meant to indicate relay activation, it should be connected so that current flows through R3 and the LED only when the relay control path is active. The anode should be at a higher potential than the cathode during the intended indicator state.

The diagnostic method is to measure the voltage across the LED and inspect the symbol orientation. If the LED is reverse biased during the intended on state, its connection must be corrected.

5.5.2 Possible Cause. R3 Too Large

R3 limits the LED current. This is necessary because an LED cannot be safely connected directly across a supply without current limitation. However, if R3 is too large, the current may be too small for visible brightness.

The LED current is calculated using

$$I_{LED} = \frac{V_{CC} - V_F}{R_3}$$

For a 5 V supply and an assumed green LED forward voltage of approximately 2 V, with R3 equal to 330 ohms, the current is

$$I_{LED} = \frac{5 - 2}{330}$$

$$I_{LED} \approx 9.09mA$$

This value is normally sufficient for an indicator LED. Therefore, if the LED does not glow under a 5 V supply with R3 equal to 330 ohms, the more likely causes are reversed polarity, wrong node connection, missing current path, or inactive indicator condition.

If the supply is different from 5 V, R3 must be recalculated.

5.5.3 Possible Cause. Wrong Indicator Node

The LED may be connected to a node that does not change state or that does not provide a complete current path. In that case, the LED may remain off even when the main circuit operates correctly.

The indicator meaning must be defined precisely. D2 may be intended to indicate supply presence, relay activation, transistor conduction, or output load activation. These are not identical meanings. If the LED is connected to the supply rail, it may indicate only that power is present. If it is connected with the relay coil path, it may indicate relay activation. If it is connected with the output load, it may indicate load activation.

A wrong indicator node creates ambiguity in the report. The report must state exactly what D2 indicates and must connect it accordingly.

5.5.4 Correction

The correction is to check LED polarity, current path, and resistor value. The LED current must be calculated, and the voltage across D2 must be measured during the intended indicator state.

A correct LED indicator path requires

$$\begin{aligned} &\text{Supply potential} \Rightarrow \text{Current limiting resistor} \\ &\Rightarrow \text{Forward biased LED} \Rightarrow \text{Return path} \end{aligned}$$

If the LED is intended to show relay activation, it must be connected so that it receives current only when the relay stage is active. If it is intended to show supply presence, it must be connected across the supply with a suitable current limiting resistor.

5.6 Transistor Gets Hot

5.6.1 Possible Cause. Relay Current Too High

The transistor Q1 may heat if the relay coil requires more current than the BC547 can safely conduct. The BC547 is a small signal NPN transistor and is not intended for high current loads. If the relay coil current is too high, the transistor may overheat, become unreliable, or fail.

The relay coil current must be calculated or measured. If the relay coil resistance is known, the current is

$$I_{relay} = \frac{V_{relay}}{R_{coil}}$$

This current is approximately the collector current of Q1. Therefore,

$$I_C \approx I_{relay}$$

The transistor power dissipation can be estimated as

$$P_Q = V_{CE}I_C$$

If Q1 is saturated, V_{CE} is low. If Q1 is not saturated, V_{CE} can become significantly higher, and the transistor power dissipation increases. Heating is therefore often a sign that Q1 is carrying too much current or is not being driven into saturation.

5.6.2 Possible Cause. Weak Saturation

Weak saturation means that Q1 is not fully acting as a closed switch. Instead, it operates partly in the active region. In this condition, the relay coil may still receive some current, but the transistor drops too much voltage and dissipates excessive power.

The saturation condition should be checked using

$$I_B \geq \frac{I_C}{10}$$

If the relay coil current is 30 mA, a conservative base current target is

$$I_B \geq 3mA$$

However, the estimated base current from the present divider and R2 values may be approximately

$$I_B \approx 0.256mA$$

This shows why weak saturation is a serious possible issue. The transistor may be commanded to turn on, but it may not receive enough base current to drive the relay

coil with a strong switching margin.

5.6.3 Possible Cause. Wrong Pin Connection

A wrong connection of collector, base, and emitter can cause abnormal transistor behavior and heating. If the emitter and collector are interchanged, the transistor may not switch properly. If the base is connected incorrectly, uncontrolled or insufficient current may flow.

The transistor should be inspected according to the intended low side switch configuration. The emitter must be connected to ground. The collector must connect to the relay coil. The base must receive the control signal through R2. The relay coil must connect from positive supply to the collector.

In physical hardware, the actual package pinout must be checked from the datasheet of the exact component used.

5.6.4 Possible Cause. Missing Flyback Diode

If D1 is missing, reversed, or not connected directly across the relay coil, the relay coil can generate high voltage spikes when Q1 turns off. These spikes can stress the transistor junctions. The transistor may not heat immediately from a single event, but repeated switching can weaken or destroy it.

The flyback diode gives the relay coil current a safe decay path after the transistor turns off. Its role is protective and necessary. The correct orientation is cathode to the positive side of the coil and anode to the transistor collector side.

5.6.5 Correction

The correction begins by measuring relay coil current and base current. If relay current is too high for the BC547, a more suitable driver must be used. If base current is too low, the base drive must be redesigned. If Q1 pinout is wrong, it must be corrected. If D1 is absent or reversed, it must be installed correctly.

A reliable correction may use a MOSFET or ULN2003 relay driver. This removes the weak base drive problem and increases relay driving reliability.

The final corrected switching condition should be

$$\begin{aligned} \text{Sufficient base drive} &\Rightarrow Q_1 \text{ saturation} \\ &\Rightarrow \text{Low transistor heating} \Rightarrow \text{Reliable relay operation} \end{aligned}$$

5.7 Relay Chattering

5.7.1 Possible Cause. No Hysteresis

Relay chattering occurs when the relay repeatedly turns on and off near the switching threshold. This can happen because the circuit has only one approximate threshold and no hysteresis. When the light level is close to the boundary between bright and dark, small changes in the LDR voltage may repeatedly push the transistor above and below the switching condition.

Without hysteresis, the turn on and turn off points are nearly the same. A better switching circuit uses two thresholds. One threshold turns the relay on, and a different

threshold turns it off. This creates stable behavior and prevents repeated switching near the boundary.

The absence of hysteresis does not make the circuit concept wrong. It means the basic design is sensitive near the threshold and should be improved for real reliability.

5.7.2 Possible Cause. Unstable Light Level

The LDR voltage can fluctuate if the light source is unstable, if the sensor receives partial shadow, or if ambient light changes slowly. Since the LDR is an analog sensor, its resistance does not jump instantly from one fixed value to another. It changes continuously with illumination.

When the LDR voltage is close to the transistor turn on condition, even small variations can change the relay state. The relay may click repeatedly, and the lamp or buzzer may flicker or pulse.

The sensing voltage must be observed during slowly changing light conditions. If V_{sense} moves around the transistor threshold, relay chatter is expected.

5.7.3 Possible Cause. Weak Base Drive

Relay chattering can also occur if the transistor barely provides enough current to the relay coil. In that case, the relay may begin to energize but fail to hold strongly. A small change in base current or supply voltage can release the relay again.

This fault is different from pure threshold instability. In weak base drive, the problem is not only the LDR signal. The relay coil current itself is marginal because Q1 is not saturated strongly enough.

The diagnostic method is to measure relay coil voltage and collector emitter voltage while the relay is supposed to be on. If the relay coil voltage is low or unstable, the driver stage is not strong enough.

5.7.4 Correction

The best correction is to add hysteresis using a comparator circuit. A comparator with positive feedback creates a stable turn on threshold and a separate stable turn off threshold. This prevents relay chatter near the light boundary.

A second correction is to improve base drive. This can be done by redesigning resistor values, using a stronger transistor stage, using a MOSFET, or using a ULN2003 relay driver.

A third correction is to add threshold adjustment with a potentiometer. This allows the switching point to be calibrated for the actual LDR and actual environment.

The strongest corrected structure is

LDR divider $\Rightarrow$ *Adjustable threshold*
 $\Rightarrow$ *Comparator with hysteresis* $\Rightarrow$ *Reliable relay driver*
 $\Rightarrow$ *Relay*

5.8 Circuit Works in Proteus but Fails in Hardware

5.8.1 Possible Cause. Real LDR Values Differ

Proteus models provide controlled behavior, but real LDRs can vary widely. Two LDRs with the same general type may have different resistance values under the same light condition. Their response can also depend on angle, light spectrum, aging, temperature, and manufacturing tolerance.

If the real LDR resistance in darkness is lower than expected, the sensing voltage may not rise enough to turn on Q1. If the real LDR resistance in light is higher than expected, the circuit may activate too early or fail to turn off.

The correction is to measure the real LDR resistance in bright light and darkness using a multimeter. Then the voltage divider must be recalculated using the measured resistance range.

5.8.2 Possible Cause. Real Relay Coil Current Is Higher

A real relay coil may require more current than the relay model used in Proteus. If the coil current is higher than expected, the BC547 may not drive it reliably. The simulation may show switching because the virtual relay model is ideal or because the current requirement is not equivalent to the physical relay.

The relay coil resistance must be measured physically. The coil current should then be calculated as

$$I_{relay} = \frac{V_{relay}}{R_{coil}}$$

The calculated relay current must be compared with the transistor current capability and available base current. If the required relay current is too high, the driver must be upgraded.

5.8.3 Possible Cause. BC547 Pinout Difference

In hardware, transistor pinout is a frequent source of failure. The symbol in Proteus may be clear, but the physical transistor package must be connected according to the exact datasheet. Different transistor families can have different pin orders, and assuming a pinout from memory can cause a complete circuit failure.

The correct hardware verification method is to identify the flat face of the transistor package, locate the pins according to the datasheet, and confirm collector, base, and emitter connections before applying power.

5.8.4 Possible Cause. Supply Voltage Drop

The relay coil draws current when energized. If the supply cannot provide enough current, the supply voltage may drop. A supply voltage drop can reduce V_{sense} , reduce base drive, weaken relay coil voltage, dim the LED, and prevent stable operation.

Supply voltage must be measured while the relay is energized, not only when the circuit is idle. A supply that appears correct without load may collapse under relay load. This is especially important when using batteries, weak adapters, or breadboard connections with high resistance.

5.8.5 Correction

The correction is to replace assumptions with measured values. The real LDR resistance must be measured. The relay coil resistance must be measured. The transistor pinout must be confirmed. The supply voltage must be measured under load. The base current and relay current must be recalculated.

If the real circuit is marginal, the design must be upgraded with proper margins. A potentiometer can calibrate the threshold. A comparator can sharpen the decision. Hysteresis can prevent chatter. A MOSFET or ULN2003 can provide reliable relay drive. A stronger flyback diode can improve protection.

The real hardware must satisfy

$$\begin{aligned} \text{Measured sensor range} &\Rightarrow \text{Correct threshold} \\ &\Rightarrow \text{Sufficient driver current} \Rightarrow \text{Stable relay operation} \\ &\Rightarrow \text{Safe output switching} \end{aligned}$$

5.9 Flyback Diode Problems

5.9.1 Possible Cause. Reversed Diode

A reversed flyback diode can create a serious fault. If D1 is connected in the wrong direction across the relay coil, it may conduct during normal operation and effectively short the supply through the relay path. This can prevent the relay from working, overload the supply, heat the diode, damage the transistor, or disturb the entire control circuit.

The flyback diode must be reverse biased while the relay is energized. This means the cathode must connect to the positive side of the relay coil, and the anode must connect to the transistor collector side.

The correct normal state is

$$\text{Relay ON} \Rightarrow \text{D1 reverse biased}$$

The correct turn off state is

$$\text{Relay OFF transition} \Rightarrow \text{D1 conducts coil current}$$

5.9.2 Possible Cause. Diode Too Weak

The diode must be able to handle the relay coil current during flyback. A 1N4148 can be suitable for small relay coils, but it is not automatically suitable for every relay. If the coil current is high, a stronger diode should be selected.

The diode must have adequate forward current capability and reverse voltage capability. The required diode strength depends on the relay coil current and switching frequency. If the relay is switched frequently, the protection diode experiences repeated energy events.

For robust hardware, a 1N400x type diode is often preferred for relay coil protection when switching speed is not critical.

5.9.3 Possible Cause. Diode Not Across Coil

The flyback diode must be connected directly across the relay coil. If it is placed elsewhere in the circuit, it may not clamp the coil voltage effectively. A diode that is near the relay in the diagram but not electrically across the coil does not provide the intended protection.

The diagnostic method is to trace both terminals of the relay coil and ensure D1 is connected across those same two terminals. The diode must not be placed across the relay contact. It protects the coil side, not the load contact side.

5.9.4 Correction

The correction is to place D1 directly across the relay coil with correct polarity. The cathode must be on the positive coil side. The anode must be on the transistor collector side. The diode current rating must match the relay coil requirements.

After correction, the transistor should be protected from the inductive voltage spike when the relay turns off.

The protection chain becomes

$$\begin{aligned} \text{Relay coil releases energy} &\Rightarrow D1 \text{ conducts} \\ &\Rightarrow \text{Voltage spike clamped} \Rightarrow Q_1 \text{ protected} \end{aligned}$$

5.10 Wrong Trigger Behavior

5.10.1 Problem

The circuit may turn on in bright light instead of darkness, or it may turn off in darkness instead of turning on. This is wrong trigger behavior. It means the circuit logic is inverted relative to the intended project function.

The intended behavior of this project is darkness activation. Therefore, the lamp and buzzer should remain off in bright light and turn on in darkness.

The required behavior is

$$\text{Bright light} \Rightarrow \text{Output OFF}$$

$$\text{Darkness} \Rightarrow \text{Output ON}$$

If the opposite occurs, the sensing divider arrangement must be inspected.

5.10.2 Possible Cause

The most likely cause is that R1 and the LDR have been placed in the opposite positions in the voltage divider. In a voltage divider, the position of the light dependent resistor determines whether the sensing voltage rises or falls when light changes.

In the intended darkness activated arrangement, the LDR is connected toward ground and R1 is connected toward the positive supply. The sensing voltage is

$$V_{sense} = V_{CC} \frac{R_{LDR}}{R_1 + R_{LDR}}$$

In this arrangement, darkness increases R_{LDR} , and this increases V_{sense} . The transistor turns on in darkness.

If R1 and the LDR are swapped, the sensing voltage becomes high in bright light and low in darkness. That creates light activated behavior instead of darkness activated behavior.

5.10.3 Correction

The correction is to place R1 and the LDR according to the desired trigger mode.

For darkness activated behavior, R1 should be connected toward the positive supply, and the LDR should be connected toward ground. The sensing node should be taken from the junction between R1 and the LDR.

For light activated behavior, the LDR and R1 positions can be swapped.

The report must explicitly define the intended trigger mode. Since this project is a darkness activated alarm and load control system, the correct arrangement is the one that raises the transistor base voltage in darkness.

5.10.4 Verification

Verification requires testing both states. In bright light, V_{sense} must be low, Q1 must be off, the relay must be off, and the lamp and buzzer must be off. In darkness, V_{sense} must rise, Q1 must turn on, the relay must energize, and the lamp and buzzer must turn on.

The verification result must satisfy

$$\begin{aligned} \text{Bright light} &\Rightarrow V_{sense} \text{ low} \Rightarrow Q_1 \text{ OFF} \Rightarrow \text{Relay OFF} \\ &\Rightarrow \text{Output OFF} \end{aligned}$$

$$\begin{aligned} \text{Darkness} &\Rightarrow V_{sense} \text{ high} \Rightarrow Q_1 \text{ ON} \Rightarrow \text{Relay ON} \\ &\Rightarrow \text{Output ON} \end{aligned}$$

5.11 Alternating Current Load Side Safety Problems

5.11.1 Possible Hazard

If the load side source represents a high voltage alternating current supply, the circuit must be treated as potentially dangerous. Simulation does not prove physical safety. Proteus can show a relay switching a lamp and buzzer, but real hardware requires insulation, correct wire spacing, proper relay contact rating, fuse protection, and safe enclosure.

The control side may be low voltage, but the relay contact side may carry a higher voltage. This separation is useful, but it is safe only if the relay rating, wiring practice, and physical construction are correct.

The main hazard is electric shock, overheating, fire, contact arcing, or insulation failure. Therefore, load side safety must be addressed in the report even if the project remains a simulation.

5.11.2 Relay Rating Requirement

The relay contact must be rated for the voltage and current of the load. The relay coil rating and relay contact rating are different. A relay may have a 5 V coil but still have contacts rated for a particular maximum voltage and current. These two ratings must not be confused.

The lamp and buzzer current must not exceed the relay contact current rating. If the lamp has a high inrush current, the relay must be chosen with enough margin. A relay that survives steady current may still be damaged by repeated inrush current if the load is not properly considered.

The safe condition is

$$I_{load} \leq I_{contact \text{ rating}}$$

and

$$V_{load} \leq V_{contact \text{ rating}}$$

If either condition is violated, the relay is not suitable.

5.11.3 Fuse Requirement

A fuse should be included in real hardware when the load side can deliver dangerous current. The fuse protects the circuit by opening the current path during a fault. Without a fuse, a short circuit may cause overheating, wire damage, component failure, or fire risk.

The fuse must be selected according to the expected load current and safety requirements. It should be placed in series with the load supply path so that a fault current is interrupted before it damages the circuit.

A fuse is not required to prove the basic Proteus logic, but it is required for responsible real hardware discussion when high voltage or high current loads are involved.

5.11.4 Insulation Requirement

High voltage load wiring must be insulated and physically separated from the low voltage control side. The relay provides electrical isolation between coil and contacts, but the physical layout must also maintain separation. Exposed conductors must not be accessible. The circuit must be placed inside a proper enclosure if used with dangerous voltage.

Insulation, spacing, enclosure, strain relief, and secure terminals are part of real circuit safety. A successful simulation does not replace these requirements.

For an academic report, the correct statement is that the Proteus simulation verifies switching behavior, but it does not certify physical safety for high voltage construction.

5.12 Troubleshooting Table

5.12.1 Fault Column

The fault column records the visible or measurable problem observed in the circuit. Examples include relay not activating, lamp not turning on, buzzer silent, LED not

glowing, transistor heating, relay chattering, wrong trigger behavior, or simulation working while physical hardware fails.

The fault must be written as an observable fact. It should not be written as a guessed cause. For example, the correct fault statement is “relay does not activate.” The cause may later be identified as low base current, wrong transistor pinout, or wrong relay voltage.

5.12.2 Possible Cause Column

The possible cause column lists the most likely technical reasons for the observed fault. Each possible cause must belong to a specific circuit domain. A relay fault may belong to the base drive domain, transistor domain, relay coil domain, or supply domain. A lamp fault may belong to the relay contact domain or load side supply domain.

This column prevents uncontrolled guessing by forcing each cause to be tied to a circuit location.

5.12.3 Test Point Column

The test point column identifies exactly where voltage or current should be measured. This is the most important part of the troubleshooting table because measurement converts opinion into evidence.

Important test points include the LDR divider output, Q1 base emitter voltage, Q1 collector voltage, relay coil voltage, relay coil current, voltage across the lamp, voltage across the buzzer, LED current, supply voltage, and load side source voltage.

5.12.4 Correction Column

The correction column states the exact action needed after diagnosis. Corrections may include reversing LED polarity, correcting transistor pinout, reducing R2, selecting a suitable relay, connecting the lamp to the normally open contact, adding a flyback diode, using a stronger diode, improving the supply, or adding comparator hysteresis.

A correct troubleshooting table does not merely list faults. It creates a complete diagnostic chain:

$$\begin{aligned} \text{Observed fault} &\Rightarrow \text{Possible cause} \Rightarrow \text{Measurement point} \\ &\Rightarrow \text{Confirmed cause} \Rightarrow \text{Correction} \end{aligned}$$

5.13 Recommended Improvements

5.13.1 Add Potentiometer

A potentiometer should be added to allow adjustment of the light and dark threshold. The fixed resistor R1 creates a fixed switching point, but real environments have different light levels. A fixed threshold may work in one condition and fail in another.

With a potentiometer, the user can calibrate the circuit so that the relay activates at the desired darkness level. This improves practical usability and compensates for LDR tolerance.

The improvement changes the sensor stage from a fixed divider to an adjustable threshold network.

5.13.2 Add Comparator

A comparator improves switching accuracy by comparing the LDR voltage with a reference voltage. In the present circuit, the transistor base receives the analog divider voltage directly. This can work, but it does not create a sharp and controlled switching threshold.

A comparator produces a cleaner output transition. When the LDR voltage crosses the reference threshold, the comparator output changes state more decisively than a simple transistor base input.

The improved structure is

$$\text{LDR voltage} \Rightarrow \text{Comparator comparison} \Rightarrow \text{Clean switching signal}$$

This makes the design more stable and more academically complete.

5.13.3 Add Hysteresis

Hysteresis prevents relay chatter by creating two different switching thresholds. One threshold turns the output on. A second threshold turns the output off. This difference creates a stable region and prevents repeated switching when the light level is near the boundary.

Without hysteresis, the circuit may switch repeatedly when V_{sense} is close to V_{BE} . With hysteresis, small noise or small light fluctuation does not immediately reverse the output state.

The corrected switching behavior becomes

$$\text{Darkness crosses upper threshold} \Rightarrow \text{Output ON}$$

$$\text{Light crosses lower threshold} \Rightarrow \text{Output OFF}$$

This is a major reliability improvement.

5.13.4 Use MOSFET or ULN2003

A MOSFET or ULN2003 relay driver improves relay driving reliability. The present BC547 stage may be limited by available base current. A MOSFET requires very little steady gate current and can switch relay current more efficiently if selected correctly. A ULN2003 provides built in transistor driver channels and is commonly used for relay driving applications.

This improvement solves weak base drive, improves switching margin, and reduces the probability of transistor overheating. It is especially useful if the relay coil current is higher than the BC547 can safely handle.

The stronger driver structure is

$$\text{LDR decision stage} \Rightarrow \text{Reliable driver} \Rightarrow \text{Relay coil}$$

5.13.5 Use Stronger Flyback Diode

The flyback diode should be selected according to relay coil current. A 1N4148 may be acceptable for small relay coils, but a stronger diode such as a 1N400x type is often more robust for relay coil protection.

A stronger diode improves reliability by giving the relay coil a safer path for stored magnetic energy during switch off. It also reduces stress on the transistor.

The protection improvement is

Relay coil energy $\Rightarrow$ Stronger flyback path $\Rightarrow$ Reduced transistor stress

5.14 Chapter Summary

5.14.1 Main Faults

The main faults identified in this chapter are weak base drive, relay wiring mistakes, wrong relay contact selection, LED polarity errors, buzzer polarity errors, diode polarity errors, unstable threshold behavior, transistor heating, relay chatter, and differences between Proteus simulation and real hardware behavior.

The most important electrical weakness is the possibility that the BC547 does not receive enough base current to saturate strongly for the selected relay coil. This weakness must be checked using base current calculation, relay coil current measurement, and transistor saturation verification.

The most important structural weakness is the absence of hysteresis. Without hysteresis, the relay can chatter near the threshold.

The most important safety weakness is the load side. If the output side uses dangerous voltage, simulation alone is not enough. Relay rating, fuse protection, insulation, and enclosure must be considered.

5.14.2 Main Corrections

The main corrections are measurement, verification, and margin improvement. The sensing voltage must be measured. The transistor base emitter voltage must be measured. The relay coil voltage and current must be measured. The load voltage must be measured. The LED current must be calculated. The diode polarity must be verified.

If the relay driver is weak, the circuit should use a stronger driver such as a MOSFET or ULN2003. If the switching point is unstable, the circuit should use a comparator with hysteresis. If the threshold is not suitable for the environment, a potentiometer should be added. If the relay coil protection is weak, a stronger flyback diode should be used. If the load side is high voltage, proper safety protections must be added.

5.14.3 Main Engineering Lesson

The main engineering lesson is that a circuit must work logically, electrically, and safely. Logical correctness means the circuit has the right sequence of operation. Electrical correctness means voltage, current, resistance, power, polarity, and component ratings are within safe and functional limits. Safety correctness means the circuit can

be handled and constructed without exposing the user or hardware to unacceptable risk.

For this project, the correct final understanding is that the design is structurally valid as a Proteus based darkness activated relay alarm system. However, its reliability depends on measured electrical margins. The circuit becomes academically strong when the troubleshooting process proves every stage:

*Light condition $\Rightarrow$ LDR resistance $\Rightarrow$ Divider voltage
 $\Rightarrow$ Base drive $\Rightarrow$ Transistor switching
 $\Rightarrow$ Relay activation $\Rightarrow$ Load operation
 $\Rightarrow$ Protection verification*

Therefore, the troubleshooting chapter is not an optional appendix. It is the chapter that proves the project can be defended beyond a visual schematic. It transforms the project from a drawn circuit into an analyzed engineering system.

6

Questions

6.1 Basic Concept Questions

This section presents the foundational conceptual questions required to understand the project. These questions establish whether the basic sensing principle, light response, and activation logic of the circuit are correctly understood. Since the project is based on an LDR-controlled switching system, the basic concept section must begin with the physical behavior of the LDR and then connect that behavior to the total circuit operation.

6.1.1 What is an LDR?

An LDR is a light-dependent resistor. It is an electronic component whose resistance changes according to the intensity of light falling on its surface. When the light intensity changes, the resistance of the LDR also changes, and this resistance variation can be used to create a changing voltage in a circuit.

In this project, the LDR is used as the main sensing component. Its function is to detect whether the surrounding condition is bright or dark. The LDR does not directly operate the relay, lamp, or buzzer. Instead, it produces a resistance change that is converted into a voltage signal by the voltage divider formed with R1. This voltage signal is then used to control the base of the BC547 transistor.

Therefore, the LDR acts as the environmental input device of the project. It converts the physical condition of light intensity into an electrical condition that can be processed by the circuit.

6.1.2 What happens to LDR resistance in bright light?

In bright light, the resistance of the LDR decreases. This means that the LDR provides a lower-resistance path when more light falls on it. In the given circuit, the LDR is connected toward ground in the voltage divider. Because of this arrangement, when the LDR resistance becomes low, the sensing node voltage also becomes low.

This low sensing voltage is not sufficient to turn on the base-emitter junction of the BC547 transistor. As a result, Q1 remains in the off state. Since Q1 is off, current cannot properly pass through the relay coil, the relay remains de-energized, and the lamp and buzzer remain inactive.

Thus, bright light causes the LDR resistance to decrease, and this keeps the output of the project turned off.

6.1.3 What happens to LDR resistance in darkness?

In darkness, the resistance of the LDR increases. This means that the LDR becomes less conductive when light intensity is reduced. In the given voltage divider arrangement, this increase in LDR resistance causes the sensing node voltage to rise.

When the sensing voltage becomes high enough, current flows through R2 into the base of Q1. If the base-emitter voltage reaches the required turn-on level, the BC547 transistor begins to conduct. Once Q1 conducts sufficiently, it allows current to flow through the relay coil. The relay then energizes and changes the state of its contact.

As a result, the load circuit is completed, and the lamp and buzzer become active. Therefore, darkness increases the LDR resistance, raises the transistor control voltage, energizes the relay, and activates the output devices.

6.1.4 Is this circuit light-activated or dark-activated?

This circuit is dark-activated. It activates its output when the light intensity on the LDR decreases. The reason is the position of the LDR in the voltage divider. Since the LDR is connected toward ground, bright light lowers the sensing voltage, while darkness raises the sensing voltage.

In bright light, the transistor remains off, so the relay is not energized. In darkness, the transistor turns on, the relay energizes, and the lamp and buzzer operate. Therefore, the circuit is correctly classified as a darkness-activated automatic alarm and load-control circuit.

6.2 Component Function Questions

This section explains the function of each major component in the circuit. The purpose of this section is to prove that every component is not randomly placed, but has a specific electrical role. A correct academic explanation must show the relationship between each component and the total function of the project.

6.2.1 What is the function of R1?

R1 forms a voltage divider with the LDR. Its main function is to help convert the resistance variation of the LDR into a voltage variation at the sensing node. Without R1, the LDR would not create a controlled voltage signal suitable for driving the transistor base.

In this circuit, R1 is connected above the sensing node, while the LDR is connected toward ground. This arrangement makes the circuit dark-activated. When the LDR resistance increases in darkness, the sensing voltage rises. When the LDR resistance decreases in bright light, the sensing voltage falls.

Therefore, R1 is not only a resistor. It is the fixed reference element that allows the LDR to produce a usable voltage signal.

6.2.2 What is the function of R2?

R2 limits the current entering the base of Q1. The base-emitter junction of a transistor behaves like a diode and must not be connected directly to a voltage source without current limitation. R2 prevents excessive base current and protects the transistor from damage.

R2 also acts as the electrical boundary between the sensor voltage divider and the transistor switching stage. It allows the sensing voltage to influence the transistor while preventing uncontrolled current flow.

In this project, R2 is essential for safe transistor operation. However, its value must be chosen carefully. If R2 is too large, the base current may become too weak to saturate the transistor strongly. If R2 is too small, excessive base current may flow. Therefore, R2 must provide both protection and sufficient drive.

6.2.3 What is the function of R3?

R3 limits the current flowing through the green LED D2. An LED must not be connected directly across a voltage source because it can draw excessive current and become damaged. R3 controls the LED current and keeps it within a safe operating range.

The current through the LED can be estimated by the formula

$$I_{LED} = \frac{V_{CC} - V_F}{R_3}$$

where V_{CC} is the supply voltage, V_F is the forward voltage of the LED, and R_3 is the current-limiting resistor.

In this project, R3 is a protection component for the indicator LED. It makes the LED usable as a visual status indicator without allowing unsafe current flow.

6.2.4 What is the function of Q1?

Q1 is the BC547 NPN transistor used as a switching device. Its function is to control the relay coil current based on the signal received from the LDR voltage divider through R2.

The LDR voltage divider cannot directly drive the relay coil because the relay requires more current than the sensor stage can reliably provide. Q1 solves this problem by acting as an electronic switch. When the base of Q1 receives sufficient voltage and current, Q1 conducts between collector and emitter. This allows current to flow through the relay coil and energize the relay.

Therefore, Q1 is the main switching bridge between the low-power sensing stage and the relay-driving stage.

6.2.5 What is the function of D1?

D1 protects Q1 from the reverse voltage spike produced by the relay coil. A relay coil is an inductive component. When current through the relay coil is suddenly interrupted, the magnetic field inside the coil collapses and generates a voltage spike in the opposite direction.

Without D1, this voltage spike could damage the transistor. D1 provides a safe path for the coil current when Q1 turns off. This process prevents excessive voltage

from appearing across the transistor.

Therefore, D1 is a flyback protection diode. Its role is essential for reliable switching and transistor protection, especially in real hardware.

6.2.6 What is the function of RL1?

RL1 is the relay used to switch the output load. It allows a low-power control circuit to operate a separate load circuit. The relay has two important parts: the coil and the contact. The coil is controlled by Q1, while the contact controls the lamp and buzzer circuit.

When the coil is energized, the relay contact changes state. This closes the load circuit and allows current to flow through the lamp and buzzer. When the coil is de-energized, the contact returns to its resting state and the load becomes inactive.

Therefore, RL1 provides electrical isolation and controlled switching between the sensor circuit and the output circuit.

6.2.7 What is the function of L1?

L1 is the lamp used as the visual output device of the project. When the relay contact closes, current flows through the lamp and the lamp turns on. This provides a visible indication that darkness has been detected and that the output stage has been activated.

The lamp gives the project a practical response. It can represent illumination, warning indication, or visible alarm output. In the structure of the project, L1 is part of the final actuation stage.

6.2.8 What is the function of BUZ1?

BUZ1 is the buzzer used as the audible output device of the project. When the relay contact closes, the buzzer receives power and produces sound. This sound indicates that the circuit has detected darkness and activated the alarm output.

The buzzer adds an audible warning function to the project. While the lamp provides visual output, the buzzer provides sound output. Together, the lamp and buzzer make the circuit more useful as an alarm or alert system.

6.3 Working Principle Questions

This section explains how the complete circuit operates under bright and dark conditions. The purpose of this section is to connect component functions into a full causal sequence. The circuit must not be explained as separate isolated parts only. It must be explained as one complete sensing, switching, relay, and output system.

6.3.1 What happens in bright condition?

In bright condition, light falls on the LDR and causes its resistance to decrease. Since the LDR is connected toward ground, the sensing node voltage becomes low. This low voltage is applied through R2 to the base of Q1, but it is not sufficient to turn on the transistor.

As a result, Q1 remains off. Since Q1 is off, the relay coil does not receive sufficient current. The relay remains de-energized, and its contact does not close the load circuit. Therefore, the lamp and buzzer remain off.

The complete bright-condition sequence is as follows:

*Bright light $\rightarrow R_{LDR}$ decreases $\rightarrow V_{sense}$ decreases
 $\rightarrow Q_1$ remains off $\rightarrow$ Relay remains off
 $\rightarrow$ Lamp and buzzer remain off*

This confirms that the circuit does not activate under bright light.

6.3.2 What happens in dark condition?

In dark condition, the light falling on the LDR is reduced. This causes the resistance of the LDR to increase. Since the LDR is part of the voltage divider, the sensing node voltage rises when the LDR resistance increases.

This higher sensing voltage is passed through R2 to the base of Q1. When the base-emitter voltage reaches approximately (0.6V) to (0.7V), Q1 begins to conduct. When Q1 conducts sufficiently, current flows through the relay coil. The relay becomes energized and changes the state of its contact.

The relay contact then completes the load circuit, allowing current to flow through the lamp and buzzer. The lamp turns on and the buzzer sounds.

The complete dark-condition sequence is as follows:

*Darkness $\rightarrow R_{LDR}$ increases $\rightarrow V_{sense}$ increases
 $\rightarrow Q_1$ turns on $\rightarrow$ Relay energizes
 $\rightarrow$ Load circuit closes $\rightarrow$ Lamp and buzzer activate*

This confirms that the circuit is a darkness-activated output system.

6.3.3 Why does the relay activate?

The relay activates because Q1 allows current to flow through the relay coil. The relay coil cannot be energized directly by the LDR voltage divider because the divider is a weak sensing stage. The transistor provides the required switching action.

When the base of Q1 receives enough voltage and current, the transistor conducts from collector to emitter. This creates a current path from the supply through the relay coil, through Q1, and then to ground. Once this current flows, the relay coil produces a magnetic field. This magnetic field moves the relay contact and activates the output circuit.

Therefore, the relay activates because the transistor converts the sensor signal into sufficient relay coil current.

6.3.4 Why does the output activate?

The output activates because the relay contact closes the load circuit. The lamp and buzzer are not controlled directly by the LDR or by the transistor base. They are controlled through the relay contact.

When the relay coil is energized, the normally open contact closes. This creates a complete electrical path for the load-side source. Current then flows through the lamp and buzzer, causing the lamp to glow and the buzzer to sound.

Therefore, the output activation is the final result of the relay contact closing after the sensor and transistor stages have detected darkness and energized the relay coil.

6.4 Calculation Questions

This section presents the mathematical basis of the circuit. The purpose of these questions is to prove that the project is not only a visual simulation, but also an electrically explainable design. The calculations connect the physical behavior of the LDR to the voltage divider, transistor switching, relay current, LED current, and saturation condition.

6.4.1 What is the voltage divider equation?

The voltage divider equation for the sensing node is

$$V_{sense} = V_{CC} \frac{R_{LDR}}{R_1 + R_{LDR}}$$

In this equation, V_{sense} is the voltage at the node between R_1 and the LDR. V_{CC} is the control supply voltage. R_1 is the fixed resistor value. R_{LDR} is the resistance of the LDR under the present light condition.

This equation is central to the project because it explains how the resistance of the LDR becomes a voltage signal. When R_{LDR} is low in bright light, V_{sense} becomes low. When R_{LDR} is high in darkness, V_{sense} becomes high.

Thus, the voltage divider equation mathematically proves why the circuit activates in darkness.

6.4.2 What voltage is needed to turn on Q1?

Q1 begins to conduct when the base-emitter junction receives approximately (0.6V) to (0.7V). This is the normal turn-on range for a silicon bipolar junction transistor such as the BC547.

The approximate condition for transistor turn-on is

$$V_{BE} \approx 0.6V \text{ to } 0.7V$$

For practical explanation, the switching condition may be written as

$$V_{sense} \geq 0.7V$$

This means that the sensing voltage must become high enough to forward-bias the base-emitter junction. However, reaching this voltage only begins transistor conduction. For strong relay operation, the transistor must also receive enough base current to saturate properly.

6.4.3 How is base current calculated?

The approximate base current can be calculated using

$$I_B \approx \frac{V_{CC} - V_{BE}}{R_1 + R_2}$$

This simplified expression assumes that the LDR resistance is very high in darkness, so the main resistance limiting base current is approximately the combination of R1 and R2. In the project, R1 is 10kΩ, R2 is 6.8kΩ, and V_{BE} is approximately (0.7V).

If $V_{CC} = 5V$, then

$$I_B \approx \frac{5 - 0.7}{10000 + 6800}$$

$$I_B \approx \frac{4.3}{16800}$$

$$I_B \approx 0.000256A$$

$$I_B \approx 0.256mA$$

This value shows that the base current is relatively small. Therefore, the relay coil current must be checked to determine whether Q1 will saturate properly.

6.4.4 How is LED current calculated?

The LED current is calculated using

$$I_{LED} = \frac{V_{CC} - V_F}{R_3}$$

In this equation, I_{LED} is the current through the LED. V_{CC} is the supply voltage. V_F is the forward voltage of the LED. R_3 is the current-limiting resistor.

If $V_{CC} = 5V$, $V_F \approx 2V$, and $R_3 = 330\Omega$, then

$$I_{LED} = \frac{5 - 2}{330}$$

$$I_{LED} = \frac{3}{330}$$

$$I_{LED} \approx 0.00909A$$

$$I_{LED} \approx 9.09mA$$

This current is suitable for many indicator LEDs. However, if the supply voltage is higher, R3 must be recalculated to prevent excessive LED current.

6.4.5 How is relay coil current calculated?

Relay coil current is calculated using Ohm's law:

$$I_{relay} = \frac{V_{relay}}{R_{coil}}$$

In this equation, I_{relay} is the current through the relay coil. V_{relay} is the voltage applied across the relay coil. R_{coil} is the resistance of the relay coil.

This calculation is necessary because the relay coil current is approximately the collector current of Q1. If the relay coil current is too high, the BC547 may not be able to drive it safely. Therefore, the relay coil current must be compared with the transistor current capability and the available base current.

The relay coil current also determines whether the transistor saturation condition is satisfied.

6.4.6 How is transistor saturation checked?

Transistor saturation is checked by comparing the base current with the collector current requirement. A conservative rule for a transistor used as a switch is

$$I_B \geq \frac{I_C}{10}$$

In this circuit, the collector current I_C is approximately the same as the relay coil current:

$$I_C \approx I_{relay}$$

Therefore, the base current must be large enough to drive the transistor into saturation. If I_B is too low, Q1 may operate in the active region instead of the saturation region. In that case, the relay may not energize reliably, and the transistor may heat because of increased voltage drop across it.

For example, if the relay coil current is (30mA), then the required base current for strong saturation is approximately

$$I_B \geq \frac{30mA}{10}$$

$$I_B \geq 3mA$$

If the available base current is only (0.256mA), the design may be marginal for that relay. This does not mean the circuit principle is wrong. It means the relay and transistor driver must be checked and possibly improved for real hardware.

6.5 Scheme Questions

This section verifies whether the circuit can be divided into correct functional sections. A proper schematic explanation must identify which components form the sensor section, switching section, protection section, output section, control side, and load side. This proves that the circuit is understood structurally, not only visually.

6.5.1 Which part is the sensor section?

The sensor section consists mainly of LDR1 and R1. The LDR senses light intensity, while R1 helps form the voltage divider that converts LDR resistance variation into a voltage signal.

The sensor section is the input stage of the circuit. It does not directly switch the relay or power the output load. Its purpose is to detect the environmental condition and produce a corresponding voltage at the sensing node.

Therefore, LDR1 and R1 form the light-sensing and signal-forming part of the scheme.

6.5.2 Which part is the switching section?

The switching section consists of R2 and Q1. R2 limits and controls base current, while Q1 acts as the transistor switch that drives the relay coil.

The switching section receives the sensing voltage from the voltage divider. When the sensing voltage is high enough, Q1 conducts and allows current through the relay coil. When the sensing voltage is too low, Q1 remains off and the relay stays de-energized.

Therefore, R2 and Q1 form the electronic control section that converts the sensor signal into relay activation.

6.5.3 Which part is the protection section?

The protection section is D1 connected across the relay coil. D1 is the flyback diode. Its purpose is to protect Q1 from the high reverse voltage generated when the relay coil is switched off.

The diode must be placed directly across the relay coil with correct polarity. During normal relay operation, it remains reverse-biased. During switch-off, it becomes forward-biased and provides a safe path for the coil current.

Therefore, D1 is the protective element that ensures safer and more reliable transistor switching.

6.5.4 Which part is the output section?

The output section consists of L1 and BUZ1. L1 is the lamp, and BUZ1 is the buzzer. These two components produce the final response of the project after the relay contact closes.

The lamp provides visual indication, while the buzzer provides audible indication. Together, they show that darkness has been detected and that the relay has successfully activated the load circuit.

Therefore, L1 and BUZ1 form the final actuation section of the scheme.

6.5.5 Which side is the control side?

The control side includes the LDR, resistors, transistor, relay coil, flyback diode, and LED indicator. This side is responsible for sensing the light condition and controlling the relay coil.

The control side operates as the decision and switching part of the circuit. It receives the light-dependent input, forms the sensing voltage, drives the transistor, energizes the relay coil, and protects the switching device from inductive kickback.

Therefore, the control side is the low-power electronic section that decides whether the output load should be activated.

6.5.6 Which side is the load side?

The load side includes the relay contact, lamp, buzzer, and load source. This side is controlled by the relay contact rather than directly by the LDR or transistor base.

When the relay coil is energized, the relay contact closes and completes the load circuit. The load source then provides current to the lamp and buzzer.

Therefore, the load side is the output power section of the project. It is separated from the control side by the relay.

6.6 Simulation Questions

This section verifies whether the circuit behavior has been tested correctly in Proteus. The purpose of simulation questions is to prove that the circuit operation was not assumed, but observed through controlled bright and dark test conditions.

6.6.1 What is tested in Proteus?

Proteus is used to test the response of the circuit to light and darkness. The simulation verifies whether the LDR changes the sensing voltage, whether Q1 responds to that voltage, whether the relay energizes correctly, and whether the lamp and buzzer activate as expected.

The simulation tests the complete chain of operation:

Light condition → *LDR resistance* → *Sensing voltage*
→ *Transistor state* → *Relay state* → *Output state*

Therefore, Proteus is used to confirm the functional behavior of the full circuit before physical construction.

6.6.2 What happens when the LDR is illuminated?

When the LDR is illuminated, its resistance decreases. This lowers the sensing voltage because the LDR is connected toward ground. Since the sensing voltage is low, Q1 does not receive sufficient base-emitter voltage to turn on.

As a result, Q1 remains off, the relay coil is not energized, and the relay contact does not close the load circuit. Therefore, the lamp and buzzer remain off.

This simulation condition confirms the off-state behavior of the project under bright light.

6.6.3 What happens when the LDR is dark?

When the LDR is dark, its resistance increases. This raises the sensing voltage at the divider output. The increased sensing voltage is applied through R2 to the base of Q1.

If the base-emitter junction receives sufficient voltage and current, Q1 turns on. Current then flows through the relay coil, energizing the relay. The relay contact closes the load circuit, causing the lamp and buzzer to activate.

This simulation condition confirms the on-state behavior of the project under darkness.

6.6.4 Which voltages should be measured?

The important voltages to measure are V_{sense} , V_{BE} , relay coil voltage, lamp voltage, and buzzer voltage.

The sensing voltage V_{sense} proves whether the LDR voltage divider is working correctly. The base-emitter voltage V_{BE} proves whether Q1 is receiving enough voltage to turn on. The relay coil voltage proves whether the relay is being energized. The lamp voltage and buzzer voltage prove whether the load circuit is receiving power after the relay contact closes.

These measurements are necessary because visual observation alone is not enough for a strong academic report. The circuit must be supported by measured electrical values.

6.6.5 Which currents should be measured?

The important currents to measure are base current, relay current, LED current, and load current.

The base current shows whether Q1 is being driven strongly enough. The relay current shows whether the relay coil current is within the transistor capability. The LED current shows whether the indicator LED is protected by R3. The load current shows whether the lamp and buzzer are operating within the relay contact and source capability.

These measurements provide quantitative evidence that the simulation is electrically consistent and not only visually successful.

6.7 Troubleshooting Questions

This section explains the possible faults that may appear in the circuit and how they should be diagnosed. Troubleshooting is an essential academic section because it proves that the student understands not only the successful state of the project, but also the failure conditions.

6.7.1 Why might the relay not switch?

The relay may not switch because the base current of Q1 is too weak, the relay voltage is incorrect, the transistor pinout is wrong, or the wiring is incorrect. The relay coil may also fail to energize if the supply voltage is insufficient or if the relay coil rating does not match the circuit supply.

The first diagnostic step is to measure the sensing voltage V_{sense} . If this voltage does not rise in darkness, the problem is in the LDR voltage divider. The second diagnostic step is to measure V_{BE} . If V_{BE} does not reach approximately (0.6V) to (0.7V), Q1 will not turn on. The third diagnostic step is to measure the relay coil voltage and current. If the coil does not receive its rated voltage and current, the relay will not activate.

Therefore, relay failure must be diagnosed through sensing voltage, transistor drive, relay coil voltage, relay coil current, and wiring verification.

6.7.2 Why might the lamp not glow?

The lamp may not glow because the relay contact is wired incorrectly, the load supply is missing, the lamp circuit is open, or the lamp model does not match the applied voltage. The relay coil may energize correctly, but if the contact side is not wired properly, the lamp will still remain off.

The correct diagnostic method is to check whether the relay contact actually closes when the relay is energized. Then the voltage across the lamp should be measured. If there is no voltage across the lamp, the fault is in the contact wiring, load source, or load path. If voltage is present but the lamp does not glow, the lamp rating or model must be checked.

Therefore, lamp failure is mainly a load-side problem unless the relay itself is not switching.

6.7.3 Why might the buzzer not sound?

The buzzer may not sound because its polarity is wrong, its required voltage is different from the applied voltage, the relay contact path is incomplete, or the load supply is not connected correctly. Some buzzers are polarity-sensitive, especially active electronic buzzers.

The correct diagnostic method is to measure the voltage across BUZ1 when the relay is activated. If the voltage is absent, the load path is incomplete. If the voltage is present but the buzzer does not sound, the buzzer polarity, rating, or model must be checked.

Therefore, buzzer failure must be diagnosed through voltage measurement, polarity verification, and load-path continuity.

6.7.4 Why might the transistor heat?

The transistor may heat because the relay current is too high, Q1 is not saturated, the transistor pinout is wrong, or the flyback diode is missing or incorrectly connected. If Q1 does not saturate, it may have a larger collector-emitter voltage drop while carrying relay current. This increases transistor power dissipation.

The transistor power dissipation can be estimated by

$$P_Q = V_{CE}I_C$$

If V_{CE} is too high during conduction, the transistor dissipates more power and may heat. The most important correction is to ensure that the base current is sufficient for saturation. The saturation condition should be checked using

$$I_B \geq \frac{I_C}{10}$$

Therefore, transistor heating indicates that current rating, saturation, wiring, or inductive protection must be reviewed.

6.7.5 Why might the relay chatter?

The relay may chatter because the circuit has no hysteresis or because the light level is unstable near the switching threshold. When the LDR voltage is close to the transistor

turn-on point, small changes in light intensity can repeatedly turn Q1 on and off. This causes the relay to rapidly energize and de-energize.

Relay chatter can also occur if the base drive is weak and the relay coil receives marginal current. In that case, the relay may not remain firmly energized.

The strongest correction is to add a comparator with hysteresis. Hysteresis creates two separate thresholds, one for switching on and one for switching off. This prevents repeated switching near the boundary condition.

Therefore, relay chatter is mainly a threshold-stability problem and should be corrected by hysteresis, stronger base drive, or adjustable threshold design.

6.7.6 Why might hardware fail even if Proteus works?

Hardware may fail even if Proteus works because real components are not always identical to simulation models. Real LDRs may have different resistance values, real relays may require more coil current, real transistors may have different pin arrangements, and real supplies may drop voltage under load.

Proteus simulation is useful for verifying logic and expected behavior, but it does not automatically prove real-hardware reliability. In physical construction, the actual relay coil resistance, actual LDR resistance, actual transistor pinout, actual supply voltage, and actual load current must be measured.

Therefore, real hardware requires verification of component values, electrical margins, wiring, ratings, and safety conditions.

6.8 Critical Thinking Questions

This section tests deeper engineering reasoning. It does not only ask what each component does. It asks why the design choices matter and how the circuit could be improved. These questions are important for academic defense because they show whether the student can evaluate the design beyond memorized explanations.

6.8.1 Why is a relay used?

A relay is used because it allows a low-power control circuit to switch a separate load circuit. The LDR and transistor stage operate at low control power, while the lamp and buzzer may require a different supply path or higher current. The relay separates these two domains.

The relay also provides electrical isolation between the sensing stage and the output stage. This makes the circuit more flexible because the control side can operate with a low-voltage signal while the load side can be powered separately.

Therefore, the relay is used to provide controlled switching, separation between circuit domains, and safe load activation.

6.8.2 Why is a transistor needed?

A transistor is needed because the LDR voltage divider cannot directly drive the relay coil. The voltage divider produces a sensing signal, but it is not designed to supply the current required by the relay coil.

Q1 solves this problem by acting as a current-switching device. A small base current controls a larger collector current through the relay coil. This allows the weak sensor signal to control the relay.

Therefore, the transistor is required to amplify the control ability of the sensor stage and provide reliable relay switching.

6.8.3 Why is a diode required across the relay coil?

A diode is required across the relay coil because the relay coil is an inductive load. When the transistor turns off, the magnetic field in the relay coil collapses and produces a voltage spike. This voltage spike can damage the transistor if it is not controlled.

The diode provides a safe path for the coil current during switch-off. It clamps the reverse voltage and protects Q1 from excessive electrical stress.

Therefore, the diode is required for transistor protection and reliable long-term switching operation.

6.8.4 Why would a comparator improve the design?

A comparator would improve the design because it would create a clearer switching decision between light and darkness. In the present circuit, the LDR voltage divider drives the transistor base directly through R2. This can produce gradual switching because transistor conduction changes continuously with base voltage.

A comparator compares the sensing voltage with a reference voltage and produces a sharper output transition. This makes the circuit more stable and predictable. It also allows the switching threshold to be set more accurately.

Therefore, a comparator improves the circuit by increasing switching precision, threshold control, and reliability.

6.8.5 Why would hysteresis improve the design?

Hysteresis would improve the design because it prevents relay chatter near the switching threshold. Without hysteresis, a small change in light level can repeatedly turn the transistor and relay on and off. This is especially likely when the LDR voltage is close to the switching point.

Hysteresis creates two different threshold levels. One threshold is used for turning the circuit on, and another threshold is used for turning it off. This separation prevents unstable repeated switching.

Therefore, hysteresis improves the design by making the transition between off state and on state more stable.

6.8.6 Why might a MOSFET be better than BC547?

A MOSFET may be better than a BC547 because it can often drive relay current with very little gate current. A BJT such as BC547 requires base current to remain saturated. If the base current is too weak, the transistor may heat or fail to drive the relay reliably.

A logic-level MOSFET can provide low-resistance switching when properly driven. This can reduce power loss and improve relay-driving reliability. For larger relay coils, a MOSFET is often a stronger switching option than a small-signal transistor.

Therefore, a MOSFET may improve the circuit by reducing control-current demand and increasing switching efficiency.

6.9 Viva and Presentation Questions

This section prepares the student for oral defense and academic presentation. It requires the student to explain the project clearly, identify components, describe operation, admit limitations, and propose improvements. This section is important because a successful project report must be defensible in spoken form.

6.9.1 Explain the project in one minute.

This project is an automatic darkness-activated alarm and load-control system designed in Proteus. It uses an LDR to detect the surrounding light level. In bright light, the LDR resistance is low, the sensing voltage remains low, Q1 stays off, the relay remains de-energized, and the lamp and buzzer remain off. In darkness, the LDR resistance increases, the sensing voltage rises, Q1 turns on, the relay energizes, and the relay contact activates the lamp and buzzer.

The project demonstrates the complete conversion of a physical environmental condition into an electrical output action. Its main stages are sensing, voltage division, transistor switching, relay control, protection, and output activation.

6.9.2 Identify every component and its function.

The LDR senses light intensity. R1 forms the voltage divider with the LDR. R2 limits the base current of Q1. Q1 is the BC547 transistor used to switch the relay coil. RL1 is the relay that controls the output load. D1 is the flyback diode that protects Q1 from relay coil kickback. D2 is the green LED used for visual indication. R3 limits the current through D2. L1 is the lamp that provides visual output. BUZ1 is the buzzer that provides audible output. The load source supplies power to the output circuit through the relay contact.

Each component has a defined function, and the circuit depends on the correct interaction of all these parts.

6.9.3 Explain the circuit operation in darkness.

In darkness, the resistance of the LDR increases. This raises the sensing voltage at the divider output. The increased voltage is applied through R2 to the base of Q1. When the base-emitter voltage becomes sufficient, Q1 conducts.

Once Q1 conducts, current flows through the relay coil. The relay energizes and its contact changes state. The closed relay contact completes the load circuit, allowing current to flow through the lamp and buzzer. The lamp turns on and the buzzer sounds.

Thus, darkness causes the complete activation chain from sensor input to final output.

6.9.4 Explain the circuit operation in light.

In bright light, the resistance of the LDR decreases. Because the LDR is connected toward ground, the sensing voltage becomes low. This low voltage is not sufficient to turn on Q1.

Since Q1 remains off, the relay coil does not receive enough current. The relay stays de-energized, and its contact does not complete the load circuit. Therefore, the lamp and buzzer remain off.

Thus, bright light keeps the project in its inactive state.

6.9.5 Explain one weakness of the project.

One important weakness of the project is that the base drive of Q1 may be weak for some real relay coils. The LDR divider and R2 may not provide enough base current to force the BC547 into strong saturation. If the transistor is not saturated, the relay may fail to activate reliably or the transistor may heat.

Another weakness is that the circuit has no hysteresis. Because of this, the relay may chatter when the light level is close to the switching threshold. The circuit also has no adjustable threshold unless a potentiometer is added.

Therefore, the project is correct in principle, but its real-hardware reliability depends on improved switching margin and threshold stability.

6.9.6 Explain one improvement.

One strong improvement is to add a comparator with hysteresis between the LDR voltage divider and the relay driver. The comparator would produce a clean switching signal, and hysteresis would prevent relay chatter near the threshold.

Another improvement is to use a potentiometer so that the darkness activation level can be adjusted. A stronger relay driver, such as a logic-level MOSFET or a relay driver integrated circuit, can also improve reliability. A stronger flyback diode may be used if the relay coil current is higher than the safe range of the existing diode.

The best improved structure is

LDR divider → *Adjustable threshold*
 → *Comparator with hysteresis* → *Strong relay driver*
 → *Relay* → *Lamp and buzzer*

This improved structure would make the circuit more stable, adjustable, and suitable for real hardware.

6.10 Final Evaluation Questions

This section evaluates whether the project achieved its aim and whether the design is ready for practical implementation. It separates simulation success from real-hardware readiness, which is essential for an honest academic conclusion.

6.10.1 Did the project achieve its aim?

The project achieves its aim if the lamp and buzzer activate automatically in darkness and remain inactive in bright light. The aim of the project is to design and simulate a

darkness-activated alarm and load-control circuit. If the Proteus simulation confirms this behavior, then the project has achieved its functional aim.

The successful operation must be proven by observing the bright condition and dark condition, and by measuring the necessary voltages and currents. The strongest proof is not only that the lamp and buzzer turn on, but that the sensing voltage, transistor state, relay state, and output state all match the expected working principle.

6.10.2 Was the Proteus simulation successful?

The Proteus simulation is successful if the circuit produces the correct response in both test conditions. In bright light, Q1 must remain off, the relay must remain de-energized, and the lamp and buzzer must remain off. In darkness, Q1 must turn on, the relay must energize, and the lamp and buzzer must activate.

A strong simulation result should include measured values for V_{sense} , V_{BE} , relay coil voltage, lamp voltage, buzzer voltage, base current, relay current, LED current, and load current. These measurements prove that the simulation is not only visually correct, but also electrically meaningful.

6.10.3 Is the design ready for real hardware?

The design is ready for real hardware only after transistor, relay, diode, supply, load, and safety margins are verified. Proteus simulation confirms the expected behavior, but physical construction requires real component checking.

Before hardware implementation, the relay coil current must be measured or taken from the relay datasheet. The BC547 must be checked to ensure that it can safely drive the relay coil. The base current must be checked to ensure transistor saturation. The flyback diode must be rated for the relay coil current. The LED resistor must be correct for the supply voltage. If the output side uses high voltage, relay contact rating, insulation, fuse protection, and enclosure safety must also be addressed.

Therefore, the design is conceptually ready, but hardware readiness requires electrical margin verification.

6.10.4 What is the best final improvement?

The best final improvement is to use an adjustable threshold, comparator hysteresis, and a stronger relay driver. The adjustable threshold allows the user to set the exact darkness level at which the circuit activates. The comparator provides a sharper switching decision. Hysteresis prevents relay chatter near the threshold. A stronger relay driver improves current capability and protects the sensing stage from overload.

The improved design would have the following structure:

LDR sensing → *Adjustable reference* → *Comparator*
→ *Hysteresis* → *MOSFET or relay driver* → *Relay*
→ *Lamp and buzzer*

This improvement would make the project more accurate, stable, reliable, and suitable for real implementation.

6.11 Chapter Summary

This section summarizes the purpose and intellectual value of the question chapter. Chapter 6 is not an optional question list. It is the academic defense system of the report. It proves that the project can be explained, calculated, simulated, criticized, and improved.

6.11.1 Question Chapter Purpose

The purpose of this chapter is to prepare the project for academic defense, viva, presentation, and evaluation. It organizes the most important questions that a teacher, examiner, or reviewer may ask about the circuit.

The chapter begins with basic concepts, then moves to component functions, working principle, calculations, scheme interpretation, simulation verification, troubleshooting, critical thinking, presentation defense, and final evaluation. This order is important because it moves from simple understanding to advanced engineering judgment.

Therefore, Chapter 6 strengthens the academic quality of the project by proving that the circuit is understood at every level.

6.11.2 Main Knowledge Areas

The main knowledge areas covered in this chapter are component function, working principle, calculation, scheme analysis, simulation, troubleshooting, critical thinking, oral presentation, and final evaluation.

Component function explains what each part does. Working principle explains how the full circuit behaves in bright and dark conditions. Calculation proves the electrical basis of the design. Scheme analysis proves structural understanding of the circuit. Simulation proves behavior in Proteus. Troubleshooting proves fault-diagnosis ability. Critical thinking proves design maturity. Presentation questions prepare the student for oral defense. Final evaluation separates simulation success from real-hardware readiness.

Therefore, this chapter connects theory, design, simulation, diagnosis, and evaluation into one complete academic defense structure.

6.11.3 Final Understanding

The final understanding is that the student must prove not only that the circuit works, but also why it works. A weak explanation only says that the lamp and buzzer turn on in darkness. A strong academic explanation shows the entire causal chain from LDR resistance to voltage divider output, from base drive to transistor switching, from relay coil energization to contact closure, and from contact closure to lamp and buzzer activation.

The complete final logic is

$$\begin{aligned} \text{Darkness} &\rightarrow R_{LDR} \text{ increases} \rightarrow V_{sense} \text{ increases} \\ &\rightarrow V_{BE} \text{ reaches turn on level} \rightarrow Q_1 \text{ conducts} \\ &\rightarrow \text{Relay coil energizes} \rightarrow \text{Relay contact closes} \\ &\rightarrow \text{Lamp and buzzer activate} \end{aligned}$$

The student must also understand that Proteus success is not the same as complete hardware proof. Real hardware requires verification of LDR behavior, base current, relay current, transistor saturation, diode rating, LED current, load rating, and safety conditions.

Therefore, Chapter 6 establishes the final academic defense of the project: the circuit is structurally valid, mathematically explainable, simulation-testable, troubleshootable, and improvable. Its strongest form is not merely that it operates, but that every operation can be explained through correct electrical reasoning.

